Information Theory & Coding



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All the figures in this set of lecture notes have either been coded using `Tikz` or in `Python`.

Chapter 1

Source Coding/Data Compression

In this chapter, we consider a source alphabet $\mathcal{U}$, such that $|\mathcal{U}| < \infty$. We first present a series of definitions for the setup of the later sections, where we will then study the expected length of a source code.

1.1 Source Codes and Kraft Inequalities

Definition 1.1.1. *A (binary)* ***source code*** *is represented by a function* $c : \mathcal{U} \longrightarrow \{0,1\}^* := \{\text{null}, 0, 1, 00, 01, 10, 11, 000, ...\}$.

In this course, for an alphabet $\mathcal{U} = \{u_1, ..., u_n\}$ and output $\{v_1, ..., v_n\}$ we will write $\mathcal{U} = \{u_1, ..., u_n\} \xrightarrow{c} \{v_1, ..., v_n\}$ to signify $u_1 \xrightarrow{c} v_1$, ..., $u_n \xrightarrow{c} v_n$, where c is a source code. Furthermore, an **injective** (or **non-singular**) code is one such that $u \neq v \Longrightarrow c(u) \neq c(v)$.

Example 1.1.2 (Injective source codes)**.** $\mathcal{U} = \{a, b, c\} \xrightarrow{c} \{\text{null}, 0, \text{null}\}$ *is <u>not</u> an injective code, whereas* $\mathcal{U} = \{a, b, c\} \xrightarrow{c} \{\text{null}, 0, 1\}$ *is an injective code.*

Now, given $c' : \mathcal{U}' \longrightarrow \{0,1\}^*$ and $c'' : \mathcal{U}'' \longrightarrow \{0,1\}^*$ we define $c = c' \times c'' : \mathcal{U}' \times \mathcal{U}'' \longrightarrow \{0,1\}^*$ such that $c(u', u'') = c'(u')c''(u'')$.

Example 1.1.3. *Let* $c' : \{a, b, c\} \xrightarrow{c} \{0, 1, 11\}$ *and* $c'' : \{\alpha, \beta\} \xrightarrow{c} \{000, 10111\}$*, then* $(c'' \times c')(\alpha b) = 0001$ *and* $(c' \times c'')(a\alpha) = 0000$.

Definition 1.1.4. *Given* $c : \mathcal{U} \longrightarrow \{0,1\}^*$ *and* $n \geq 1$*, we define* $\boldsymbol{c^n}$ *as* $c^n = c \times c \times ... \times c$ *(n times) where* $c^n : \mathcal{U}^n \longrightarrow \{0,1\}^*$ *and* $c^n(u_1...u_n) = c(u_1)...c(u_n)$*. Also, we may consider* $\boldsymbol{c^*}$ *to be* $c^* : \mathcal{U}^* \longrightarrow \{0,1\}^*$ *such that* $c^*(u_1...u_n) = c^n(u_1...u_n)$.

Definition 1.1.5. *A code is* ***uniquely decodable*** *if* c^* *is injective.*

Example 1.1.6. *Let* $\mathcal{U} = \{a, b, c\} \xrightarrow{c} \{0, 10, 11\}$ *is uniquely decodable.*

Definition 1.1.7. *A sequence $u_1...u_n$ is said to be a **prefix** of a sequence $v_1...v_m$ if $m \geq n$ and $u_1...u_n = v_1...v_n$.*

Example 1.1.8. *Let $v = \text{coding}$, then v has prefixes $\{\text{null}, \text{c}, \text{co}, \text{cod}, \text{codi}, \text{codin}, \text{coding}\}$.*

Definition 1.1.9. *A code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ is said to be **prefix-free** if $u \neq v$ and where $c(u)$ is not a prefix of $c(v)$.*

Example 1.1.10. *$\{a,b,c\} \xrightarrow{c} \{0,10,11\}$ is prefix-free.*

Theorem 1.1.11. *If c is prefix-free, then c is uniquely decodable.*

Proof. Suppose c is prefix-free but not uniquely decodable. Then there exists $u^n := u_1...u_n$ and v^m such that $c^*(u^n) = c^*(v^m)$. Among these examples, pick one for which $n+m$ is the smallest. In this example, $u_1 \neq v_1$ otherwise we may construct a shorter example. Also, $c(u_1)c(u_2)...c(u_n) = c(v_1)c(v_2)...c(v_n)$. Thus, either $c(u_1)$ is a prefix of $c(v_1)$ or $c(v_1)$ is a prefix of $c(u_1)$. We have a contradiction. □

Example 1.1.12 (Uniquely decodable $\not\Longrightarrow$ Prefix-free)**.** *$\{a,b,c\} \xrightarrow{c} \{0,01,11\}$ is uniquely decodable but not prefix-free.*

As a result, we may picture the universe of source codes. We therefore have the following sets of inclusions: prefix-free $\subset$ uniquely decodable $\subset$ injective $\subset$ all source codes.

Definition 1.1.13. *Given a code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ we define the **Kraft sum** of c, $\text{KS}(c)$, as:*

$$\text{KS}(c) := \sum_{u \in \mathcal{U}} 2^{-\text{length}(c(u))}. \tag{1.1}$$

Example 1.1.14. *As an application we have:*
- *$\mathcal{U} = \{a,b,c\} \xrightarrow{c} \{\text{null}, \text{null}, \text{null}\} \Longrightarrow \text{KS}(c) = 3$.*
- *$\mathcal{U} = \{a,b,c\} \xrightarrow{c} \{\text{null}, 0, 1\} \Longrightarrow \text{KS}(c) = 2$.*

Theorem 1.1.15. *If $c : \mathcal{U} \longrightarrow \{0,1\}^*$ is injective, then:*

$$\text{KS}(c) \leq \log_2(1 + |\mathcal{U}|). \tag{1.2}$$

Proof. The injective code c with largest $\text{KS}(c)$ will have the first $|\mathcal{U}|$ elements of $\{0,1\}^*$ as its codewords. We have that $|\mathcal{U}| = \sum_{k=0}^{j-1} 2^k + r = 2^j - 1 + r$, where $0 \leq r < 2^j$. Thus, $\max_{c:\text{ injective}} \text{KS}(c) = j + r2^{-j}$. As a result:

$$1+|\mathcal{U}| = 2^j(1+r2^{-j}) \Longrightarrow \log_2(1+|\mathcal{U}|) = j+\log_2(1+r2^{-j}). \tag{1.3}$$

Now, since $0 \leq r < 2^j$ we remark that $0 \leq r2^{-j} < 1$. Thus, since $\log_2(1+x) \geq x$ for $0 \leq x \leq 1$, we conclude:

$$\log_2(1+|\mathcal{U}|) = j+\log_2(1+r2^{-j}) \geq j+r2^{-j} = \max_{c:\text{ injective}} \text{KS}(c). \tag{1.4}$$

□

Lemma 1.1.16.

$$\text{KS}(c' \times c'') = \text{KS}(c')\text{KS}(c'') \tag{1.5}$$

Proof. This is a simple computation:

$$\begin{aligned}\text{KS}(c' \times c'') = \sum_{u',u''} 2^{-[\text{length}(c'(u'))+\text{length}(c''(u''))]} &= \left(\sum_{u'} 2^{-\text{length}(c'(u'))}\right)\left(\sum_{u''} 2^{-\text{length}(c''(u''))}\right) \\ &= \text{KS}(c')\text{KS}(c'').\end{aligned}$$

□

Corollary 1.1.17.

$$\text{KS}(c^n) = \left[\text{KS}(c)\right]^n \tag{1.6}$$

Proof. This holds using Lemma (1.1.16). □

Corollary 1.1.18. *If c is uniquely decodable, then* $\text{KS}(c) \leq 1$.

Proof. Since c is uniquely decodable, then for all $n \geq 1$ we have that c^n is injective. Now:

$$\left[\text{KS}(c)\right]^n = \text{KS}(c^n) \leq \log_2(1+|\mathcal{U}^n|) = \log_2(1+|\mathcal{U}|^n). \tag{1.7}$$

This inequality holds only when $\text{KS}(c) \leq 1$. □

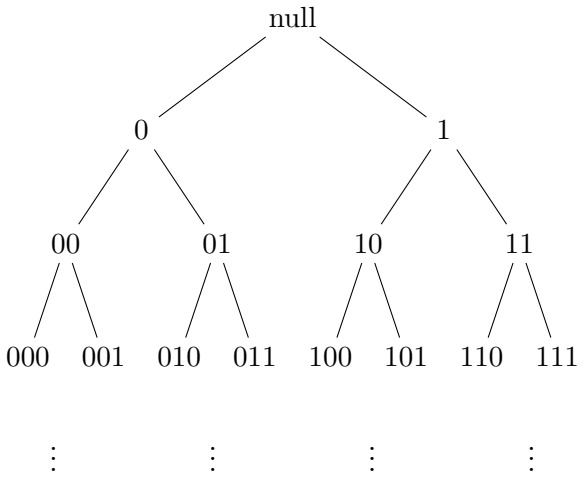
Theorem 1.1.19. *Suppose $L : \mathcal{U} \longrightarrow \{0,1,2,...\}$ satisfies $\sum_{u\in\mathcal{U}} 2^{-L(u)} \leq 1$. Then, there exists a prefix-free code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ such that:*

$$\text{length}(c(u)) = L(u), \qquad \forall u \in \mathcal{U}. \tag{1.8}$$

Before presenting the proof of this Theorem, we refer one last corollary in the case where we are considering a uniquely decodable code.

Corollary 1.1.20. *If $c : \mathcal{U} \longrightarrow \{0,1\}^*$ is uniquely decodable, then there exists a prefix-free code c' such that* $\text{length}(c'(u)) = \text{length}(c(u))$ *for all $u \in \mathcal{U}$.*

Proof of Theorem (1.1.19). We consider the following binary tree:



We now follow the lines of the following algorithm to pick nodes in the above binary tree (we consider $A_d = 2^d$ free nodes at depth d):

1: order $\mathcal{U} = \{1, 2, ..., k\}$ such that $L(1) \leq L(2) \leq ... \leq L(k)$
2: set $i = 0$ and start with an empty tree
3: **while** $i < k$ **do**
4: • pick an available node at depth $L(i+1)$
5: • assign $c(i+1) =$ label of the node
6: • mark the node and all its descendants as unavailable
7: • $i \leftarrow i+1$
8: **end while**

Now, the number of available nodes at depth $d \geq L(i)$ just before the **while** loop is:

$$2^d - \sum_{j=1}^{i} 2^{d-L(j)} = 2^d \left(1 - \sum_{j=1}^{i} 2^{-L(j)}\right) > 0, \tag{1.9}$$

where we have used the fact that $\sum_{j=1}^{i} 2^{-L(j)} < 1$ since $\mathrm{KS}(c) \leq 1$ by Corollary (1.1.18). Thus, this algorithm always works since there is without fail a strictly positive number of available nodes. □

1.2 Expected Codeword Length

Suppose $\mathcal{U}$ is an alphabet and U is a random variable taking values in $\mathcal{U}$. Consider also a code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ and define the expected number of bits in the representation of U as:

$$\mathbb{E}\big[\mathrm{length}(c(U))\big] = \sum_{u \in \mathcal{U}} \mathbb{P}(U = u)\mathrm{length}(c(u)). \tag{1.10}$$

In this setup, an interesting problem to solve would be to minimize the above expectation, i.e.

$$\min_{c:\text{ uniquely decodable}} \mathbb{E}\big[\mathrm{length}(c(U))\big]. \tag{1.11}$$

We also note that using some statements from the previous section, we may rewrite this problem in two other ways:

$$\min_{c:\text{ uniquely decodable}} \mathbb{E}\big[\text{length}(c(U))\big] \overset{(1.1.11)}{=} \min_{c:\text{ prefix-free}} \mathbb{E}\big[\text{length}(c(U))\big] \overset{(1.1.18)}{=} \min_{\sum_{u\in\mathcal{U}} 2^{-L(u)}\leq 1} \left(\sum_{u\in\mathcal{U}} p(u)L(u)\right),$$

where L are integer valued functions of $\mathcal{U}$ and $p(u) := \mathbb{P}(U=u)$. We now present a lower-bound on the expected length of a codeword.

Theorem 1.2.1. *For any uniquely decodable code $c:\mathcal{U}\longrightarrow\{0,1\}^*$ we have:*

$$\mathbb{E}\big[\text{length}(c(U))\big] \geq H(U), \tag{1.12}$$

where $H(U)=\sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{1}{p(u)}\right) = -\sum_{u\in\mathcal{U}} p(u)\log_2(p(u))$ is the entropy of U, and $p(u) := \mathbb{P}(U=u)$.

Proof. We study the difference,

$$H(U)-\mathbb{E}\big[\text{length}(c(U))\big] = \sum_{u\in\mathcal{U}} p(u)\left[\log_2\left(\frac{1}{p(u)}\right) - L(u)\right], \tag{1.13}$$

where $L(u)=\text{length}(c(U))$ satisfies $\sum_{u\in\mathcal{U}} 2^{-L(u)}\leq 1$ (since c is uniquely decodable). Now, for convenience set $q(u) := 2^{-L(u)}$ so that $\sum_{u\in\mathcal{U}} q(u)=1$ and $L(u)=\log_2(1/q(u))$. Then:

$$H(U)-\mathbb{E}\big[\text{length}(c(U))\big] = \sum_{u\in\mathcal{U}} p(u)\left[\log_2\left(\frac{1}{p(u)}\right) - \log_2\left(\frac{1}{q(u)}\right)\right] = \sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{q(u)}{p(u)}\right). \tag{1.14}$$

Recall that $\log_2$ is a constant multiple of the natural logarithm, thus it suffices to show that $\sum_{u\in\mathcal{U}} p(u)\ln\left(\frac{q(u)}{p(u)}\right)\leq 0$. To this end, consider the Taylor expansion of $\ln(\xi)$ around $\xi=1$ and we find $\ln(\xi)=\xi-1-(2\alpha^2)^{-1}(\xi-1)^2\leq\xi-1$. Thus:

$$\sum_{u\in\mathcal{U}} p(u)\ln\left(\frac{q(u)}{p(u)}\right) \leq \sum_{u\in\mathcal{U}} p(u)\left(\frac{q(u)}{p(u)}-1\right) = \sum_{u\in\mathcal{U}} (q(u)-p(u)) = 0. \tag{1.15}$$

□

We may now also present an upper-bound for the expected length of a codeword.

Theorem 1.2.2. *Given a random variable U taking values in $\mathcal{U}$, there exists a prefix-free code c such that:*

$$\mathbb{E}\big[\text{length}(c(U))\big] \leq H(U)+1. \tag{1.16}$$

Proof. Choose $L(u)=\left\lceil\log_2\left(\frac{1}{p(u)}\right)\right\rceil < \log_2\left(\frac{1}{p(u)}\right)+1$ so that $2^{-L(u)}\leq p(u)$ which gives $\sum_{u\in\mathcal{U}} 2^{-L(u)}\leq 1$. Hence, by Theorem (1.1.19), there exists a prefix-free code c with $\text{length}(c(u))=L(u)<\log_2\left(\frac{1}{p(u)}\right)+1$. As a result, $\mathbb{E}\big[\text{length}(c(U))\big]\leq\sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{1}{p(u)}\right)+1 = H(U)+1$. □

As an even more general result, we may provide a lower-bound on the expected length of a codeword for a general code.

Theorem 1.2.3 (Expected codeword length for a general code)**.** *For **any** code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ we have:*

$$\mathbb{E}\big[\text{length}(c(U))\big] \geq H(U) - \log_2\big(\text{KS}(c)\big), \tag{1.17}$$

where $H(U)$ is the entropy of U and $\text{KS}(c)$ is the Kraft sum of the code c.

Proof. Let $q(u) = 2^{-L(u)}$ so that $\sum_{u\in\mathcal{U}} q(u) = \text{KS}(c)$ and $L(u) = \log_2\left(\frac{1}{q(u)}\right)$. Thus we have:

$$\begin{aligned} H(U) - \mathbb{E}\big[\text{length}(c(U))\big] &= \sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{q(u)\cdot\text{KS}(c)}{p(u)\cdot\text{KS}(c)}\right) \\ &= \log_2(\text{KS}(c)) + \underbrace{\sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{q(u)/\text{KS}(c)}{p(u)}\right)}_{\leq 0} \leq \log_2(\text{KS}(c)). \end{aligned}$$

□

Corollary 1.2.4. *From the above Theorem, we have the following results:*
(i) Theorem (1.2.1).
(ii) For an injective code c: $\mathbb{E}\big[\text{length}(c(U))\big] \geq H(U) - \log_2\log_2\big(1+|U|\big)$.

Proof. For (ii), from Theorem (1.1.15), we know that an injective code satisfies $\text{KS}(c) \leq \log_2(1+|\mathcal{U}|)$. Thus:

$$\mathbb{E}\big[\text{length}(c(U))\big] \geq H(U) - \log_2\big(\text{KS}(c)\big) \geq H(U) - \log_2\log_2\big(1+|U|\big). \tag{1.18}$$

□

Consider now a code $c_n : \mathcal{U}^n \longrightarrow \{0,1\}^*$ which is injective. Then, the number of hits per letter is lower-bounded by,

$$\frac{1}{n}\mathbb{E}\big[\text{length}(c_n(u_1...u_n))\big] \geq \frac{1}{n}\left[H(u_1...u_n) - \log_2\log_2\big(1+|\mathcal{U}|^n\big)\right] = \frac{1}{n}H(u_1...u_n) - \epsilon_n, \tag{1.19}$$

for a small ϵ_n as n gets large. On the other hand, there exists a prefix-free code $c_n : \mathcal{U}^n \longrightarrow \{0,1\}^n$ such that,

$$\frac{1}{n}\mathbb{E}\big[\text{length}(c_n(u_1...u_n))\big] \leq \frac{1}{n}H(u_1...u_n) + \frac{1}{n}. \tag{1.20}$$

Thus, we have introduced useful bounds on $\mathbb{E}\big[\text{length}(c(U))\big]$ where $\mathbb{E}\big[\text{length}(c(U))\big] \overset{n\to\infty}{\sim} \frac{1}{n}H(u_1...u_n)$.

1.3 Prefix-free Codes with Minimal Expected Codeword Length

In this section, we consider the following problem: given a probability density $\{p(u) : u \in \mathcal{U}\}$ we find $\{L(u) : u \in \mathcal{U}\}$ satisfying $\sum_{u\in\mathcal{U}} 2^{-L(u)} \leq 1$ to minimize $\sum_{u\in\mathcal{U}} p(u)L(u)$. This problem is actually

equivalent to solve the following: given p, we find q such that $\sum_{u\in\mathcal{U}} q(u) \leq 1$ for $q(u) \in \left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, ...\right\}$ to minimize $\sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{1}{q(u)}\right)$. As a first approach, consider the *relaxed* problem:

$$\min_{q:\ \sum_{u\in\mathcal{U}} q(u)=1 \atop q(u)\geq 0} \left(\sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{1}{q(u)}\right)\right). \tag{1.21}$$

Using Lagrange multipliers, it is not difficult to show that the minimizer of q is equal to p with value equal to $H(U)$. Now, let us look at four properties of optimal prefix-free codes.

Proposition 1.3.1. *If $p(u) > p(v)$ then $L^*(u) \leq L^*(v)$ where $L^*(u)$ is the length of the optimal code $c(u)$.*

Proof. Suppose $p(u) > p(v)$ and $\text{length}(c(u)) > \text{length}(c(v))$, let us show that c cannot be optimal. Consider a new code c' for which $c'(v) = c(u)$ and $c'(u) = c(v)$. Now,

$$\mathbb{E}\big[\text{length}(c'(U))\big] = \sum_{\substack{w\in\mathcal{U}\\ w\neq u,v}} p(w)\text{length}(c'(w)) + p(u)\text{length}(c(v)) + p(v)\text{length}(c(u)), \tag{1.22}$$

and also,

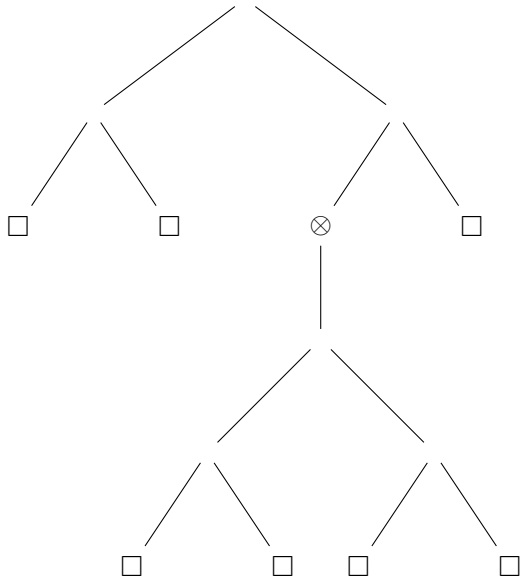
$$\mathbb{E}\big[\text{length}(c(U))\big] = \sum_{\substack{w\in\mathcal{U}\\ w\neq u,v}} p(w)\text{length}(c(w)) + p(u)\text{length}(c(u)) + p(v)\text{length}(c(v)). \tag{1.23}$$

Taking the difference, we obtain:

$$\mathbb{E}\big[\text{length}(c'(U))\big] - \mathbb{E}\big[\text{length}(c(U))\big] = \underbrace{\big(p(u)-p(v)\big)}_{>0}\underbrace{\big(\text{length}(c(v)) - \text{length}(c(u))\big)}_{<0} < 0. \tag{1.24}$$

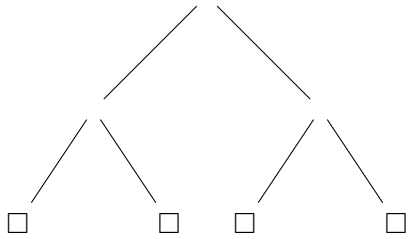
We have thus constructed a code c' being more optimal than c. □

Consider now the following binary tree, where $\otimes$ represents a single offspring and $\square$ is a codeword:



Proposition 1.3.2. *In the binary tree representation of an optimal prefix-free code c, each node is either a codeword (□) or it has two children. Thus we must have a complete binary tree (i.e. a tree in which all the nodes have two children except possibly for the nodes in the last level).*

In other words, if we consider the tree represented above [↘ is 0 and ↙ is 1], we may move up the block given below in place of ⊗:



Corollary 1.3.3. *There are at least two longest codewords.*

Corollary 1.3.4. *There is an optimal prefix-free code c such that the two least likely codewords are assigned as sibling codewords.*

We now go back to our original problem, presented at the beginning of this section. We recall the *design problem*:

Given $k = |\mathcal{U}|$ and $p_1 \geq p_2 \geq ... \geq p_k$, find $\{L_1, ..., L_k\}$ (integers) with $\sum_{i=1}^{k} 2^{-L_i} \leq 1$ to minimize $\sum_{i=1}^{k} p_i L_i$.

$$\Updownarrow$$

We know that $L_{k-1} = L_k$ in the optimal solution. Then, find $\{L_1, ..., L_{k-1}\}$ such that:

$$\sum_{i=1}^{k} 2^{-L_i} = \sum_{i=1}^{k-2} 2^{-L_i} + 2^{-L_{k-1}} + 2^{-L_k} = \sum_{i=1}^{k-2} 2^{-L_i} + 2 \cdot 2^{-L_{k-1}} = \sum_{i=1}^{k-2} 2^{-L_i} + 2^{-(L_{k-1}-1)} \leq 1,$$

to minimize $\sum_{i=1}^{k} p_i L_i = \sum_{i=1}^{k-2} p_i L_i + (p_{k-1} + p_k)(L_{k-1} - 1) + (p_{k-1} + p_k)$.

$$\Updownarrow$$

Given $(k-1)$ and $(p'_1, p'_2, ..., p'_{k-2}, p'_{k-1}) := (p_1, p_2, ..., p_{k-2}, p_{k-1} + p_k)$, find $\{L'_1, ..., L'_{k-1}\}$ such that $\sum_{i=1}^{k-1} 2^{-L'_i} \leq 1$ and minimize $\sum_{i=1}^{k-1} p'_i L'_i$.

Now, note that we have transformed a problem of size k to a problem of size $k-1$. As a result we could continue this process to end up with a problem of size 1. This is known as the **Huffman Procedure**.

Example 1.3.5. *Consider an alphabet* $\mathcal{U} = \{a, b, c, d, e, f\}$ *such that* $p(a) = 0.3$, $p(b) = p(c) = 0.2$, $p(d) = p(e) = p(f) = 0.1$. *The idea is, at each step, to pair the two least likely codewords (they become siblings). Following this procedure, we find: (1) at first level in the binary tree* $p(ef) = 0.2$ *and* $p(cd) = 0.3$, *(2) at second level in the binary tree* $p(b[ef]) = 0.4$ *and* $p(a[cd]) = 0.6$, *(3) at third level (and the last) in the binary tree we have* $p\big([a[cd]]\,[b[ef]]\big) = 1$.
Thus, if we set $\uparrow$ *as 1 and* $\downarrow$ *as 0 in the tree, an optimal code in this case is given by:* $c(a) = 11$, $c(b) = 01$, $c(c) = 101$, $c(d) = 100$, $c(e) = 001$ *and* $c(f) = 000$ *(draw the binary tree to see this). Also, note that* $\mathbb{E}\big(\text{length}(c(U))\big) = 1 + 0.6 + 0.4 + 0.3 + 0.2 = 2.5$.

Chapter 2

Information Measures

We have seen previously that the entropy of U, $H(U)$, and the entropy of the random vector $H(u_1...u_n)$ playing an important role in describing the efficiency of uniquely decodable codes. In this chapter we present the tools used in Information Theory, such as the notions of Entropy and KL-divergence.

2.1 Discrete Probability Distributions

Definition 2.1.1. *Given a random variable U taking values in $\mathcal{U}$ with $p(u) = \mathbb{P}(U = u)$, we define the **entropy** as:*

$$H(U) = \sum_{u \in \mathcal{U}} p(u) \log_2 \left(\frac{1}{p(u)} \right) = - \sum_{u \in \mathcal{U}} p(u) \log_2 (p(u)), \tag{2.1}$$

with the convention $0 \log_2(1/0) = 0$.

Lemma 2.1.2. *$H(U) \geq 0$, with equality if and only if there exists $u_0 \in \mathcal{U}$ such that $\mathbb{P}(U = u_0) = 1$ (i.e. U is deterministic).*

Proof. Since $0 \leq p(u) \leq 1$ we have the following series of implications:

$$\log_2 \left(\frac{1}{p(u)} \right) \geq 0 \Longrightarrow p(u) \log_2 \left(\frac{1}{p(u)} \right) \geq 0 \Longrightarrow H(U) \geq 0.$$

□

Lemma 2.1.3. *$H(U) \leq \log_2(|\mathcal{U}|)$, with equality if and only if U is uniformly distributed on $\mathcal{U}$.*

Proof. We consider the difference and use the fact that $\ln(\xi) \leq \xi - 1$ to find:

$$\begin{aligned} H(U) - \log_2(|\mathcal{U}|) &= \left(\sum_{u \in \mathcal{U}} p(u) \log_2 \left(\frac{1}{p(u)} \right) \right) - \left(\sum_{u \in \mathcal{U}} p(u) \log_2 (|\mathcal{U}|) \right) \\ &= \sum_{u \in \mathcal{U}} p(u) \log_2 \left(\frac{1}{|\mathcal{U}| p(u)} \right) \end{aligned}$$

$$
\begin{aligned}
&= \log_2(e) \sum_{u \in \mathcal{U}} p(u) \ln\left(\frac{1}{|\mathcal{U}| p(u)}\right) \\
&\leq \log_2(e) \sum_{u \in \mathcal{U}} p(u) \left(\frac{1}{|\mathcal{U}| p(u)} - 1\right) \\
&= \log_2(e) \left(\sum_{u \in \mathcal{U}} \frac{1}{|\mathcal{U}|} - \sum_{u \in \mathcal{U}} p(u)\right) = 0.
\end{aligned}
$$

□

Suppose now that U and V are random variables, then $W = (U,V)$ is another random variable taking values in $\mathcal{W} = \mathcal{U} \times \mathcal{V}$ if $w = (u,v) \in \mathcal{W}$. Then set $\mathbb{P}(W = w) = \mathbb{P}(U = u, V = v) := p(uv)$ to be the joint distribution of (U,V). We furthermore consider the definition of the **joint entropy** to be:

$$
H(U,V) = \sum_{u,v} p(uv) \log_2\left(\frac{1}{p(uv)}\right) = -\sum_{u,v} p(uv) \log_2(p(uv)). \tag{2.2}
$$

Theorem 2.1.4. *$H(U,V) \leq H(U) + H(V)$, with equality if and only if U and V are independent.*

Before we prove the previous Theorem, we present a result that will be useful for the proof.

Theorem 2.1.5 (Log-Sum Inequality)**.** *If $a_1, ..., a_k \geq 0$ and $b_1, ..., b_k \geq 0$ with $A = \sum_i a_i$ and $B = \sum_i b_i$, then:*

$$
\sum_{i=1}^{k} a_i \log\left(\frac{a_i}{b_i}\right) \geq A \log\left(\frac{A}{B}\right), \tag{2.3}
$$

with the conventions $0 \log(0/*) = 0$ and $* \log(*/0) = +\infty$ if $* \neq 0$.

Proof. Assume without loss of generality that $A, B > 0$. We seek for,

$$
\sum_{i=1}^{k} \frac{a_i}{A} \log\left(\frac{a_i}{b_i}\right) \geq \log\left(\frac{A}{B}\right) \iff \sum_{i=1}^{k} \frac{a_i}{A} \log\left(\frac{a_i}{b_i}\right) - \log(A) \geq \log\left(\frac{A}{B}\right) - \log(A), \tag{2.4}
$$

Now, let $p_i = \frac{a_i}{A}$ such that $\sum_{i=1}^{k} p_i = 1$ and so,

$$
\sum_{i=1}^{k} p_i \log\left(\frac{p_i}{b_i}\right) \geq \log\left(\frac{1}{B}\right). \tag{2.5}
$$

Similarly, we add $\log(B)$ on both sides and set $q_i = \frac{b_i}{B}$ such that $\sum_{i=1}^{k} q_i = 1$ and so,

$$
\sum_{i=1}^{k} p_i \log\left(\frac{p_i}{q_i}\right) \geq 0 \iff \sum_{i=1}^{k} p_i \log\left(\frac{q_i}{p_i}\right) \leq 0. \tag{2.6}
$$

But using $\ln(\xi) \leq \xi - 1$ we know that (2.6) holds since:

$$
\sum_{i=1}^{k} p_i \log\left(\frac{q_i}{p_i}\right) \leq \sum_{i=1}^{k} p_i \left(\frac{q_i}{p_i} - 1\right) = \sum_{i=1}^{k} q_i - \sum_{i=1}^{k} p_i = 0. \tag{2.7}
$$

□

We are now ready to prove Theorem (2.1.4) using the Log-Sum Inequality.

Proof of Theorem (2.1.4). We want to show,

$$\sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{1}{p_{U,V}(u,v)}\right) \leq \sum_u p_U(u)\log_2\left(\frac{1}{p_U(u)}\right) + \sum_v p_V(v)\log_2\left(\frac{1}{p_V(v)}\right). \tag{2.8}$$

Now, note that $\{U=u\} = \bigcup_v \{U=u, V=v\}$ is a disjoint union and thus $\mathbb{P}(U=u) = \sum_v \mathbb{P}(U=u, V=v)$ and thus $p_U(u) = \sum_v p_{U,V}(u,v)$ (this is just marginalization). As a result, we may write:

$$\sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{1}{p_{U,V}(u,v)}\right) \leq \sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{1}{p_U(u)}\right) + \sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{1}{p_V(v)}\right), \tag{2.9}$$

which is equivalent to the inequality,

$$\sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{p_{U,V}(u,v)}{p_U(u)p_V(v)}\right) \geq 0. \tag{2.10}$$

However this inequality is true by the Log-Sum Inequality, thus we are done. □

Definition 2.1.6. *Given a pair of random variables U and V, the **mutual information** between U and V is defined as:*

$$I(U;V) = H(U) + H(V) - H(U,V) = H(U) - H(U|V) = H(V) - H(V|U). \tag{2.11}$$

Corollary 2.1.7. *$I(U;V) \geq 0$ with equality if and only if U and V are independent.*

Proof. The proof is immediate with Theorem (2.1.4). □

Observe that $H(U) = \mathbb{E}(f(U))$ with $f(u) = \log_2\left(\frac{1}{p(u)}\right)$ for fixed $u \in \mathcal{U}$. Also recall that $p_{U,V}(u,v) = p_U(u)p_{V|U}(v|u)$ and so $-\log_2(p_{U,V}(u,v)) = -\log_2(p_U(u)) - \log_2(p_{V|U}(v|u))$ giving us:

$$\mathbb{E}\left[\log_2\left(\frac{1}{p_{U,V}(u,v)}\right)\right] = \mathbb{E}\left[\log_2\left(\frac{1}{p_U(u)}\right)\right] + \mathbb{E}\left[\log_2\left(\frac{1}{p_{V|U}(v|u)}\right)\right]. \tag{2.12}$$

This remark leads to the following definition.

Definition 2.1.8. *Given a pair of random variables U and V, the **conditional entropy** of V given U is defined as:*

$$H(V|U) = \sum_{u,v} p_{U,V}(u,v)\log_2\left(\frac{1}{p_{V|U}(v|u)}\right) = H(U,V) - H(U). \tag{2.13}$$

More generally, we may consider a larger joint probability mass function for which the framework of conditional probabilities yields,

$$p_{U_1,\ldots,U_n}(u_1,\ldots,u_n) = p_{U_1}(u_1)p_{U_2|U_1}(u_2|u_1)\ldots p_{U_n|U_1,\ldots,U_{n-1}}(u_n|u_1,\ldots,u_n). \tag{2.14}$$

Proposition 2.1.9 (Chain rule for entropy)**.** *Denoting $U^n := U_1, ..., U_n$ we have that:*

$$H(U^n) = H(U_1) + H(U_2|U_1) + ... + H(U_n|U_1, ..., U_{n-1}) = \sum_{k=1}^{n} H\left(U_k|U^{k-1}\right). \tag{2.15}$$

In a similar way, including another random variable V to obtain:

$$H(V, U_1, ..., U_n) = H(V) + \underbrace{H(U_1, ..., U_n|V)}_{(*)} = H(V) + \underbrace{H(U_1|V) + ... + H(U_n|U_1, ..., U_n, V)}_{(**)}, \tag{2.16}$$

gives a relation for the chain rule of the conditional entropy, simply by equating $(*) = (**)$.

Proposition 2.1.10 (Chain rule for conditional entropy)**.** *Denoting $U^n := U_1, ..., U_n$ we have that:*

$$H(U^n|V) = \sum_{k=1}^{n} H\left(U_k|U^{k-1}V\right). \tag{2.17}$$

Let us now give an interpretation of conditional entropy: for each $u \in \mathcal{U}$, we design a code $c_u : \mathcal{V} \longrightarrow \{0,1\}^*$. Thus, the distribution of V is now $p_{V|U=u}(v)$ and the number of bits to describe V is $\sum_v p(v|u) \log_2\left(\frac{1}{p(v|u)}\right)$. Thus, overall we will use $H(V|U) = \sum_{u,v} p_U(u) p_{V|U}(v|u) \log_2\left(\frac{1}{p(v|u)}\right)$ bits.

Theorem 2.1.11 (Conditioning reduces entropy (1))**.** *$H(V|U) \leq H(V)$ with equality if and only if U and V are independent.*

Proof. This follows simply from the facts that $H(U,V) = H(U) + H(V|U)$ and $H(U,V) \leq H(U) + H(V)$. □

Considering the mutual information, $I(U;V) := H(U) + H(V) - H(U,V) = H(U) - H(U|V) = H(V) - H(V|U)$, we observe a curious symmetry when looking at the last two inequalities. Namely, interchanging U and V will not change the value of the mutual information. As a result, since $H(U) - H(U|V)$ represents the saving in bits to describe U by the common knowledge of V, we have that the saving in $H(V) - H(V|U)$ is the same as in $H(U) - H(U|V)$.

Definition 2.1.12. *Given random variables U, V and W, the **conditional mutual information** is defined as:*

$$I(U;V|W) = H(U|W) - H(U|V,W). \tag{2.18}$$

Theorem 2.1.13. *$I(U;V|W) \geq 0$ with equality if and only if conditional on W, U and V are independent (i.e. U, V, W form a Markov chain $U \leftrightarrow W \leftrightarrow V$).*

Proof. We start with the definition of the conditional mutual information, and use the fact that $H(U|W) = H(U,W) - H(W)$ to write:

$$I(U;V|W) = H(U|W) - H(U|V,W) = H(U,W) - H(W) - H(U,V,W) + H(V,W). \tag{2.19}$$

Remark now that this last expression is invariant upon changing the order $U \longleftrightarrow V$. Thus:

$$I(U;V|W) = H(U,W) - H(W) - H(U,V,W) + H(V,W) = H(V|W) - H(V|U,W). \tag{2.20}$$

Finally, we expand explicitly the entropy and use $p_{U,V,W}(u,v,w) = p_W(w)p_{U,V|W}(u,v|w)$ to find:

$$\begin{aligned} I(U;V|W) = H(V|W) - H(V|U,W) &= \sum_{u,v,w} p_{U,V,W}(u,v,w) \log_2 \left[\frac{p_W(w)p_{U,V,W}(u,v,w)}{p_{U,W}(u,w)p_{V,W}(v,w)} \right] \\ &= \sum_w p_W(w) \underbrace{\sum_{u,v} p_{U,V|W}(u,v|w) \log_2 \left[\frac{p_{U,V|W}(u,v|w)}{p_{U|W}(u|w)p_{V|W}(v|w)} \right]}_{=I(U;V|W=w)\geq 0 \text{ by Corollary (2.1.7)}} \geq 0. \end{aligned}$$

□

Corollary 2.1.14 (Conditioning reduces entropy (2))**.** *$H(U|W) \geq H(U|W,V)$.*

Proposition 2.1.15 (Chain rule for mutual information)**.** *Denoting $U^n := U_1,...,U_n$ we have that:*

$$I\left(U^n;V|W\right) = \sum_{k=1}^{n} I\left(U_k;V|U^{k-1},W\right). \tag{2.21}$$

Also: $I(X;Y,Z) = I(X;Z) + I(X;Y|Z) = I(X;Y) + I(X;Z|Y)$.

Proof. We use the chain rule for entropy [Proposition (2.1.9)] to find:

$$\begin{aligned} I\left(U^n;V|W\right) = H\left(U^n|W\right) - H\left(U^n|V,W\right) &= \sum_{k=1}^{n} \left[H\left(U_k|U^{k-1},W\right) - H\left(U_k|U^{k-1},V,W\right) \right] \\ &= \sum_{k=1}^{n} I\left(U_k;V|U^{k-1},W\right). \end{aligned}$$

□

Example 2.1.16. *Consider a code $\{XYZ\} = \{000,011,101,110\}$ such that $p(000) = p(011) = p(101) = p(110) = \frac{1}{4}$. Then, we provide below several examples of calculations related to entropy and information (note that we abuse the notation and write $H(XYZ)$ to mean $H(X,Y,Z)$):*

- $H(XYZ) = \log_2(4) = 2$.
- $H(XY) = H(XZ) = H(YZ) = \log_2(4) = 2$.
- $H(X) = H(Y) = H(Z) = \log_2(2) = 1$.
- *X and Y are independent, thus* $H(X|Y) = 1$.
- *Since we have pairwise independence it also holds that* $I(X;Y) = I(X;Z) = I(Y;Z) = 0$.
- $H(Z|XY) = H(X|YZ) = H(Y|XZ) = 0$.
- $I(X;Y|Z) = 1 > I(X;Y) = 0$.

Remark 2.1.17. *From the last point in the example we mention that conditioning may reduce, increase, or leave unchanged mutual information (in that sense it differs from entropy).*

Proposition 2.1.18 (Data Processing Inequality)**.** *If $X \leftrightarrow Y \leftrightarrow Z$ is a Markov chain (i.e. the conditional distribution of Z depends only on Y and is conditionally independent of X) then:*

$$I(X;Y) \geq I(X;Z). \tag{2.22}$$

Proof. By the chain rule for the mutual information, we can expand the latter in two different ways:

$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z) = I(X;Y) + I(X;Z|Y). \tag{2.23}$$

Since X and Z are conditionally independent given Y, we have that $I(X;Z|Y) = 0$. Thus since $I(X;Y|Z) \geq 0$ we find $I(X;Y) \geq I(X;Z)$, with equality if and only if $I(X;Y|Z) = 0$ (i.e. $X \leftrightarrow Z \leftrightarrow Y$ forms a Markov chain). □

Corollary 2.1.19. *If $X \leftrightarrow Y \leftrightarrow g(Y)$ is a Markov chain then:*

$$I(X;Y) \geq I(X;g(Y)). \tag{2.24}$$

We finally present one last definition, for the sole purpose of some notation.

Definition 2.1.20. *We define the* ***binary entropy function*** *to be the function $h_2 : [0,1] \to [0,1]$ given by $h_2(\alpha) := \alpha \log_2\left(\frac{1}{\alpha}\right) + (1-\alpha)\log_2\left(\frac{1}{1-\alpha}\right)$.*

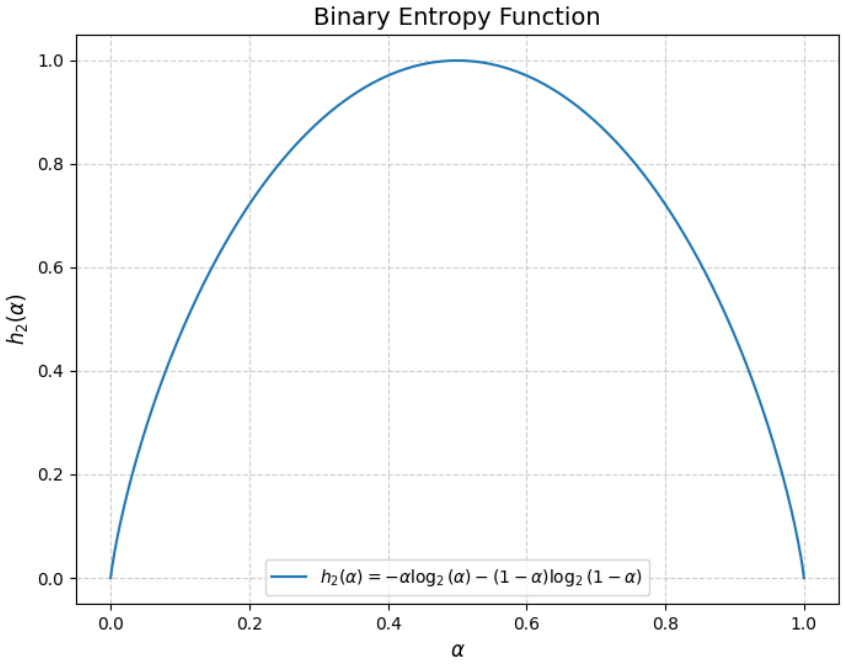


Figure 2.1: Plot of the Binary Entropy Function $h_2(\alpha)$. The maximum of h_2 is attained for $\alpha = \frac{1}{2}$.

2.2 Continuous Probability Distributions

In this section, we review the results and definitions from the previous section by considering continuous probability distributions. Suppose $U \in \mathbb{R}$ with probability density function f_U.

Definition 2.2.1. *Given a random variable U taking values in $\mathcal{U}$, we define the **differential entropy** as:*

$$h(U) = \int_{\mathbb{R}} f_U(x) \log_2 \left(\frac{1}{f_U(x)} \right) dx = - \int_{\mathbb{R}} f_U(x) \log_2 (f_U(x)) dx, \tag{2.25}$$

with the convention $0 \log_2(1/0) = 0$.

Example 2.2.2. *For $U \in [0,1]$ a uniform distribution, $h(U) = 0$. Suppose $a > 0$, consider $V = aU$. Then:*

$$h(V) = \int_0^a \frac{1}{a} \log(a) dx = \log(a).$$

Remark 2.2.3. *Depending on the value of a, we can have that $h(\cdot) < 0$ (which is allowed). Also note that even though $U = f(V)$ and $V = g(U)$ we have $h(U) \neq h(V)$ (for $a \neq 1$) which would give equality in the discrete setting.*

For a collection $U_1, U_2, ..., U_n$ of $\mathbb{R}$-valued random variables with joint density $f_{U^n}(u^n)$, we define the **joint differential entropy** as:

$$h(U^n) = \int_{\mathbb{R}} ... \int_{\mathbb{R}} f_{U^n}(u^n) \log \left(\frac{1}{f_{U^n}(u^n)} \right) du_1 ... du_n. \tag{2.26}$$

Note that by the Bayes' rule,

$$f_{U^n}(u^n) = f_{U_1}(u_1) f_{U_2|U_1}(u_2|u_1) ... f_{U_n|U^{n-1}}(u_n|u^{n-1}), \tag{2.27}$$

and as a result the chain rule holds also for the differential entropy.

Proposition 2.2.4 (Chain rule for differential entropy)**.** *Denoting $U^n := U_1, ..., U_n$ we have that:*

$$h(U^n) = h(U_1) + h(U_2|U_1) + ... + h(U_n|U_1, ..., U_{n-1}) = \sum_{k=1}^{n} h\left(U_k|U^{k-1}\right). \tag{2.28}$$

Definition 2.2.5. *Given a pair of random variables U and V, the **conditional differential entropy** of U given V is defined as:*

$$h(U|V) = \int_{\mathbb{R}} \int_{\mathbb{R}} f_{U,V}(u,v) \log_2 \left(\frac{1}{f_{U|V}(u|v)} \right) = h(U,V) - h(V). \tag{2.29}$$

Theorem 2.2.6 (Conditioning reduces entropy)**.** *$h(U|V) \leq h(U)$ with equality if and only if U and V are independent.*

Proof. • Note that, using the convexity of the function $y \mapsto -\log(y)$ we obtain:

$$\begin{aligned} h(U) - h(U|V) = h(U) + h(V) - h(U,V) &= \int_{\mathbb{R}}\int_{\mathbb{R}} f_{U,V}(u,v)\log\left(\frac{f_{U,V}(u,v)}{f_U(u)f_V(v)}\right)dudv \\ &= -\int_{\mathbb{R}}\int_{\mathbb{R}} f_{U,V}(u,v)\log\left(\frac{f_U(u)f_V(v)}{f_{U,V}(u,v)}\right)dudv \\ &\geq \log\left(\int_{\mathbb{R}}\int_{\mathbb{R}} f_U(u)f_V(v)dudv\right) = \log(1) = 0 \end{aligned}$$

• We can also prove this using $\ln(\xi) \leq \xi - 1$ (with equality if and only if $\xi = 1$) since:

$$\int_{\mathbb{R}}\int_{\mathbb{R}} f_{U,V}(u,v)\ln\left(\frac{f_U(u)f_V(v)}{f_{U,V}(u,v)}\right)dudv \leq \int_{\mathbb{R}}\int_{\mathbb{R}}\left(f_{U,V}(u,v)\frac{f_U(u)f_V(v)}{f_{U,V}(u,v)} - f_{U,V}(u,v)\right)dudv = 0.$$

□

Definition 2.2.7. *Given a pair of random variables U and V, the **mutual information** between U and V is defined as:*

$$I(U;V) = h(U) + h(V) - h(U,V) = h(U) - h(U|V) = h(V) - h(V|U). \tag{2.30}$$

Let us now provide an explanation for the relationship between discrete and differential entropies. Suppose $U \in \mathbb{R}$ with a density function f. Let us define $U_\delta := \delta\lfloor\frac{U}{\delta}\rfloor$ (i.e. a left continuous step function such that $U_\delta(k\delta) = k\delta$ for $k \in \mathbb{Z}$). Then:

$$H(U_\delta) = \sum_{i=-\infty}^{\infty} p_i \log\left(\frac{1}{p_i}\right), \quad \text{where } p_i := \mathbb{P}(U_\delta = i\delta) = \mathbb{P}(i\delta \leq U_\delta \leq (i+1)\delta).$$

On the other hand,

$$h(U) = \int_{\mathbb{R}} f(u)\log\left(\frac{1}{f(u)}\right)du = \sum_{i=-\infty}^{\infty}\int_{i\delta}^{(i+1)\delta} f(u)\log\left(\frac{1}{f(u)}\right)du.$$

To approximate f we consider squares under the function of length δ [i.e. create a square between $i\delta$ and $(i+1)\delta$] whose probability of ending in this square is exactly p_i. Thus:

$$h(U) \approx \sum_{i=-\infty}^{\infty}\int_{i\delta}^{(i+1)\delta} f(u)\log\left(\frac{\delta}{p_i}\right)du = \sum_i p_i \log\left(\frac{\delta}{p_i}\right) = H(U_\delta) + \log(\delta).$$

As a result, we have the following (without proof):

Theorem 2.2.8. *Assuming the limit exists and for $U_\delta := \delta\lfloor\frac{U}{\delta}\rfloor$:*

$$h(U) = \lim_{\delta \searrow 0}\left[H\left(\frac{1}{\delta}U_\delta\right) + \log(\delta)\right]. \tag{2.31}$$

In a similar way as above, one can show this result for $h(U,V)$ by considering the quantity $H\left(\frac{1}{\delta}U_\delta, \frac{1}{\epsilon}V_\epsilon\right)$ for $U_\delta := \delta\lfloor\frac{U}{\delta}\rfloor$ and $V_\epsilon := \epsilon\lfloor\frac{V}{\epsilon}\rfloor$, and so we obtain the following result:

Theorem 2.2.9. *Assuming the limit exists and for* $U_\delta := \delta\lfloor \frac{U}{\delta} \rfloor$ *and* $V_\epsilon := \epsilon\lfloor \frac{V}{\epsilon} \rfloor$*:*

$$h(U,V) = \lim_{\delta \searrow 0} \lim_{\epsilon \searrow 0} \left[H\left(\frac{1}{\delta}U_\delta, \frac{1}{\epsilon}V_\epsilon\right) + \log(\delta) + \log(\epsilon) \right]. \tag{2.32}$$

Consequently, for the mutual information we deduce:

$$\begin{aligned} I(U;V) &= h(U) + h(V) - h(U,V) \\ &= \lim_{\delta \searrow 0} \lim_{\epsilon \searrow 0} H\left(\frac{1}{\delta}U_\delta\right) + H\left(\frac{1}{\epsilon}V_\epsilon\right) - H\left(\frac{1}{\delta}U_\delta, \frac{1}{\epsilon}V_\epsilon\right) \\ &= \lim_{\delta \searrow 0} \lim_{\epsilon \searrow 0} I\left(\frac{1}{\delta}U_\delta; \frac{1}{\epsilon}V_\epsilon\right). \end{aligned}$$

Definition 2.2.10. *Given a pair of random variables* U *and* V *with density functions* f *and* g *respectively, the* ***KL-Divergence*** *between* U *and* V *is defined as:*

$$D(f\|g) := \int_{\mathbb{R}} f(u) \log\left(\frac{f(u)}{g(u)}\right) du = \lim_{\delta \searrow 0} D(p_\delta \| q_\delta), \tag{2.33}$$

where $p_\delta(i) = \int_{i\delta}^{(i+1)\delta} f(u)du$ *and* $q_\delta(i) = \int_{i\delta}^{(i+1)\delta} g(u)du$*.*

Proposition 2.2.11. $D(f\|g) \geq 0$ *with equality if and only if* $f = g$*.*

Proof. Use $\ln(\xi) \leq \xi - 1$ on $D(f\|g)$ and the result follows immediately. □

Proposition 2.2.12. $I(U;V) = D(f_{UV}\|f_U f_V)$*.*

Proof. The proof is obvious by starting with $I(U;V) = h(U) + h(V) - h(U,V)$. □

Theorem 2.2.13. *Suppose* $U \in [0,1]$*. Then* $h(U) \leq 0$*, with equality if and only if* U *is uniform.*

Proof. Let $g(u) = \mathbb{1}_{\{0 \leq u \leq 1\}}(u)$ be the density of a uniform random variable on $[0,1]$. Let f be the density of U. Then:

$$0 \geq -D(f\|g) = \int_0^1 f(u) \log\left(\frac{1}{f(u)}\right) du = h(U),$$

with equality if and only if $D(f\|g) = 0$, which happens if and only if $f = g$ (i.e. U is uniformly distributed). □

Theorem 2.2.14. *Suppose* $U \in [0,\infty)$ *and* $\mathbb{E}(U) = 1$*. Then* $h(U) \leq \log(e)$*, with equality if and only if* U *is exponentially distributed.*

Proof. Let f be the density of U. Thus $f(u) \geq 0$, $\int_0^\infty f(u)du = 1$ and $\int_0^\infty uf(u)du = 1$. Let $g(u) = e^{-u}$ for $u \geq 0$. Now:

$$0 \geq -D(f\|g) = \int_0^\infty f(u) \log\left(\frac{g(u)}{f(u)}\right) du = \int_0^\infty f(u) \log\left(\frac{e^{-u}}{f(u)}\right) du = h(U) - \log(e) \int_0^\infty uf(u)du$$
$$= h(U) - \log(e).$$

□

Theorem 2.2.15. *Suppose $U \in \mathbb{R}$ and $\mathbb{E}(U^2) = 1$. Then $h(U) \leq \frac{1}{2}\log(2\pi e)$, with equality if and only if $U \sim \mathcal{N}(0,1)$.*

Proof. Let f be the density of U. Thus $f(u) \geq 0$, $\int_{\mathbb{R}} f(u)du = 1$ and $\int_{\mathbb{R}} u^2 f(u)du = 1$. Let $g(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ for $u \in \mathbb{R}$. Now:

$$\begin{aligned} 0 \geq -D(f\|g) = \int_{\mathbb{R}} f(u) \log\left(\frac{g(u)}{f(u)}\right) du &= \int_{\mathbb{R}} f(u) \log\left(\frac{\frac{1}{\sqrt{2\pi}} e^{-u^2/2}}{f(u)}\right) du \\ &= h(U) - \int_{\mathbb{R}} \left(\frac{u^2}{2}\log(e) + \frac{1}{2}\log(2\pi)\right) f(u)du \\ &= h(U) - \frac{1}{2}\log(2\pi e). \end{aligned}$$

□

Recall that,

$$f_U(u) = \frac{d}{du}\mathbb{P}(U \leq u) = \lim_{\epsilon \to 0} \frac{\mathbb{P}(U \leq u+\epsilon) - \mathbb{P}(U \leq u)}{\epsilon} = \lim_{\epsilon \to 0} \frac{\mathbb{P}(u < U \leq u+\epsilon)}{\epsilon}.$$

Then, suppose that $V = aU$ for $a > 0$, thus:

$$f_V(v) = \lim_{\epsilon \to 0} \frac{\mathbb{P}(v < V \leq v+\epsilon)}{\epsilon} = \lim_{\epsilon \to 0} \frac{(1/a)\mathbb{P}\left(\frac{v}{a} < U \leq \frac{v}{a} + \frac{\epsilon}{a}\right)}{(\epsilon/a)} = \frac{1}{a} f_U\left(\frac{v}{a}\right).$$

Thus, for a general $a \neq 0$ and $V = aU$, we find $f_V(v) = \frac{1}{|a|} f_U\left(\frac{v}{a}\right)$.

Theorem 2.2.16. *(i) If $a \in \mathbb{R}$ then $h(U+a) = h(U)$ and (ii) if $a \neq 0$ then $h(aU) = h(U) + \log|a|$.*

Proof. First note that if $V = a+U$ we have $f_V(v) = f_U(v-a)$ and so $\mathbb{E}[-\log(f_V(v))] = \mathbb{E}[-\log(f_U(v-a))]$. We conclude $h(a+U) = h(U)$. Furthermore, if $V = aU$ then by the above remark we have $-\log(f_V(v)) = \log|a| - \log(f_U(v/a))$. Thus, $\mathbb{E}[-\log(f_V(v))] = \log|a| + \mathbb{E}[-\log(f_U(v/a))]$ which concludes on $h(V) = \log|a| + h(U)$. □

Corollary 2.2.17. *If $U \in [a,b]$, then $h(U) \leq \log(b-a)$ with equality if and only if U is uniform.*

Proof. Define $V = \frac{1}{b-a}(U-a)$. So by Theorem (2.2.16) we have that,

$$h(V) = h\left(\frac{U-a}{b-a}\right) = h(U-a) - \log(b-a) = h(U) - \log(b-a).$$

Furthermore, we have that $V = (U-a)/(b-a) \in [0,1]$, and by Theorem (2.2.13) we know $h(V) \leq 0$. We thus conclude on $h(U) \leq \log(b-a)$ with equality if and only if U is uniform (see (2.2.13)). □

Corollary 2.2.18. *If $U \geq 0$ and $\mathbb{E}(U) = \mu$ then $h(U) \leq \log(e\mu)$ with equality if and only if U is exponential.*

Proof. Let $V := U/\mu$. Then $V \geq 0$ and $\mathbb{E}(V) = 1$ which by Theorem (2.2.14) and Theorem (2.2.16) gives us $h(V) = h(U/\mu) = h(U) - \log(\mu) \leq \log(e)$. Hence $h(U) \leq \log(\mu) + \log(e) = \log(e\mu)$ with equality if and only if U is exponential (see Theorem (2.2.14)). □

Corollary 2.2.19. *For $\mathrm{Var}(U) = \sigma^2$ we have $h(U) \leq \frac{1}{2}\log(2\pi e\sigma^2)$ with equality if and only if U is Gaussian.*

Proof. Let $V := (U-\mu)/\sigma$ and $\mu = \mathbb{E}(U)$, then $V \sim \mathcal{N}(0,1)$ and by Theorem (2.2.15) we know that $h(V) := h((U-\mu)/\sigma) \leq \frac{1}{2}\log(2\pi e)$. From Theorem (2.2.16) we conclude $h(U-\mu) - \log(\sigma) = h(U) - \log(\sigma) \leq \frac{1}{2}\log(2\pi e)$ giving us $h(U) \leq \log(\sigma) + \frac{1}{2}\log(2\pi e) = \frac{1}{2}\log(2\pi e\sigma^2)$ with equality if and only if U is Gaussian (see (2.2.15)). □

Let us now study the generalisation of these bounds to a multidimensional setting. To this end, suppose $U^n \in \mathbb{R}^n$ and let $K_{ij} = \mathbb{E}(U_i U_j)$ be the second-moment matrix. We are looking to know how large can $h(U^n)$ be. Observe first that the matrix $K \succeq 0$ is non-negative definite (i.e. for any vector $a \in \mathbb{R}^n$, $a^T K a \geq 0$). Indeed,

$$a^T K a = \sum_{i,j} a_i K_{ij} a_j = \mathbb{E}\left[\sum_{i,j} a_i U_i U_j a_j\right] = \mathbb{E}\left[\left(\sum_i a_i U_i\right)^2\right] \geq 0.$$

Suppose now that $K \succ 0$ is positive definite (i.e. for any vector $a \in \mathbb{R}^n$, $a^T K a > 0$). Then we can write $K = LL^T$ with L lower-triangular and invertible (i.e. $L_{ii} \neq 0$).

Theorem 2.2.20. *Let $K = \mathbb{E}(UU^T)$ then $h(U^n) \leq \frac{1}{2}\log\det(2\pi eK)$ with equality if and only if $U^n \sim \mathcal{N}(0,K)$.*

Proof - First method. Consider $V = L^{-1}U$ so that $VV^T = L^{-1}UU^T(L^T)^{-1}$. As a result $\mathbb{E}(VV^T) = L^{-1}K(L^T)^{-1} = L^{-1}LL^T(L^T)^{-1} = \mathbb{1}$ and thus $\mathbb{E}(V_iV_j) = \mathbb{1}_{\{i=j\}}$. Hence by Theorem (2.2.15),

$$h(V^n) \leq \sum_{i=1}^n h(V_i) \leq \frac{n}{2}\log(2\pi e).$$

Note now that $U = LV$, namely

$$\begin{pmatrix} U_1 \\ \vdots \\ U_n \end{pmatrix} = \begin{pmatrix} L_{11} & 0 & \dots & 0 \\ \vdots & \ddots & & 0 \\ L_{1j} & \dots & \dots & L_{nn} \end{pmatrix} \begin{pmatrix} V_1 \\ \vdots \\ V_n \end{pmatrix}.$$

Then:

$$\begin{aligned} h(U^n) &= h(U_1) + h(U_2|U_1) + ... + h(U_n|U^{n-1}) \\ &= h(L_{11}V_1) + h(L_{22}V_2 + L_{21}V_1|U_1) + ... + h(L_{nn}V_n + L_{n1}V_1 + ... + L_{nn-1}V_{n-1}|U^{n-1}) \\ &= h(L_{11}V_1) + h(L_{22}V_2|U_1) + ... + h(L_{nn}V_n|U^{n-1}) \\ &\overset{(*)}{\leq} \sum_{i=1}^{n} h(L_{ii}V_i) = \sum_{i=1}^{n} \log|L_{ii}| + h(V_i) \overset{(**)}{\leq} \log|\det(L)| + \frac{n}{2}\log(2\pi e), \end{aligned}$$

where in the third equality we have used the fact that knowing U_i gives us V_i. Hence dropping the conditioning gives us the above result. Therefore:

$$h(U^n) \leq \frac{1}{2}\left[\log(\det(L)^2) + \log((2\pi e)^n)\right] = \frac{1}{2}\left[\log(\det(K)) + \log((2\pi e)^n)\right] = \frac{1}{2}\log\det(2\pi e K),$$

using the fact that $\det(aU) = a^n \det(U)$. For equality to hold we require by $(*)$ that $V_i \sim \mathcal{N}(0,1)$ and by $(**)$ that the V_i's are independent. This is equivalent to the requirement that U^n is a zero mean Gaussian vector. □

As an aside, given a Gaussian random vector U^n with zero mean and non-singular covariance matrix, we can find a lower-triangular L such that $U^n = LV^n$ with $V_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ such that: $U_1 = V_1\beta_{11}$, $U_2 = \alpha_{21}U_1 + V_2\beta_{22}, U_3 = \alpha_{31}U_1 + \alpha_{32}U_2 + V_3\beta_{33}, ...$ where the $\{V_i\beta_{ii}\}_i$ is the innovation sequence. This idea extends to a more general framework where the $V_i's$ are uncorrelated zero-mean random variables.

Proof - Second method. Let $f(u^n)$ be the density function of U^n. Let $g(u^n)$ be the Gaussian density function given by,

$$g(u^n) = [\det(2\pi K)]^{-\frac{1}{2}} \exp\left(-\frac{1}{2}u^T K^{-1} u\right).$$

Now,

$$\begin{aligned} 0 \geq -D(f\|g) &= \int ... \int f(u^n) \log\left(\frac{g(u^n)}{f(u^n)}\right) du^n \\ &= h(U^n) + \int f(u^n) \log(g(u^n)) du^n \\ &= h(U^n) + \int f(u^n) \left[-\frac{1}{2}\det(2\pi K) - \frac{1}{2}\log(e)\sum_{i,j} u_i (K^{-1})_{ij} u_j\right] \\ &= h(U^n) - \frac{1}{2}\log\det(2\pi K) - \frac{1}{2}\log(e)\sum_{i,j} K_{ji}(K^{-1})_{ij} \\ &= h(U^n) - \frac{1}{2}\log\det(2\pi K) - \frac{1}{2}\log(e)\sum_{i}(KK^{-1})_{ii} = h(U^n) - \frac{1}{2}\log\det(2\pi K) - \frac{n}{2}\log(e), \end{aligned}$$

where we have used the fact that $\sum_{i,j} A_{ij}B_{ji} = \sum_i (AB)_{ii}$. As a result:

$$\begin{aligned} 0 \geq h(U^n) - \frac{1}{2}\log\det(2\pi K) - \frac{n}{2}\log(e) &= h(U^n) - \frac{1}{2}\log\det(2\pi K) - \frac{1}{2}\log(e^n) \\ &= h(U^n) - \frac{1}{2}\log\det(2\pi e K). \end{aligned}$$

□

Corollary 2.2.21. *Suppose $U \in \mathbb{R}^n$ with $K = \mathrm{Cov}(U) = \mathbb{E}(UU^T) - \mathbb{E}(U)\mathbb{E}(U^T)$ (i.e. the second-moment matrix of $U - \mathbb{E}(U)$), then:*

$$h(U) \leq \frac{1}{2}\log\det(2\pi e K), \tag{2.34}$$

where we have equality if and only if $U \sim \mathcal{N}(\mu, K)$ such that

$$f_U(u) = [\det(2\pi K)]^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(u-\mu)^T K^{-1}(u-\mu)\right). \tag{2.35}$$

Theorem 2.2.22. *Suppose $A \in \mathbb{R}^{n\times n}$ is a deterministic matrix and $U \in \mathbb{R}^n$ is a random vector. Then,*

$$h(AU) = h(U) + \log|\det(A)|. \tag{2.36}$$

Proof. We have already prove this fact for $A = L$ a lower-triangular matrix,

$$h(LU^n) = \log|\det(L)| + h(U^n).$$

Now let $V^n = LU^n$ such that:

$$\begin{aligned} h(V^n) &= h(V_1) + h(V_2|V_1) + ... + h(V_n|V^{n-1}) \\ &= h(L_{11}U_1) + h(L_{22}U_2|U_1) + ... + h(L_{nn}U_n|U^{n-1}) \\ &= \sum_{i=1}^{n} \log|L_{ii}| + h(U^n) \\ &= \log|\det(L)| + h(U^n). \end{aligned}$$

Since this holds for the lower-triangular matrix, it must also hold for the upper-triangular one. For a general A, just use the LU-decomposition. □

Chapter 3

Entropy Rate of a Stochastic Process

Recall, we have seen that $\frac{1}{n}H(U_1...U_n)$ is a key quantity in data compression. Indeed, for any injective $c_n : \mathcal{U}^n \to \{0,1\}^*$:

$$\frac{1}{n}H(U^n) - \frac{1}{n}\log_2\log_2\left(1+|\mathcal{U}|^n\right) \leq \frac{1}{n}\mathbb{E}\big[\text{length}(c_n(U^n))\big], \tag{3.1}$$

and also there exists a prefix-free code c_n such that:

$$\frac{1}{n}\mathbb{E}\big[\text{length}(c_n(U^n))\big] \leq \frac{1}{n}H(U^n) + \frac{1}{n}. \tag{3.2}$$

This framework motivates the following definition.

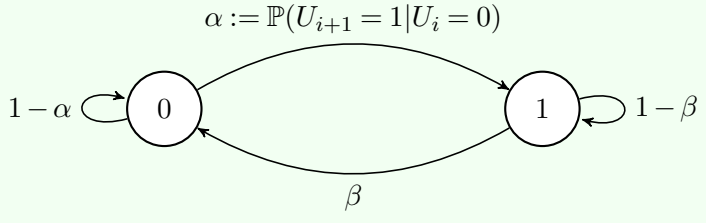
Definition 3.0.1. *Given a stochastic process $\{U_1, U_2, ...\}$, the **entropy rate** is defined as $\mathcal{H}(\{U_i : i \in \mathbb{Z}^+\}) := \lim_{n\to\infty}\frac{1}{n}H(U^n)$, provided that the limit exists.*

Example 3.0.2 (IID random variables)**.** *Suppose $U_1, U_2, ...$ are i.i.d. then:*

$$\begin{aligned} H(U^n) &= H(U_1) + H(U_2|U_1) + H(U_3|U_1U_2) + ... + H\big(U_n|U^{n-1}\big) \\ &= H(U_1) + H(U_2) + H(U_3) + ... + H(U_n) \\ &= nH(U_1), \end{aligned}$$

where we have used the chain rule in the first equality, and the i.i.d. property in the second and third equalities. Thus, the entropy rate is given by $\lim_{n\to\infty}\frac{1}{n}H(U^n) = H(U_1)$.

Example 3.0.3 (Stationary binary Markov process)**.** *Suppose $U_1, U_2, ...$ is a stationary binary Markov process. Consider the following Markov transition diagram:*



Suppose now that $\mathbb{P}(U_1 = 0) = \pi_0$ and $\mathbb{P}(U_1 = 1) = \pi_1$. Then:

$$\mathbb{P}(U_2 = 0) = \pi_0(1-\alpha) + \pi_1\beta \overset{\text{Stationarity}}{=} \pi_0. \tag{3.3}$$

Thus $\alpha\pi_0 = \beta\pi_1$ (it is not difficult to show that the same conditions hold for $\mathbb{P}(U_2 = 1)$). Hence:

$$(\pi_0, \pi_1) = \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right). \tag{3.4}$$

Finally, from the chain rule we can write:

$$\begin{aligned} H(U^n) &= H(U_1) + H(U_2|U_1) + H(U_3|U_1U_2) + ... + H\big(U_n|U^{n-1}\big) \\ &\overset{\text{Markov}}{=} H(U_1) + H(U_2|U_1) + H(U_3|U_2) + ... + H(U_n|U_{n-1}) \\ &\overset{\text{Stationarity}}{=} H(U_1) + (n-1)H(U_2|U_1). \end{aligned}$$

As a result, we have that the entropy rate is given by:

$$\frac{1}{n}H(U^n) = \frac{H(U_1)}{n} + \frac{(n-1)}{n}H(U_2|U_1) \xrightarrow{n\to\infty} H(U_2|U_1) = \pi_0 h_2(\alpha) + \pi_1 h_2(\beta) = \frac{\beta h_2(\alpha) + \alpha h_2(\beta)}{\alpha+\beta}.$$

Theorem 3.0.4. *Suppose $U_1, U_2, ...$ is a stationary process. Then, the entropy rate exists and,*

$$\lim_{n\to\infty}\frac{1}{n}H(U^n) = \lim_{n\to\infty} H\big(U_n|U^{n-1}\big). \tag{3.5}$$

Proof. We first prove that the limit of $H\big(U_n|U^{n-1}\big)$ exists, then show that the limit of the entropy rate also exists and finally that they are equal.

(i) Let $\alpha_n := H\big(U_n|U^{n-1}\big) = H(U_n|U_1...U_n)$. Then:

$$\alpha_n = H(U_{n+1}|U_2...U_n) \geq H(U_{n+1}|U_1U_2...U_n) = \alpha_{n+1}, \tag{3.6}$$

where we have used the fact that conditioning reduces entropy and also stationarity. Thus, (α_n) is monotone decreasing and bounded below, therefore $\alpha := \lim_n \alpha_n = \inf_n \alpha_n$ exists.

(ii) Let $\beta_n := \frac{1}{n}H(U^n) = \frac{1}{n}(\alpha_1 + ... + \alpha_n)$, where this last equality follows from the chain rule. Note again that (β_n) is monotone decreasing and bounded below and thus $\lim_n \beta_n$ exists.
We now present the following Lemma:

Lemma 3.0.5 (Cesàro)**.** *If $\{\alpha_n\}_{\{n=1,2,...\}}$ is such that $\alpha := \lim_n \alpha_n$, then $\beta_n := \frac{\alpha_1+...+\alpha_n}{n}$ has a limit given by $\lim_n \beta_n = \alpha$.*

Proof. Without loss of generality, assume $\alpha = 0$ (otherwise, just subtract $\alpha \neq 0$ from the result). Then, for all $\epsilon > 0$, there exists $n(\epsilon) > 0$ such that for all $n > n(\epsilon)$ we have $|\alpha_n| < \epsilon$. Then, choose $N(\epsilon) = \left\lceil \frac{S(\epsilon)}{\epsilon} \right\rceil \vee n(\epsilon)$ where $S(\epsilon) := |\alpha_1| + ... + |\alpha_{n(\epsilon)}|$. Now, for all $n > N(\epsilon)$:

$$|\beta_n| \leq \frac{|\alpha_1| + ... + |\alpha_{n(\epsilon)}| + |\alpha_{n(\epsilon)+1}| + ... + |\alpha_n|}{n} < \frac{S(\epsilon) + n\epsilon}{n} = \epsilon + \frac{S(\epsilon)}{n} < 2\epsilon. \tag{3.7}$$

□

The conclusion to this second point (ii) is immediate from the Cesàro Lemma we have just proved. □

Chapter 4

Typicality

The aim of this chapter is to present the notion of a type function, state some interesting properties and study the general framework of typicality.

Definition 4.0.1. *For* $\{U_1, U_2, ...\} \overset{iid}{\sim} p_U$ *and given* $u^n = (u_1, ..., u_n) \in \mathcal{U}^n$*, the* ***type function*** *is defined as:*

$$\text{type}_{(u^n)}(u) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}}, \qquad u \in \mathcal{U}. \tag{4.1}$$

In fact, $\text{type}_{(u^n)}(u)$ is a probability distribution on $\mathcal{U}$ representing a form of empirical distribution.

Remark 4.0.2. *Note that not all probability distributions are type functions. Indeed for example* $p_1 = \frac{1}{\sqrt{2}}$ *and* $p_2 = 1 - \frac{1}{\sqrt{2}}$ *(which is a valid distribution), however we do not consider irrational numbers in the context of type functions.*

Definition 4.0.3. *Given* p_U*,* n *and* $\epsilon > 0$*, define the* ***ϵ-typical sequences of length*** **n** *with respect to* p_U *to be:*

$$T(n, \epsilon, p_U) := \left\{ u^n \in \mathcal{U}^n : \text{type}_{(u^n)}(\cdot) \sim p_U \right\}, \tag{4.2}$$

where the condition $\sim$ *(for the choice of the probability law) is:*

$$(1 - \epsilon) p_U(u) \leq \text{type}_{(u^n)}(u) \leq (1 + \epsilon) p_U(u), \qquad \forall u \in \mathcal{U}. \tag{4.3}$$

Note that we could also define the condition $\sim$ in three other ways:
(i) For all $u \in \mathcal{U}$:

$$\left| \text{type}_{(u^n)}(u) - p_U(u) \right| < \epsilon. \tag{4.4}$$

(ii) For all $u \in \mathcal{U}$:

$$\sum_{u \in \mathcal{U}} \left| \text{type}_{(u^n)}(u) - p_U(u) \right| < \epsilon. \tag{4.5}$$

(iii) For all $u \in \mathcal{U}$:

$$\left| \sum_{u \in \mathcal{U}} \text{type}_{(u^n)}(u) \log \left(\frac{1}{p_U(u)} \right) - \sum_{u \in \mathcal{U}} p_U(u) \log \left(\frac{1}{p_U(u)} \right) \right| < \epsilon. \tag{4.6}$$

Example 4.0.4. *For $\mathcal{U} \in \{0,1\}$ with $p_U = (\frac{1}{2}, \frac{1}{2})$ and $\epsilon = 0.1$:*

$$T(1000, 0.1, p_U) = \{u^n \in \mathcal{U}^n : \#\text{ of 0's in } u^n \in [450, 550]\} \tag{4.7}$$

$$T(1000, 0.01, p_U) = \{u^n \in \mathcal{U}^n : \#\text{ of 0's in } u^n \in [495, 505]\} \tag{4.8}$$

$$T(1000, 0.001, p_U) = \{u^n \in \mathcal{U}^n : \#\text{ of 0's in } u^n = 500\} \tag{4.9}$$

4.1 Entropic Bounds for Type Functions

With the setup above, we may present several results relating the ϵ-typical sequences of length n with respect to p_U and a sequence of i.i.d. $U_1, ..., U_n \sim p$ for some probability distribution p.

Theorem 4.1.1. *For all p_U, all $\epsilon > 0$, there exists $n(\epsilon, p_U)$ such that for all $n > n(\epsilon, p_U)$ (i.e. for n large enough) we have: for $U_1, ..., U_n \overset{iid}{\sim} p_U$:*

$$\mathbb{P}(U^n \notin T(n, \epsilon, p_U)) < \epsilon. \tag{4.10}$$

Proof. The event $\{U^n \notin T(n, \epsilon, p_U)\}$ is equivalent to the fact that there exists $u \in \mathcal{U}$ such that:

$$\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u). \tag{4.11}$$

Then we have that:

$$\begin{aligned} \mathbb{P}(U^n \notin T(n, \epsilon, p_U)) &= \mathbb{P}\left(\exists u : \left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u) \right) \\ &\leq \sum_{u \in \mathcal{U}} \mathbb{P}\left(\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u) \right). \end{aligned}$$

Observe that if $p_U(u) = 0$ then $\mathbb{P}\left(\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u) \right) = 0$ and thus:

$$\mathbb{P}(U^n \notin T(n, \epsilon, p_U)) \leq \sum_{u:\ p_U(u) > 0} \mathbb{P}\left(\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u) \right). \tag{4.12}$$

Now fix $u \in \mathcal{U}$ so that $p_U(u) > 0$, and let $X_i := \mathbb{1}_{\{u_i = u\}}$. These are i.i.d. such that $\mathbb{P}(X_i = 1) = p_U(u)$ and $\mathbb{P}(X_i = 0) = 1 - p_U(u)$. Consequently, $\mathbb{E}(X_i) = p_U(u)$ and $\mathrm{Var}(X_i) = p_U(u)(1 - p_U(u))$. Thus:

$$\left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{\{u_i = u\}} - p_U(u) \right| > \epsilon \cdot p_U(u) \iff \left| \frac{1}{n} \sum_{i=1}^{n} X_i - p_U(u) \right| > \epsilon \cdot p_U(u), \tag{4.13}$$

where $\mathbb{E}\left(\frac{1}{n} \sum_{i=1}^{n} X_i \right) = p_U(u)$ and $\mathrm{Var}\left(\frac{1}{n} \sum_{i=1}^{n} X_i \right) = \frac{1}{n} p_U(u)(1 - p_U(u))$. As a result, using Chebyshev's Inequality, we obtain:

$$\mathbb{P}\left(\left| \frac{1}{n} \sum_{i=1}^{n} X_i - p_U(u) \right| > \epsilon \cdot p_U(u) \right) \leq \frac{\frac{1}{n} p_U(u)(1 - p_U(u))}{\epsilon^2 \cdot p_U(u)^2} = \frac{1 - p_U(u)}{n \epsilon^2 \cdot p_U(u)}. \tag{4.14}$$

Hence, from equation (4.12), for n large enough:

$$\mathbb{P}(U^n \notin T(n,\epsilon,p_U)) \leq \underbrace{\left(\sum_{u:\ p_U(u)>0} \frac{1-p_U(u)}{p_U(u)}\right)}_{<\infty} \frac{1}{n\epsilon^2} < \epsilon. \tag{4.15}$$

□

We note that $|T(n,\epsilon,p_U)|$ is approximately exponentially smaller that $|\mathcal{U}^n|$, and in fact $\mathcal{U}^n$ is also approximately uniformly distributed in $T(n,\epsilon,p_U)$ such that $|T(n,\epsilon,p_U)| \approx 2^{nH(U)}$. In other words, if we pick $U_1,...,U_n$ i.i.d. then it will almost surely land in $T(n,\epsilon,p_U)$.

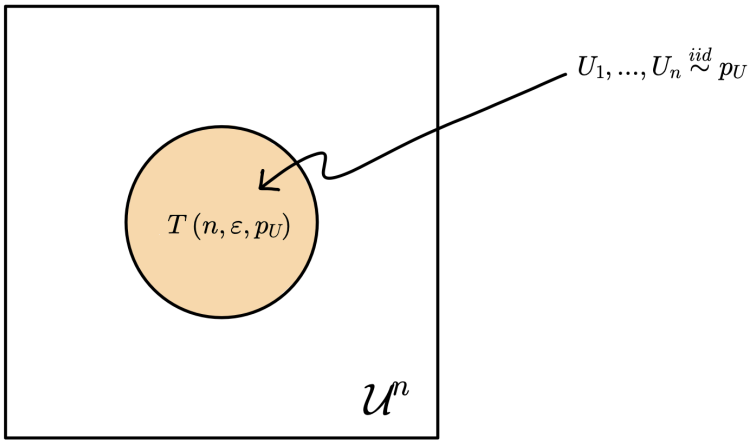


Figure 4.1: Picking $U_1,...,U_n \overset{\text{iid}}{\sim} p_U$ then it will almost surely land in $T(n,\epsilon,p_U)$.

Theorem 4.1.2. *For all p_U, all n, all $\epsilon > 0$, for every $U^n \in T(n,\epsilon,p_U)$ and where $U_1,...,U_n \overset{iid}{\sim} p_U$, the following holds:*

(i)

$$2^{-n(1+\epsilon)H(U)} \leq \mathbb{P}(U^n = u^n) \leq 2^{-n(1-\epsilon)H(U)}. \tag{4.16}$$

(ii)

$$|T(n,\epsilon,p_U)| \leq 2^{n(1+\epsilon)H(U)}. \tag{4.17}$$

(iii) For n large enough:

$$|T(n,\epsilon,p_U)| \geq (1-\epsilon)2^{n(1-\epsilon)H(U)}. \tag{4.18}$$

Proof. We show each statement in order.
(i) Recall from the definition of the type of u^n given by $\text{type}_{(u^n)}(u) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{u_i=u\}}$, we know from Definition (4.3) that $\text{type}_{(u^n)}(u) = p_U(u)(1\pm\epsilon)$. Now:

$$\mathbb{P}(U^n = u^n) \overset{\text{iid}}{=} \prod_{i=1}^n p_U(u_i) = \prod_{u\in\mathcal{U}} p_U(u)^{n\cdot\text{type}_{(u^n)}(u)} \Longrightarrow \frac{1}{n}\log_2\left(\mathbb{P}(U^n = u^n)\right) = \sum_{u\in\mathcal{U}} \text{type}_{(u^n)}(u)\log_2(p_U(u)).$$

Now, we may conclude on the bounds:

$$(1+\epsilon)\sum_{u\in\mathcal{U}} p_U(u)\log_2(p_U(u)) \leq \sum_{u\in\mathcal{U}} \text{type}_{(u^n)}(u)\log_2(p_U(u)) \leq (1-\epsilon)\sum_{u\in\mathcal{U}} p_U(u)\log_2(p_U(u)).$$

Using the definition of the entropy finally yields:

$$-(1+\epsilon)H(U) \leq \sum_{u\in\mathcal{U}} \text{type}_{(u^n)}(u)\log_2(p_U(u)) \leq -(1-\epsilon)H(U).$$

(ii) We bound above the probability by the largest value possible, i.e. the value 1, and find:

$$1 \geq \mathbb{P}(U^n \in T(n,\epsilon,p_U)) = \sum_{u^n\in T(n,\epsilon,p_U)} \mathbb{P}(U^n = u^n) \overset{\text{(i)}}{\geq} \sum_{u^n\in T(n,\epsilon,p_U)} 2^{-n(1+\epsilon)H(U)} = \big|T(n,\epsilon,p_U)\big|2^{-n(1+\epsilon)H(U)}.$$

(iii) For n large enough we have:

$$\begin{aligned}(1-\epsilon) \overset{(4.1.1)}{\leq} \mathbb{P}(U^n \in T(n,\epsilon,p_U)) = \sum_{u^n\in T(n,\epsilon,p_U)} \mathbb{P}(U^n = u^n) &\overset{\text{(i)}}{\leq} \sum_{u^n\in T(n,\epsilon,p_U)} 2^{-n(1-\epsilon)H(U)} \\ &= \big|T(n,\epsilon,p_U)\big|2^{-n(1-\epsilon)H(U)}.\end{aligned}$$

□

A general fact (that can be proven) is that if $S_n \subseteq \mathcal{U}^n$ for $n = 1,2,...$ such that $\mathbb{P}(U^n \in S_n) \geq \delta > 0$ for $U_1,...,U_n \overset{\text{iid}}{\sim} p_U$ then:

$$\liminf_n \left(\frac{1}{n}\log_2(|S_n|)\right) \geq H(U). \tag{4.19}$$

As a result, $|S_n|$ cannot be larger than $2^{nH(U)}$, where for equality we require $S_n = T(n,\epsilon,p_U)$ for $\delta = 1$. Suppose now that we consider $U_1,...,U_n \overset{\text{iid}}{\sim} p$ and consider $U^n \in T(n,\epsilon,q)$, for some probability densities p and q.

Theorem 4.1.3. *For all probability densities p and q, all n, all $\epsilon > 0$, for every $U^n \in T(n,\epsilon,q)$ and where $U_1,...,U_n \overset{iid}{\sim} p$, the following holds:*
(i)

$$2^{-n(1+\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \leq \mathbb{P}(U^n = u^n) \leq 2^{-n(1-\epsilon)\left(H(q)+D_{KL}(q\|p)\right)}. \tag{4.20}$$

(ii) For n large enough in the lower bound:

$$(1-\epsilon)2^{-n\left[D_{KL}(q\|p)-\epsilon\left(2H(q)+D_{KL}(q\|p)\right)\right]} \leq \mathbb{P}\big(U^n \in T(n,\epsilon,q)\big) \leq 2^{-n\left[D_{KL}(q\|p)-\epsilon\left(2H(q)+D_{KL}(q\|p)\right)\right]}. \tag{4.21}$$

Morally, for $U_1,..,U_n \overset{\text{iid}}{\sim} p$, we have the rough estimate $\mathbb{P}(U^n \in T(n,\epsilon,q)) \approx 2^{-n\cdot D_{KL}(q\|p)}$.

Proof. We prove the first statement in a similar way as in the previous Theorem.
(i) We have that:

$$\mathbb{P}(U^n = u^n) = \prod_{i=1}^n p(u_i) = \prod_{u\in\mathcal{U}} p(u)^{n\cdot\text{type}_{(u^n)}(u)} \Longrightarrow \frac{1}{n}\log_2\left(\mathbb{P}(U^n = u^n)\right) = \sum_{u\in\mathcal{U}} n\cdot\text{type}_{(u^n)}(u)\log_2(p(u)).$$

Now, we may conclude on the bounds:

$$(1+\epsilon)\sum_{u\in\mathcal{U}} q(u)\log_2(p(u)) \leq \frac{1}{n}\log_2\left(\mathbb{P}(U^n=u^n)\right) \leq (1-\epsilon)\sum_{u\in\mathcal{U}} q(u)\log_2(p(u)).$$

Dividing by $q(u)$ in the logarithm yields the required result:

$$-(1+\epsilon)\big(H(q)+D_{KL}(q\|p)\big) \leq \frac{1}{n}\log_2\left(\mathbb{P}(U^n=u^n)\right) \leq -(1-\epsilon)\big(H(q)+D_{KL}(q\|p)\big),$$

where $D_{KL}(q\|p)$ is the KL-divergence from p to q.

(ii) We start with the upper-bound to find:

$$\begin{aligned}
\mathbb{P}\big(U^n\in T(n,\epsilon,q)\big) = \sum_{u^n\in T(n,\epsilon,q)} \mathbb{P}(U^n=u^n) &\overset{(i)}{\leq} \sum_{u^n\in T(n,\epsilon,q)} 2^{-n(1-\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \\
&= |T(n,\epsilon,q)|2^{-n(1-\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \\
&\overset{(4.17)}{\leq} 2^{-n(1-\epsilon)\left(H(q)+D_{KL}(q\|p)\right)}2^{n(1+\epsilon)H(q)} \\
&= 2^{-n\left[D_{KL}(q\|p)-\epsilon\left(2H(q)+D_{KL}(q\|p)\right)\right]}.
\end{aligned}$$

For the lower-bound now, we find in a similar way:

$$\begin{aligned}
\mathbb{P}(U^n\in T(n,\epsilon,q)) = \sum_{u^n\in T(n,\epsilon,q)} \mathbb{P}(U^n=u^n) &\overset{(\mathrm{i})}{\geq} \sum_{u^n\in T(n,\epsilon,q)} 2^{-n(1+\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \\
&\geq |T(n,\epsilon,q)|2^{-n(1+\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \\
&\overset{(4.18)}{\geq} (1-\epsilon)2^{n(1-\epsilon)H(q)}2^{-n(1+\epsilon)\left(H(q)+D_{KL}(q\|p)\right)} \\
&= (1-\epsilon)2^{-n\left[D_{KL}(q\|p)-\epsilon\left(2H(q)+D_{KL}(q\|p)\right)\right]}.
\end{aligned}$$

□

Remark 4.1.4. *Suppose we have a binary hypothesis test given by:*

$$\begin{cases} (1) & U_1,U_2,\ldots \overset{\text{iid}}{\sim} p \\ (2) & U_1,U_2,\ldots \overset{\text{iid}}{\sim} q \end{cases} \tag{4.22}$$

We now consider the decision rule in $S_n\in\mathcal{U}^n$ to be:

$$\begin{cases} (1) & U^n\in S_n \Longrightarrow \text{decide } q \\ (2) & U^n\notin S_n \Longrightarrow \text{decide } p \end{cases} \tag{4.23}$$

And we note that if $\mathbb{P}(\text{error}|(2))\leq\epsilon$ we obtain as a result $\mathbb{P}(\text{error}|(1))\gtrapprox 2^{-n\cdot D_{KL}(q\|p)}$.

4.2 Data Compression of IID Sources via Typicality

To describe $U_1, ..., U_n \overset{iid}{\sim} p$ using nR bits, we take $T(n, \epsilon, p)$ knowing that $|T(n, \epsilon, p)| \leq 2^{n(1+\epsilon)H(p)}$. Thus, if $R > (1+\epsilon)H(p)$ we can assign distinct binary sequences of lengths nR to each number of $T(n, \epsilon, p)$ (Call this condition $(*)$). Then, reveal these sequences to the encoder and decoder:

- Encoder $\Longrightarrow$ read U^n and if $U^n \in T(n, \epsilon, p)$ then transmit the nR bits given by $(*)$.
- Decoder $\Longrightarrow$ read the nR bits and produce the u^n to which the nR bits are assigned.

Note that this is not a system that is practical, since it usually has a large running time along with a heavy computational power. Now, if $U^n \notin T(n, \epsilon, p)$ we then make an error, but as n gets large $\mathbb{P}(U^n \notin T(n, \epsilon, p)) \to 0$. The other option is to assign a binary sequence depending on the outcome of whether U^n is in $T(n, \epsilon, p)$ or not. Specifically:

- $U^n \in T(n, \epsilon, p) \Longrightarrow$ transmit 1 followed by nR bits.
- $U^n \notin T(n, \epsilon, p) \Longrightarrow$ transmit 0 followed by $n \log_2(|\mathcal{U}|)$ bits.

Hence with this scenario, depending on U^n, $\#\{\text{bits/letter}\} \approx R + \frac{1}{n}$ when $U^n \in T(n, \epsilon, p)$ and $\#\{\text{bits/letter}\} \approx \log_2(|\mathcal{U}|) + \frac{1}{n}$ when $U^n \notin T(n, \epsilon, p)$.

Chapter 5

Universal Data Compression

In this chapter, we investigate data compression under different point of views and assumptions. In particular, we will concentrate first on source coding for sequences under distribution uncertainty, and in a second instance we will study the Lempel-Ziv algorithm for universal source coding.

5.1 Data Compression for Sequences under Distribution Uncertainty

Consider $\mathcal{P}$ to be a class of stochastic processes on an alphabet $\mathcal{U}$ and suppose we want to solve the $\min_q \max_{p\in\mathcal{P}} D_{KL}(p\|q)$ problem described in Appendix (A.4). Thus, we must find a stochastic process q where we suppose further that $\frac{1}{n}\max_{p\in\mathcal{P}} D_{KL}(p_n\|q_n) \to 0$ as $n\to\infty$. Here, p_n denotes p restricted to the first n letters (same idea for q_n). Let us now investigate what it means. Here,

$$D(p_n\|q_n) = \sum_{u^n} p(u^n)\log_2\left(\frac{p(u^n)}{q(u^n)}\right), \tag{5.1}$$

with $p_n(u^n) = p(u_1)p(u_2|u_1)...p(u_n|u^{n-1})$ and $q_n(u^n) = q(u_1)q(u_2|u_1)...q(u_n|u^{n-1})$. Thus:

$$\begin{aligned}\frac{1}{n}D_{KL}(p_n\|q_n) = \frac{1}{n}\sum_{u^n} p_n(u^n)\sum_{i=1}^{n}\log_2\left(\frac{p(u_i|u^{i-1})}{q(u_i|u^{i-1})}\right) &= \frac{1}{n}\sum_{i=1}^{n}\sum_{u^i} p_i(u^i)\log_2\left(\frac{p(u_i|u^{i-1})}{q(u_i|u^{i-1})}\right)\\ &= \frac{1}{n}\sum_{i=1}^{n}\sum_{u^{i-1}} p_{i-1}(u^{i-1})\sum_{u^i} p_i(u_i|u^{i-1})\log_2\left(\frac{p(u_i|u^{i-1})}{q(u_i|u^{i-1})}\right)\\ &= \frac{1}{n}\sum_{i=1}^{n}\underbrace{\sum_{u^{i-1}} p_{i-1}(u^{i-1})D_{KL}(p(\cdot|u^{i-1})\|q(\cdot|u^{i-1}))}_{(\Delta)}.\end{aligned}$$

Then, if $\frac{1}{n}D_{KL}(p_n\|q_n) = \epsilon_n$ then for at most $n\sqrt{\epsilon_n}$ of the i's in (Δ) can be strictly greater than $\sqrt{\epsilon_n}$. Indeed, assuming that $\sum_{i=1}^{n} p_i a_i \le \epsilon$, Markov's Inequality gives us:

$$\mathbb{P}(a_i > \sqrt{\epsilon}) = \sum_{i=1}^{n}\mathbb{1}_{\{a_i>\sqrt{\epsilon}\}}p_i \le \frac{\mathbb{E}(a_i)}{\sqrt{\epsilon}} = \frac{1}{\sqrt{\epsilon}}\sum_{i=1}^{n} a_i p_i \le \frac{\epsilon}{\sqrt{\epsilon}} = \sqrt{\epsilon}. \tag{5.2}$$

For the i's for which $(\Delta)\le\sqrt{\epsilon_n}$, the total probability $p(\cdot)$ of the (u^{i-1})'s for which $D_{KL}(p(\cdot|u^{i-1})\|q(\cdot|u^{i-1}))\ge\sqrt{\sqrt{\epsilon_n}}$ is at most $\sqrt{\sqrt{\epsilon_n}}$. This implies that for most i's and $(u^{i-1})'s$ we have that $D_{KL}(p(\cdot|u^{i-1})\|q(\cdot|u^{i-1}))\le\sqrt{\sqrt{\epsilon_n}}$. This procedure represents a learning process.

Remark 5.1.1. *Suppose $\mathcal{U}=\{0,1\}$ and $\mathcal{P}$ is the set of i.i.d. processes on $\{0,1\}$. Thus:*

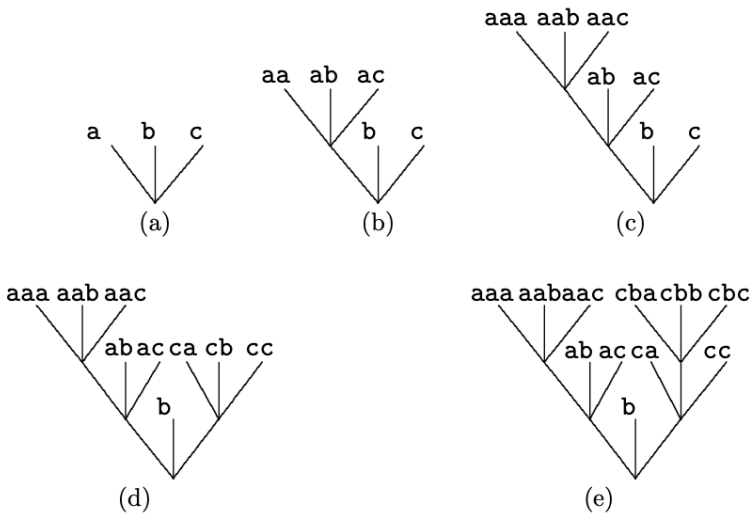
$$(*)=\begin{cases} q_{i|i-1}(0|u^{i-1})=\frac{\#\left\{0\text{'s in } u^{i-1}\right\}+\frac{1}{2}}{i} \\ q_{i|i-1}(1|u^{i-1})=\frac{\#\left\{1\text{'s in } u^{i-1}\right\}+\frac{1}{2}}{i} \end{cases} \quad , \quad i=1,2,... \tag{5.3}$$

Thus, the q_n we obtain via $(*)$ has the property that:

$$\frac{1}{n}D_{KL}(p_\theta^n\|q_n)\leq\frac{1}{n}\log_2(n+1). \tag{5.4}$$

5.2 Universal Data Compression by the Lempel-Ziv Algorithm

[**This section is taken from the notes of Prof. Emre Telatar**] Let us describe the Lempel-Ziv algorithm. The algorithm proceeds by generating a dictionary for the source and constantly updating it. It starts up with a dictionary just consisting of the words of length 1, and operates in the following manner: when the dictionary has D words, each of its words is assigned a binary codeword of length $\lceil\log_2(D)\rceil$ in lexicographic order. When a word in the dictionary is recognized in the input sequence, the encoder generates the binary codeword of that word on its output, and enlarges the dictionary by replacing the just recognized word with all its single letter extensions. The dictionary can be represented as a tree, whose leaves are the current dictionary entries. The figure below shows an example of the operation of the algorithm. Since the recognized words are encoded before the dictionary is modified, the decoder can keep track of the encoder's operation. Finally note that each parse extends the dictionary by $|\mathcal{U}|-1$ entries.

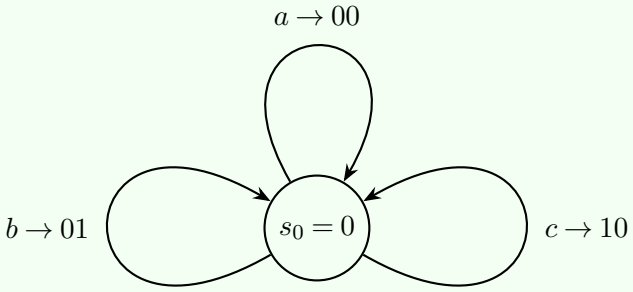


The parsing of the sequence `aaaccb` with the Lempel-Ziv algorithm. The figure shows the evolution of the dictionary. The sequence is parsed into the phrases `a`, `aa`, `c` and `cb`. Figure 2(a) shows the initial dictionary. In 2(b) we see the dictionary after reading `a`, 2(c) shows after `aaa` has been read, etc. At each stage one might assign each dictionary entry a fixed length binary codeword. If the assignment is done in lexicographic order, at stage (a) it will be {`a` → 00, `b` → 01, `c` → 10}, at stage (b) {`aa` → 000, `ab` → 001, ..., `c` → 100}, at stage (c) {`aaa` → 000, `aab` → 001, ..., `cc` → 110}, and at stage (d) {`aaa` → 0000, `aab` → 0001, `aac` → 0010, ..., `cc` → 1000}, and the output sequence will be 00,000,110,0111. (Commas are put in to aid the reader, they will not appear at the output.)

5.3 Finite-State Machine (FSM) Compression

Consider a set $\mathcal{S}$ of states such that $|\mathcal{S}| < \infty$, with a starting state $s_0 \in \mathcal{S}$. We also set a function $\texttt{next-state} : \mathcal{S} \times \mathcal{U} \to \mathcal{S}$ and an $\texttt{output} : \mathcal{S} \times \mathcal{U} \to \{0,1\}^*$ where for $i = \{0,1,2,...\}$ we have that $s_{i+1} = \texttt{next-state}(s_i, u_{i+1})$ and $y_{i+1} = \texttt{output}(s_i, u_{i+1})$. Now, the state the machine is in after reading u_n is given by the function $\texttt{final-state}(s, u_1u_2...u_n) \in \mathcal{S}$ where $u_1u_2...u_n \in \mathcal{U}^*$ and $n < \infty$. Then, the concatenation of the outputs while reading $u_1...u_n$ is $\texttt{output}(s, u_1...u_n) \in \{0,1\}^*$. From now on, we will assume that every $s \in \mathcal{S}$ may be reached starting from s_0 (i.e. $\mathcal{S} = \text{range}(\texttt{final-state})$).

Example 5.3.1 (Injective machine)**.** *Let $\mathcal{U} = \{a, b, c\}$. Then, if we read each letter we may encode it in the following way (reading $\to$ encoding):*



Now, the class of injective FSMs are the competitors to Lempel-Ziv, however we will enlarge this class to those machines which are **information lossless (IL)**.

Definition 5.3.2. *A FSM is **information lossless (IL)** if for all $u^n \neq v^m \in \mathcal{U}^*$ such that $u^n, v^m \in \mathcal{U}^*$ then either $\texttt{output}(s_0, u^n) \neq \texttt{output}(s_0, v^m)$, where $s_0 \in \mathcal{S}$ is the starting state of the machine, or $\texttt{final-state}(s_0, u^n) \neq \texttt{final-state}(s_0, v^m)$.*

Proposition 5.3.3. *Under the assumption that all $s \in \mathcal{S}$ is reachable from s_0 (i.e. $\forall s$, $\exists w \in \mathcal{U}^*$ such that $s = \texttt{final-state}(s_0, w)$), then a FSM is IL if and only if $\forall s, u \in \mathcal{U}^*, v \in \mathcal{U}^*, u \neq v$ either $\texttt{output}(s, u) \neq \texttt{output}(s, v)$ or $\texttt{final-state}(s, u) \neq \texttt{final-state}(s, v)$.*

Proof. ($\Longleftarrow$) Let $s = s_0$, then the statement holds by definition.

($\Longrightarrow$) Let $u^n = wu$ and $v^n = wv$, then by definition either $\texttt{output}(s_0, wu) \neq \texttt{output}(s_0, wv)$, where $s_0 \in \mathcal{S}$ is the starting state of the machine, or $\texttt{final-state}(s_0, wu) \neq \texttt{final-state}(s_0, wv)$. □

Definition 5.3.4. *Given $u^n = (u_1...u_n) \in \mathcal{U}^*$, $\{w_1, w_2, ..., w_m\} \in \mathcal{U}^*$ is called a **distinct parsing of u** if $u = w_1w_2...w_m$ and $i \neq j \Longrightarrow w_i \neq w_j$.*

Example 5.3.5. *For $u^n = abbaac$ then $\{\text{null}, a, b, ba, ac\}$ is a distinct parsing of u.*

Definition 5.3.6. *Define* $m^*(u^n) = \max\{m : u^n = w_1...w_m \text{ such that } i \neq j \implies w_i \neq w_j\}$. *In other words,* $\mathbf{m^*(u^n)}$ *is the largest* m *for which there is a distinct* m*-word parsing of* u^n.

Example 5.3.7. *For* $u^n = aaa$ *then* $\{\text{null}, a, aa\}$ *is a distinct parsing of* u. *Hence,* $m^*(u^n) = 3$.

Lemma 5.3.8. *Given* $u_n \in \mathcal{U}^*$:

$$\lim_{n\to\infty} \frac{m^*(u^n)}{n} = 0. \tag{5.5}$$

Proof. Suppose $m = m^*$ and $w_1, ..., w_m$ is a distinct parsing of $u^n = w_1...w_m$. Now, pick a positive integer k. The number of $(w_i)'s$ having $\text{length}(w_i) < k$ is at most $1 + \sum_{j=1}^{k-1} |\mathcal{U}|^j =: F_k$. Consequently, $(m - F_k)$ of the $(w_i)'s$ have length at least k. Thus, $n \geq (m - F_k)k$ if and only if $m = \frac{n}{k} + F_k$. Hence:

$$\frac{m^*(u^n)}{n} \leq \frac{1}{k} + \frac{F_k}{n} \implies \limsup_{n\to\infty} \frac{m^*(u^n)}{n} \leq \frac{1}{k}, \quad \forall k.$$

Since k is arbitrary we obtain $\limsup_{n\to\infty} \frac{m^*(u^n)}{n} = 0$. It is also evident that $\liminf_{n\to\infty} \frac{m^*(u^n)}{n} = 0$. Therefore, $\limsup_{n\to\infty} \frac{m^*(u^n)}{n} = \liminf_{n\to\infty} \frac{m^*(u^n)}{n} = \lim_{n\to\infty} \frac{m^*(u^n)}{n} = 0$. □

Lemma 5.3.9. *If* $y_1, ..., y_m$ *with* $m = \sum_{j=0}^{k-1} 2^j + r$ *for* $0 \leq r < 2^k$ *are distinct binary sequences (i.e.* $y_i \in \{0,1\}^*$, $i \neq j \implies y_i \neq y_j$*) then:*

$$\sum_{i=1}^{m} \text{length}(y_i) \geq \sum_{j=0}^{k-1} j2^j + rk. \tag{5.6}$$

Proof. Set the binary string $\{0,1\}^* = \{\text{null}, 0, 1, 00, 01, 10, 11, 000, ...\}$. Picking the first m elements of $\{0,1\}^*$ yields the smallest total length, giving us the result. □

Corollary 5.3.10. *If* $y_1, ..., y_m$ *are distinct binary sequences then:*

$$\sum_{i=1}^{m} \text{length}(y_i) \geq m \log_2\left(\frac{m}{8}\right). \tag{5.7}$$

Proof. We use the previous lemma. Of course, $m = \sum_{j=0}^{k-1} 2^j + r = 2^k - 1 + r$ for $0 \leq r < 2^k$. Now observe:

$$\sum_{j=0}^{k-1} \alpha^j = \frac{\alpha^k - 1}{\alpha - 1} \implies \frac{\partial}{\partial\alpha} \sum_{j=0}^{k-1} \alpha^j = \sum_{j=0}^{k-1} j\alpha^{j-1} = \frac{1}{\alpha} \sum_{j=0}^{k-1} j\alpha^j = \frac{\partial}{\partial\alpha}\left(\frac{\alpha^k - 1}{\alpha - 1}\right) = \frac{\alpha^{k-1}((k-1)\alpha - k) + 1}{(\alpha - 1)^2}.$$

As a result:

$$\sum_{j=0}^{k-1} j\alpha^j = \frac{\alpha^k((k-1)\alpha - k) + \alpha}{(\alpha - 1)^2} \quad \overset{\alpha=2}{\implies} \quad \sum_{j=0}^{k-1} j2^j = 2^k(k-2) + 2.$$

As a result, from Lemma (5.3.9):

$$\sum_{i=1}^{m}\text{length}(y_i) \geq 2^k(k-2)+2+rk = (k-2)(m+1-r)+2+rk = m(k-2)+2r+k \geq m(k-2).$$

Also, since $0 \leq r < 2^k$ we have that $m < 2^{k+1}$ and thus $\log_2(m) < k+1 \Longrightarrow \log_2\left(\frac{m}{8}\right) < k-2$. Overall:

$$\sum_{i=1}^{m}\text{length}(y_i) \geq m(k-2) \geq m\log_2\left(\frac{m}{8}\right).$$

□

Theorem 5.3.11. *Suppose u^n is fed to a IL-FSM with s states, and consider $u^n = w_1...w_m$ such that we have $m = m^*(u^n)$ distinct words. Then:*

$$\text{length(output of the machine having read } u^n) \geq m^*(u^n)\log_2\left(\frac{m^*(u^n)}{8s^2}\right). \tag{5.8}$$

Proof. Suppose we consider a set of states $\{\xi_1,\xi_2,...,\xi_m\}$ such that any pair of operations for each state (input,output) $:= \{(w_i,t_i)\}_{i=1}^m$ is being made in successive order of i. This simply means:

$$s_0 = \xi_0 \xrightarrow{\text{step 1}} (w_1,t_1) \xrightarrow{\text{step 2}} \xi_1 \xrightarrow{\text{step 3}} (w_2,t_2) \xrightarrow{\text{step 4}} \xi_2 \xrightarrow{\text{step 5}} \ldots \longrightarrow \xi_{m-1} \longrightarrow (w_m,t_m) \longrightarrow \xi_m$$

For $\alpha,\beta \in \mathcal{S}$, $I_{\alpha\beta} := \{i : \xi_{i-1} = \alpha, \xi_i = \beta\}$. Observe now that $\{w_i : i \in I_{\alpha\beta}\}$ is a distinct collection of words implying that $\{t_i : i \in I_{\alpha\beta}\}$ are distinct binary sequences (since the machine is IL). Now, using Corollary (5.3.10):

$$\sum_{i\in I_{\alpha\beta}}\text{length}(t_i) \geq |I_{\alpha\beta}|\log_2\left(\frac{|I_{\alpha\beta}|}{8}\right) \Longrightarrow \sum_{i=1}^{m}\text{length}(t_i) \geq \sum_{\alpha,\beta}|I_{\alpha\beta}|\log_2\left(\frac{|I_{\alpha\beta}|}{8}\right), \quad m = \sum_{\alpha,\beta}|I_{\alpha\beta}|.$$

We have $k = s^2$ non-negative numbers $(x_1,...,x_k)$ which sum to m. By symmetry and convexity of $x \mapsto x\log_2(x)$ we finally obtain:

$$\sum_{i=1}^{k} x_i\log_2\left(\frac{x_i}{8}\right) \geq \sum_{i=1}^{k}\left(\frac{m}{k}\right)\log_2\left(\frac{m}{8k}\right) = m\log_2\left(\frac{m}{8s^2}\right).$$

□

5.4 Lempel-Ziv and Information Lossless FSM (IL-FSM)

• Let us re-examine the Lempel-Ziv (LZ) Algorithm by considering $u^n = w_1w_2...w_mw_{m+1}$ for distinct $w_1,...,w_m$ and such that w_{m+1} has not yet been parsed (i.e. analysis of a string into logical syntactic components). Recall that Lempel-Ziv never uses {null} thus $u^n = \text{null}, w_1,...,w_{m-1}$ is a distinct parsing of u^n, where w_m is a leftover term. As a result, $(m+1) \leq m^*(u^n)$. Now:

$$\begin{aligned}\#\{\text{bits produced by LZ that allows recovery of } u_1...u_n\} &\leq \lceil \log_2(|\mathcal{U}|)\rceil + \lceil \log_2(2|\mathcal{U}|)\rceil + ... \\ &\quad + \lceil \log_2(m|\mathcal{U}|)\rceil + \lceil \log_2((m+1)|\mathcal{U}|)\rceil \\ &\leq (m+1)\lceil \log_2((m+1)|\mathcal{U}|)\rceil \\ &\leq m^*(u^n)\log_2\left(2|\mathcal{U}|m^*(u^n)\right).\end{aligned}$$

Note that, as a result, Lempel-Ziv produces more bits than IL-FSM where this excess of LZ above IL-FSM is at most $m^*(u^n)\log_2\left(16s^2|\mathcal{U}|\right)$ (taking a subtraction of the bound above with the one in Theorem (5.3.11)). As a result:

$$\{\text{excess source per letter of LZ above IL-FSM}\} \leq \frac{1}{n}m^*(u^n)\log_2\left(16s^2|\mathcal{U}|\right). \tag{5.9}$$

However, recall that (Lemma (5.3.8)) $\lim_{n\to\infty}\frac{1}{n}m^*(u^n) = 0$. Thus this difference in excess of source per letter of LZ above IL-FSM is less than or equal to 0, as $n \to \infty$.

• Let us continue with some further remarks. Suppose $U_1, U_2, ...$ is a stationary stochastic process with entropy rate $\mathcal{H}$. Now, given $\epsilon > 0$, we can find n such that $\frac{H(U^n)+1}{n} \leq \mathcal{H} + \epsilon$. As a result, there exists a prefix-free code c_n such that,

$$\frac{1}{n}\mathbb{E}\big[\text{length}(c_n(U_1...U_n))\big] \leq \mathcal{H} + \epsilon,$$

and in fact by stationarity for $i \in \mathbb{N}$,

$$\frac{1}{n}\mathbb{E}\big[\text{length}(c_n(U_{in+1}...U_{(i+1)n}))\big] \leq \mathcal{H} + \epsilon.$$

We can then implement any set of codes c_n using $\{U_1...U_n\}$, $\{U_{n+1}...U_{2n}\}$,$\{U_{2n+1}...U_{3n}\}$,... by a FSM, and so for this FSM:

$$\frac{1}{k}\mathbb{E}[\#\text{bits after seeing } u_1...u_k] \leq \mathcal{H} + \epsilon.$$

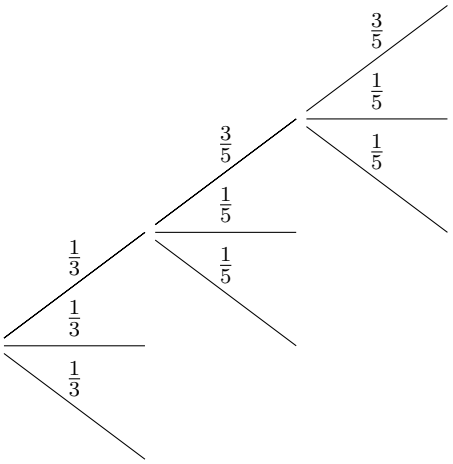
Overall we get that,

$$(*) := \lim_{n\to\infty}\frac{1}{n}\mathbb{E}[\#\text{bits produced by LZ that allows recovery of } U_1...U_n] \leq \mathcal{H}.$$

On the other hand we also have that $(*) \geq \mathcal{H}$. Thus $(*) = \mathcal{H}$ as n gets large, meaning that LZ compresses stationary processes well. Then, we might as well pick the Lempel-Ziv since in the long run (i.e. for long strings) the Lempel-Ziv algorithm will do as well as the best finite state machine.

• For LZ by considering $\mathcal{U} = \{a, b, c\}$ and the dictionary $\mathcal{D} = \{a, b, c\}$, we had that the update of the dictionary after a was given by,

$$\mathcal{D} = \{a, b, c\} \xrightarrow{\text{after } a} \{aa, ab, ac, b, c\}.$$



In this tree, $\nearrow$ represents a, $\rightarrow$ represents b and $\searrow$ represents c. Thus, it is clear that we can infer on the letter a by using $q_{LZ}(u_n|u_1...u_{n-1})$. Also note that for a stationary process, we have for most times i, with probability 1 and set of sequences $(u_1,...,u_{i-1})$: $D_{KL}(p_{U_i|U^{i-1}}(\cdot|u^{i-1})\|q_{LZ}(\cdot|u^{i-1})) \approx 0$. As a result, Lempel-Ziv could serve as a good predictive algorithm using the probability measure q_{LZ}.

5.5 Predicting Realizations of Random Variables

The goal is to predict the realization of a random variable U in the form of a probability distribution q on $\mathcal{U}$. To this end, suppose we observe U with the value $u \in \mathcal{U}$. As a result, we must pay a penalty given by the quantity $\texttt{penalty}(q(u))$. Here the $\texttt{penalty}$ function is defined as $\texttt{penalty} : [0,1] \to \mathbb{R}$ and it should be a decreasing function. Now suppose $U \sim p$ has true distribution p, then:

$$\mathbb{E}(\text{penalty}) = \sum_{u \in \mathcal{U}} p(u)\texttt{penalty}(q(u)). \tag{5.10}$$

Example 5.5.1. *Consider the function* $\texttt{penalty}(z) = 1 - z$ *on a binary outcome* $\{0,1\}$. *We require* $p_0 + p_1 = 1$ *and* $q_0 + q_1 = 1$ *to solve the minimization problem:*

$$\min_{0 \le q_0 \le 1} \mathbb{E}(\text{penalty}) = \min_{0 \le q_0 \le 1} p_0(1-q_0) + p_1(1-q_1) = \min_{0 \le q_0 \le 1} (p_1 - p_0)q_0 + p_0.$$

By differentiation on q_0, *it is not difficult to see that the optimal* q_0 *is given by* $q_0^* = \mathbb{1}_{\{p_1 < p_0\}}$.

Example 5.5.2. *Consider the function* $\texttt{penalty}(z) = (1-z)^2$ *on a binary outcome* $\{0,1\}$. *We require* $p_0 + p_1 = 1$ *and* $q_0 + q_1 = 1$ *to solve the minimization problem:*

$$\min_{0 \le q_0 \le 1} \mathbb{E}(\text{penalty}) = \min_{0 \le q_0 \le 1} p_0(1-q_0)^2 + p_1(1-q_1)^2 = \min_{0 \le q_0 \le 1} p_0(1-q_0)^2 + p_1 q_0^2.$$

By differentiation on q_0, *it is not difficult to see that the optimal* q_0 *is given by* $q_0^* = p_0$.

Remark 5.5.3. *The two examples above only work on the binary alphabet* $\{0,1\}$. *In general, we would expect* $q^* = p$ *for general* $\texttt{penalty}$ *functions on arbitrary alphabets. These functions usually take the form* $\texttt{penalty}(z) = -\log_2(\xi)$.

Chapter 6

Transmission of Information

In this chapter, we leave aside the notion of data compression and move forward to the problem of transmission of information. To this end, we consider a **channel** taking the following form:



This channel is governed by a probability kernel $\{\mathbb{P}(Y = y|x) : x \in \mathcal{X}, y \in \mathcal{Y}\}$. Consequently, the use of a channel over multiple time instances will similarly be given by:



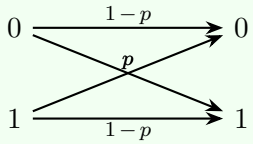
In this case, the channel is governed by the kernel $\mathbb{P}(Y_n = y_n|x_n, x_{n-1}, ..., x_1, Y_{n-1} = y_{n-1}, ..., Y_1 = y_1)$ which we will denote by $f_n(y^n, x^n)$. We remark that this kernel depends on two sets of quantities, namely $\{Y_n = y_n\}$ and x_n being the present values, and $\{x_{n-1}, ..., x_1, Y_{n-1} = y_{n-1}, ..., Y_1 = y_1\}$ being the past values.

6.1 Memoryless and Stationary Channels

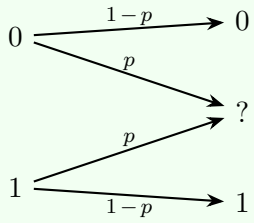
Definition 6.1.1. *A channel is said to be* ***memoryless*** *if* $f_n(y^n, x^n) = p_n(y_n|x_n)$.

Definition 6.1.2. *A channel is said to be* ***memoryless and stationary*** *if* $p_n(y_n, x_n) = p(y_n|x_n)$.

Example 6.1.3 (Binary symmetric channel)**.** *Below,* $x \in \{0,1\}$ *and* $y \in \{0,1\}$.



Example 6.1.4 (Binary erasure channel)**.** *Below, $x \in \{0,1\}$ and $y \in \{0,1,?\}$ where ? represents an erasure of a symbol. For example, sequences may look like $u_1??u_2u_3u_4?u_5???....$*



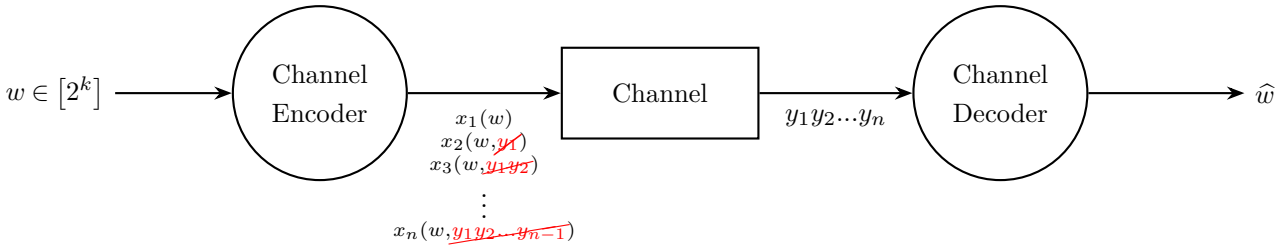
Example 6.1.5 (Gaussian channel)**.** *A Gaussian channel is one in which $x \in \mathbb{R}$ and $Y \sim \mathcal{N}(x, \sigma^2)$, i.e. $Y = x + \text{Gaussian}(0, \sigma^2)$.*

Definition 6.1.6. *A memoryless channel is said to have **no feedback** if:*

$$\text{No feedback} \implies \mathbb{P}(X_n = x_n | Y^{n-1} = y^{n-1}, X^{n-1} = x^{n-1}) = \mathbb{P}(X_n = x_n | X^{n-1} = x^{n-1}), \quad (6.1)$$

i.e. we have no backward information flowing in the system. And,

$$\text{Memoryless} \implies \mathbb{P}(Y_n = y_n | X^n = x^n, Y^{n-1} = y^{n-1}) = \mathbb{P}(Y_n = y_n | X_n = x_n). \quad (6.2)$$



Note that in the above figure, we have crossed out any knowledge of $y_1y_2...y_{n-1}$ as a result of the no feedback property. Also remark that at the output stage, $\mathbb{P}(\widehat{w} \neq w | w) \approx 0$.

Proposition 6.1.7. *For a memoryless and no feedback channel we have that:*

$$\mathbb{P}(Y^n = y^n | X^n = x^n) = \prod_{i=1}^{n} \mathbb{P}(Y_i = y_i | X_i = x_i) \equiv \prod_{i=1}^{n} p_i(y_i | x_i). \quad (6.3)$$

Proof. First, by Bayes' Theorem we observe that:

$$\mathbb{P}(Y^n = y^n | X^n = x^n) = \frac{\mathbb{P}(X^n = x^n, Y^n = y^n)}{\mathbb{P}(X^n = x^n)}.$$

Now, we use properties of conditioning to find:

$$\mathbb{P}(X^n = x^n, Y^n = y^n) = \mathbb{P}(X_1 = x_1)\mathbb{P}(Y_1 = y_1|X_1 = x_1)\mathbb{P}(X_2 = x_2|X_1 = x_1, \underbrace{\cancel{Y_1 = y_1}}_{\text{no feedback}}) \times$$

$$\times \mathbb{P}(Y_2 = y_2|X_2 = x_2, \underbrace{\cancel{X_1 = x_1}, \cancel{Y_1 = y_1}}_{\text{memoryless}}) \times \ldots \times \mathbb{P}(X_n = x_n|x^{n-1}, \cancel{y^{n-1}})\mathbb{P}(Y_n = y_n|x_n, \cancel{x^{n-1}}, \cancel{y^{n-1}})$$

$$= \mathbb{P}(X^n = x^n) \prod_{i=1}^{n} \mathbb{P}(Y_i = y_i|X_i = x_i).$$

□

Corollary 6.1.8. *For a stationary, memoryless and no feedback channel we have that:*

$$\mathbb{P}(Y^n = y^n|X^n = x^n) = \prod_{i=1}^{n} p(y_i|x_i). \tag{6.4}$$

Proof. This corollary holds immediately from the previous result and the definition of stationarity. □

Note that from $p_{Y^n|X^n}(y^n|x^n) = \prod_{i=1}^n p_{Y_i|X_i}(y_i|x_i)$ we have that $H(Y^n|X^n) = \sum_{i=1}^n H(Y_i|X_i)$ under memoryless, stationary and no feedback channels. In generality, without any such assumptions, we expect $H(Y^n|X^n) \leq \sum_{i=1}^n H(Y_i|X_i)$.

Definition 6.1.9. *Given a memoryless, stationary channel described by the probability kernel $p_{Y|X}$, we define the quantity $\boldsymbol{c(p_{Y|X})}$ to be:*

$$c(p_{Y|X}) := \max_{p_X} I(X;Y). \tag{6.5}$$

Example 6.1.10. *Consider the binary symmetric channel defined in Example (6.1.3) $BSC(p)$. Now, $H(Y|X=0) = h_2(p) = H(Y|X=1)$ and so $H(Y|X) = h_2(p)$. Then,*

$$I(X;Y) = H(Y) - H(Y|X) = H(Y) - h_2(p).$$

We can then make $H(Y) = 1$ by choosing $p_X = \left(\frac{1}{2}, \frac{1}{2}\right)$ implying that $c(BSC(p)) = 1 - h_2(p)$.

Example 6.1.11. *Consider the binary erasure channel defined in Example (6.1.4) $BEC(p)$. First note that $H(X|Y=0) = 0 = H(X|Y=1)$. Also, considering $p_X = (p_0, p_1)$ we find,*

$$\mathbb{P}(Y = ?) = \mathbb{P}(Y = ?|X = 0)p_0 + \mathbb{P}(Y = ?|X = 1)p_1 = p$$

Moreover, $\mathbb{P}(X = 0|Y = ?) = \frac{\mathbb{P}(X=0, Y=?)}{\mathbb{P}(Y=?)} = \frac{p_0}{p}\mathbb{P}(Y = ?|X = 0) = p_0$ and similarly $\mathbb{P}(X = 1|Y = ?) = p_1$. Therefore, $I(X;Y) = H(X) - H(X|Y) = (1-p)H(X)$ and hence $c(BEC(p)) = 1 - p$ is achieved for $X \sim Ber(p)$.

Remark 6.1.12 (**Important**). *We note that:*

$$I(X;Y) = \texttt{function}(p_X, p_{Y|X}) = H(Y) - H(Y|X) = \texttt{function}(p_Y) - \underbrace{\sum_x p_X(x) \underbrace{H(Y|X=x)}_{\text{function of } p_{Y|X}}}_{\text{linear in } p_X}$$

Now, $p_Y(y) = \sum_x p_{Y|X}(y|x)p_X(x)$ *and so* p_Y *is a linear function of* p_X*. Also,* $H(Y)$ *is a concave function of* p_Y *and so:*

$$p_X \xrightarrow{\text{linear}} p_Y \xrightarrow{\text{concave}} H(Y).$$

As a result:

$$\lambda p_{X_1} + (1-\lambda)p_{X_2} \longrightarrow \lambda p_{Y_1} + (1-\lambda)p_{Y_2} \longrightarrow \lambda H(p_{Y_1}) + (1-\lambda)H(p_{Y_2}).$$

This implies that $p_X \longrightarrow H(Y)$ *is concave and so* $p_X \longrightarrow I(X;Y) = H(Y) - H(Y|X)$ *is a concave function.*

Theorem 6.1.13 (Fano's Inequality). *Suppose* U *and* V *are random variables taking values in a common alphabet* $\mathcal{U}$*. Let* $p := \mathbb{P}(U \neq V)$*. Then:*

$$H(U|V) \leq h_2(p) + p\log_2\left(|\mathcal{U}| - 1\right). \tag{6.6}$$

Proof. Consider $W = \mathbb{1}_{\{U=V\}} = 1 - \mathbb{1}_{\{U\neq V\}}$. Now, on the one hand by the chain rule, we have that $H(UW|V) = H(U|V) + \underbrace{H(W|UV)}_{=0} = H(U|V)$. On the other hand, we may also write the chain rule in the following way:

$$H(UW|V) = H(W|V) + H(U|WV) \leq H(W) + H(U|WV) \leq h_2(p) + H(U|WV).$$

Here, $H(U|WV) = \sum_{w\in\{0,1\}} \sum_v p_{WV}(w,v) H(U|W=w, V=v)$ and thus noting that $H(U|W=1, V=v) = 0$ and $H(U|W=0, V=v) \leq \log_2\left(|\mathcal{U}| - 1\right)$. Therefore:

$$H(U|WV) \leq \sum_v \mathbb{P}(W=0, V=v)\log_2\left(|\mathcal{U}|-1\right) = \mathbb{P}(W=0)\log_2\left(|\mathcal{U}|-1\right) = p\log_2\left(|\mathcal{U}|-1\right).$$

Combining the two chain rules we finally find:

$$H(U|V) = H(UW|V) \leq h_2(p) + H(U|WV) \leq h_2(p) + p\log_2\left(|\mathcal{U}|-1\right).$$

□

Corollary 6.1.14. *Suppose* $U_1, ..., U_k$ *and* $V_1, ..., V_k$ *are sequences of random variables taking values in a common alphabet* $\mathcal{U}$*. Let* $p_i := \mathbb{P}(U_i \neq V_i)$ *and* $p = \frac{1}{k}\sum_{i=1}^k p_i$*. Then:*

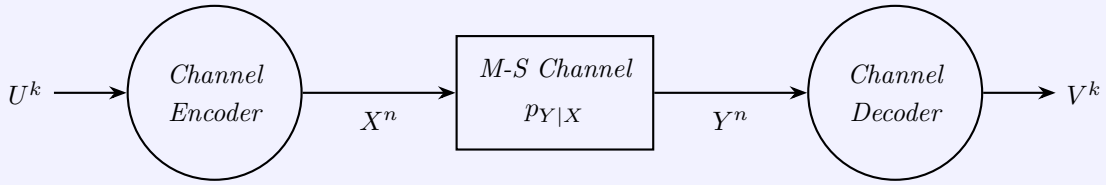
$$\frac{1}{k} H\left(U^k | V^k\right) \leq h_2(p) + p\log_2\left(|\mathcal{U}| - 1\right). \tag{6.7}$$

Proof. We start with the chain rule to find:

$$\begin{aligned}
\frac{1}{k}H(U^k|V^k) &= \frac{1}{k}\sum_{i=1}^{k} H(U_i|U^{i-1}V^k) \\
&\leq \frac{1}{k}\sum_{i=1}^{k} H(U_i|V_i) && \rightsquigarrow \text{Conditioning reduces entropy} \\
&\leq \frac{1}{k}\sum_{i=1}^{k} \left(h_2(p_i) + p_i \log_2\left(|\mathcal{U}|-1\right)\right) && \rightsquigarrow \text{Fano's Inequality} \\
&= \left(\frac{1}{k}\sum_{i=1}^{k} h_2(p_i)\right) + p\log_2\left(|\mathcal{U}|-1\right) \\
&\leq h_2\left(\frac{1}{k}\sum_{i=1}^{k} p_i\right) + p\log_2\left(|\mathcal{U}|-1\right) && \rightsquigarrow h_2 \text{ is concave} \\
&= h_2(p) + p\log_2\left(|\mathcal{U}|-1\right).
\end{aligned}$$

□

Theorem 6.1.15. *Suppose we consider the following diagram, for a memoryless and stationary channel (M-S channel):*



Let $p_i := \mathbb{P}(U_i \neq V_i)$ *and* $p = \frac{1}{k}\sum_{i=1}^{k} p_i$. *Thus:*

$$h_2(p) + p\log_2\left(|\mathcal{U}|-1\right) \geq \frac{1}{k}H(U^k) - \frac{n}{k}c(p_{Y|X}). \tag{6.8}$$

Proof. Observe from the diagram that it must be that $U^k \leftrightarrow X^n \leftrightarrow Y^n \leftrightarrow V^k$ forms a Markov chain. Thus in particular, $(U^k X^n) \leftrightarrow Y^n \leftrightarrow V^k$ is also a Markov chain. Then, by the Data Processing Inequality [Proposition (2.1.18)] we have $I(U^k X^n; V^k) \leq I(U^k X^n; Y^n)$ and also $I(U^k X^n; V^k) \geq I(U^k; V^k)$. Finally by the chain rule:

$$\begin{aligned}
I(U^k;V^k) \leq I(U^k X^n;V^k) \leq I(U^k X^n;Y^n) &\leq I(X^n;Y^n) \\
&= H(Y^n) - H(Y^n|X^n) \\
&\leq \underbrace{\sum_{i=1}^{n} H(Y_i)}_{\substack{\text{upper-bound} \\ \text{on } H(Y^n)}} - \underbrace{\sum_{i=1}^{n} H(Y_i|X_i)}_{\substack{\text{memoryless channel} \\ \text{with no feedback}}} = \sum_{i=1}^{n} I(X_i;Y_i).
\end{aligned}$$

Here, $I(X_i;Y_i)$ is a function of p_{X_i} and of $p_{Y|X}$ where $p_{XY} = p_X p_{Y|X}$ such that $p_{Y|X}$ represents the channel and $p_X = \frac{1}{n}\sum_{i=1}^{n} p_i$. We then have:

$$\sum_{i=1}^{n} I(X_i;Y_i) = n\left(\frac{1}{n}\sum_{i=1}^{n} I(X_i;Y_i)\right) \leq nI(X;Y) \leq n \cdot c(p_{Y|X}),$$

where the first inequality holds by the concavity of $I(X;Y)$ in p_{X_i} [see Remark (6.1.12)]. Thus, from Fano's Inequality [Theorem (6.1.13)] we finally conclude:

$$h_2(p) + p\log_2(|\mathcal{U}|-1) \geq \frac{1}{k}H(U^k|V^k) = \frac{1}{k}[H(U^k) - I(U^k;V^k)] \geq \frac{1}{k}[H(U^k) - n \cdot c(p_{Y|X})].$$

□

Theorem 6.1.16. *Suppose we consider the diagram from Theorem (6.1.15), where we have a stationary source U^k producing ρ_S letter/unit-time and we have ρ_C channel uses/unit-time. Let $p_i := \mathbb{P}(U_i \neq V_i)$ and $p = \frac{1}{k}\sum_{i=1}^{k} p_i$. Then:*

$$h_2(p) + p\log_2(|\mathcal{U}|-1) \geq \mathcal{H} - \frac{\rho_C}{\rho_S}c(p_{Y|X}). \tag{6.9}$$

Proof. Recall that for a stationary source U^k, $\frac{1}{k}H(U^k)$ is a decreasing sequence with limit $\mathcal{H} = \lim_k \frac{1}{k}H(U^k)$. Then:

$$\frac{1}{k}H(U^k|V^k) \geq \mathcal{H} - \frac{n}{k}c(p_{Y|X}) = \mathcal{H} - \frac{\rho_C}{\rho_S}c(p_{Y|X}).$$

Now, from Fano's Inequality, we have:

$$h_2(p) + p\log_2(|\mathcal{U}|-1) \geq \frac{1}{k}H(U^k|V^k).$$

As a result, we may conclude that:

$$h_2(p) + p\log_2(|\mathcal{U}|-1) \geq \mathcal{H} - \frac{\rho_C}{\rho_S}c(p_{Y|X}).$$

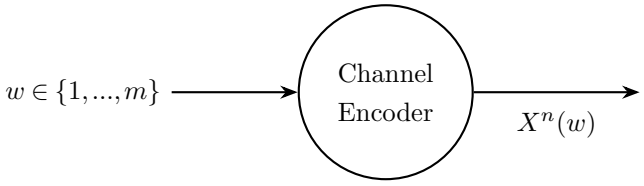
□

Corollary 6.1.17. *If $\rho_S\mathcal{H} > \rho_C c(p_{Y|X})$, then it is impossible to design arbitrarily reliable data transmission systems.*

So far, we have seen some bad news results. We now start a program to show good news counterparts, where we will use the process of random construction.

6.2 Random Coding

We consider an encoder of the form,



The encoder is also described by a table,

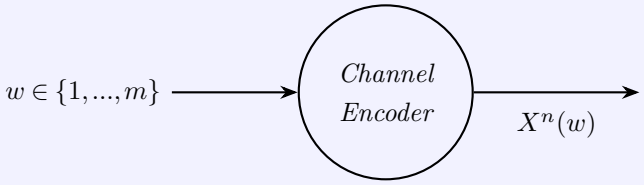
$$\begin{pmatrix} X_1(1) & \dots & X_n(1) \\ \vdots & & \vdots \\ X_1(m) & \dots & X_n(m) \end{pmatrix}.$$

The idea is to populate this table by choosing each of its nm entries as i.i.d. $\sim p_X$. Now, for the decoder, given $(Y_1,...,Y_n)$ and for each row of the table, pick $\delta > 0$ and check if for each $w \in \{1,...,m\}$

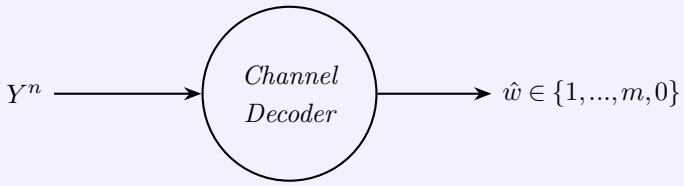
$$((X_1(w),Y_1),(X_2(w),Y_2),...,(X_n(w),Y_n)) \in T(n,\delta,p_{XY}).$$

If only this specific w satisfies the check then declare success for this w. In other words, set $\hat{w} = w_0$ if $(X^n(w_0),Y^n) \in T(n,\delta,p_{XY})$ and thus for all $w \neq w_0$, $(X_n(w),Y^n) \notin T(n,\delta,p_{XY})$. If no such w_0 exist, we set $\hat{w} = 0$. Let us now analyse the probability of error:

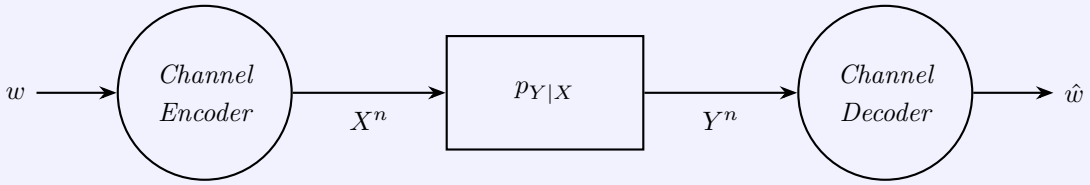
Corollary 6.2.1. *[Probability of error] For any $p_{Y|X}, p_X, \epsilon > 0, m, n$, there exists an encoder*



and a decoder



such that, when



we have as a result:

$$\frac{1}{m}\sum_{w=1}^{m} \mathbb{P}(\hat{w} \neq w | w) \leq \frac{\epsilon}{2} + (m-1)2^{-n(1-\epsilon)I(X;Y)}. \tag{6.10}$$

Proof. Suppose $w \in \{1,...,m\}$ is the sent message, then for n large we have

$$\mathbb{P}[(X_1(w),Y_1),(X_2(w),Y_2),...,(X_n(w),Y_n) \notin T(n,\delta,p_{XY})] < \frac{\epsilon}{2},$$

where all the elements $(X_i(w),Y_i) \sim p_{XY}$. Now, for $w \neq w'$, note that $\mathbb{P}(X_1(1) = x, X_1(2) = x', Y_1 = y) = p_X(x)p_X(x')p_{Y|X}(y|x)$, since y and x' never interact. Thus, for $w \neq w'$, since for all i we

have $X_i(w') \sim p_X$ and $Y_i \sim p_Y$, we know that $(X_i(w'), Y_i) \sim p_X p_Y$. Also, by Theorem (4.1.3),

$$\mathbb{P}\big((X^n(w'), Y^n) \in T(n, \delta, p_{XY})\big) \leq 2^{-n[(1-\delta)D(p_{XY}\|p_X p_Y) - 2\delta H(p_{XY})]}$$

and in fact we will choose a small enough $\delta > 0$ such that $(1-\delta)D(p_{XY}\|p_X p_Y) - 2\delta H(p_{XY}) \geq (1-\epsilon)D(p_{XY}\|p_X p_Y)$. Thus,

$$\{\hat{w} \neq w\} = \{(X^n(w), Y^n) \notin T(n, \delta, p_{XY})\} \cup \left(\bigcup_{w' \neq w} \{(X^n(w'), Y^n) \in T(n, \delta, p_{XY})\} \right),$$

and consequently,

$$\begin{aligned} \mathbb{P}(\hat{w} \neq w | w) &\leq \mathbb{P}\left((X^n(w), Y^n) \notin T(n, \delta, p_{XY})\right) + \sum_{w' \neq w} \mathbb{P}\left((X^n(w'), Y^n) \in T(n, \delta, p_{XY})\right) \\ &\leq \frac{\epsilon}{2} + (m-1)2^{-n(1-\epsilon)D(p_{XY}\|p_X p_Y)}. \end{aligned}$$

Using the fact that $I(X;Y) = D(p_{XY}\|p_X p_Y)$ we conclude

$$\frac{1}{m}\sum_{w=1}^{m} \mathbb{P}(\hat{w} \neq w | w) \leq \frac{\epsilon}{2} + (m-1)2^{-n(1-\epsilon)I(X;Y)}.$$

□

Let us now make an observation: if $2m$ non-negative numbers $a_1, ..., a_{2m}$ satisfy $\frac{1}{2m}(a_1 + ... + a_{2m}) \leq \eta$ then $\sum_{j=1}^{2m} \mathbb{1}_{\{a_j > 2\eta\}} \leq m$.

Corollary 6.2.2. *Considering the same setup as in Corollary (6.2.1), for all $w \in \{1, ..., m\}$:*

$$\mathbb{P}(\hat{w} \neq w | w) \leq 2\left(\frac{\epsilon}{2} + (2m-1)2^{-n(1-\epsilon)I(X;Y)}\right). \tag{6.11}$$

Proof. Consider a system with $2m$ numbers and apply the previous Corollary. Then, discard the m numbers with the highest probability. □

Corollary 6.2.3. *For any $p_{Y|X}, \delta > 0, R < c(p_{Y|X})$. Then, there exists an encoder/decoder with $m = \lceil 2^{nR} \rceil$ (so that $\frac{1}{n}\log_2(m) \geq R$) and for all $w \in \{1, ..., m\}$, $\mathbb{P}(\hat{w} \neq w | w) < \delta$.*

Proof. Choose p_X such that $I(X;Y) > R$, then by the previous Corollary with $m = \lceil 2^{nR} \rceil$ we have,

$$\begin{aligned} \mathbb{P}(\hat{w} \neq w) &< \epsilon + 2\left(2\lceil 2^{nR} \rceil - 1\right)2^{-n(1-\epsilon)I(X;Y)} \\ &\leq \epsilon + 2(2 \cdot 2^{nR} + 1)2^{-n(1-\epsilon)I(X;Y)} && \rightsquigarrow \lceil x \rceil \leq x + 1 \\ &\leq \epsilon + 6 \cdot 2^{nR} 2^{-n(1-\epsilon)I(X;Y)} && \rightsquigarrow 1 \leq 2^{nR} \end{aligned}$$

Choose $\epsilon < \delta/2$ and $(1-\epsilon)I(X;Y) > R$ so that

$$\mathbb{P}(\hat{w} \neq w|w) \leq \frac{\delta}{2} + \underbrace{6 \cdot 2^{-n[(1-\epsilon)I(X;Y)-R]}}_{(*)}.$$

Since $(1-\epsilon)I(X;Y) - R > 0$ we may choose n large enough so that $(*) < \delta/2$. □

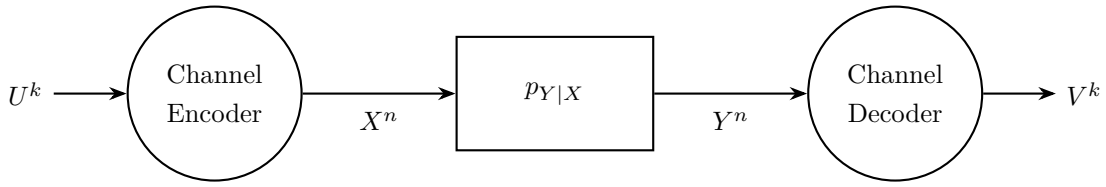
The moral of the story is that if $R < c(p_{Y|X})$ then reliable communication is possible at a rate $\geq R$ bits/channel-uses.

Remark 6.2.4. *Instead of the decoding rule $\hat{w} = \arg\max_{1\leq j\leq m} p_{Y|X}(Y^n|X^n(j))$ (the maximum likelihood rule) which minimises $(1/m)\sum_{i=1}^{m}\mathbb{P}(\hat{w} \neq w|w)$, we have used the typicality rule which is suboptimal (i.e. bounds are not as tight as for the ML rule), and still we obtained $\mathbb{P}(\hat{w} \neq w|w) \searrow 0$ as $n \to \infty$ as long as $m = \lceil 2^{nR} \rceil$ with $R < c(p_{Y|X})$. We could show with the ML rule that $\mathbb{P}(\hat{w} \neq w|w)$ decays exponentially small in n.*

For the practical aspects: For $R = 1/2$ and $n = 100$, we have to consider $2^{nR} = 2^{50}$ rows and so the decoding effort is greater than 2^{50} operations only for 50 bits of information. All of this theory has more of a conceptual aspect rather than practical. We could however find practical codes in which the decoding effort of m bits is not of order $\mathcal{O}(m)$ anymore but $\mathcal{O}(\log(m)\log\log(m))$ which is of course much better.

6.3 Cost Constraints in Channels

Suppose we have p_X and also a cost $b(x)$ of using letter x where $b : \mathcal{X} \to \mathbb{R}$. Suppose also that the average cost is given by $\mathbb{E}[\frac{1}{n}\sum_{i=1}^{n} b(x_i)] =: \beta$. We can show that if



then we have,

$$I(U^k;V^k) \leq n \cdot \underbrace{\max_{\substack{p_X \\ \mathbb{E}(b(x))=\beta}} I(X;Y)}_{=:c(p_{Y|X},\beta)}. \tag{6.12}$$

Remark 6.3.1. *Note the following:*

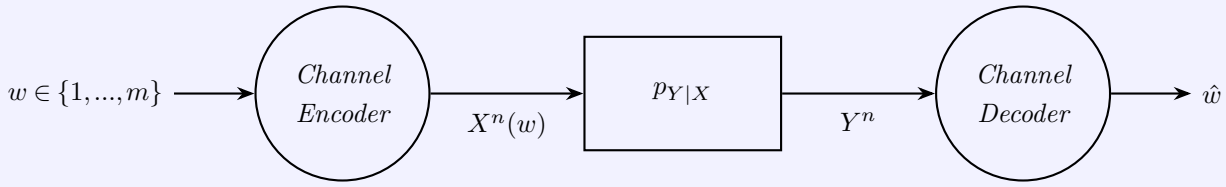
$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(b(x_i)) = \frac{1}{n}\sum_{i=1}^{n}\sum_{x} p_{X_i}(x)b(x) = \sum_{x}\left(\frac{1}{n}\sum_{i=1}^{n} p_{X_i}(x)\right)b(x) \equiv \sum_{x} p_X(x)b(x) = \mathbb{E}(b(X)).$$

Consequently, we can now give the two following theorems.

Theorem 6.3.2. *If (i) $U_1, U_2, \ldots$ is a stationary source with entropy rate $\mathcal{H}$ producing ρ_S letters/unit-time, (ii) if the channel $p_{Y|X}$ is used ρ_C times/unit-time, (iii) if $\rho_S \mathcal{H} > \rho_C c(p_{Y|X}, \beta)$, and (iv) if the time-average expected cost is equal to β, then with $p = \frac{1}{k}\sum_{i=1}^{k} \mathbb{P}(U_i \neq V_i)$ we have:*

$$h_2(p) + p\log(|\mathcal{U}| - 1) \geq \mathcal{H} - \frac{\rho_C}{\rho_S} c(p_{Y|X}) > 0. \tag{6.13}$$

Theorem 6.3.3. *For any $p_{Y|X}, b : \mathcal{X} \to \mathbb{R}, R, \beta$ satisfying $R < c(p_{Y|X}, \beta)$, $\epsilon > 0$, there exists an encoder/decoder such that*



where:

(i)

$$\frac{1}{n}\log_2(m) \geq R. \tag{6.14}$$

(ii)

$$\mathbb{P}(\hat{w} \neq w | w) < \epsilon \quad \text{for all } w. \tag{6.15}$$

(iii)

$$\frac{1}{n}\sum_{i=1}^{n} b(X_i(w)) = \beta \pm \epsilon \quad \text{for all } w. \tag{6.16}$$

Proof. We use the random encoder construction with the corresponding typicality decoder, i.e. we pick p_X with $\mathbb{E}(b(X)) = \beta$ and $I(X;Y) > R$. Pick n and $\tilde{m} = 2\lceil 2^{nR} \rceil - 1$, fill the encoding table with i.i.d. $\sim p_X$ entries. Decoding is done by declaring j_0 if $(X^n(j_0), Y^n) \in T(n, \delta, p_{XY})$ and for all $j \neq j_0$, $(X^n(j), Y^n) \notin T(n, \delta, p_{XY})$.
Then as we saw for small enough δ and large enough n,

$$\frac{1}{\tilde{m}}\sum_{w=1}^{\tilde{m}} \mathbb{P}(\hat{w} \neq w | w) < \frac{\epsilon}{2}.$$

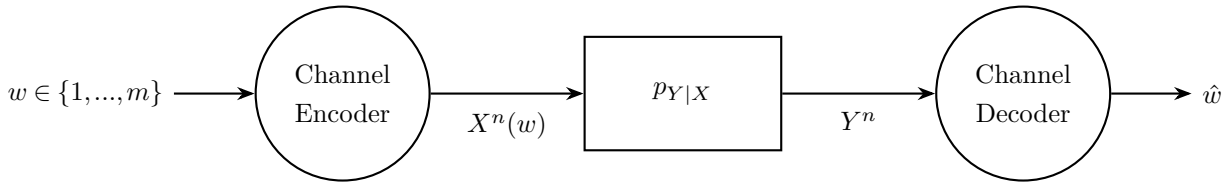
Now pick a subset of w's such that $\mathbb{P}(\hat{w} \neq w | w) < \epsilon$. This subset has a size greater than $\lceil \tilde{m}/2 \rceil = \lceil 2^{nR} \rceil$. Keep on those rows in the encoding table. This encoder/decoder satisfies (i) $(1/n)\log_2(m) \geq R$ and (ii) $\mathbb{P}(\hat{w} \neq w | w) < \epsilon$ for all w by construction.
Also note, the decoder never declares a w for which $X^n(w) \notin T(n, \delta, p_X)$. Indeed, recall (4.3), giving us that $u^n \in T(n, \delta, p_U)$ we have that for all $u \in \mathcal{U}$, $(1/n)\sum_{i=1}^{n} \mathbb{1}_{\{U_i = u\}} = p_U(u)(1 \pm \delta)$. Then for a pair of sequences $(u^n, v^n) \in T(n, \delta, p_{UV})$ it holds that $(1/n)\sum_{i=1}^{n} \mathbb{1}_{\{U_i = u, V_i = v\}} = p_{UV}(u, v)(1 \pm \delta)$. By summing over $v \in \mathcal{V}$, $(1/n)\sum_{i=1}^{n} \mathbb{1}_{\{U_i = u\}} = p_U(u)(1 \pm \delta)$ and thus $u^n \in T(n, \delta, p_U)$, i.e. typicality of the pair (u^n, v^n) implies typicality of u^n and typicality of v^n.

Thus because of (ii), for all w, $X^n(w) \in T(n, \delta, p_X)$ (i.e. a specific row of the table). Hence:

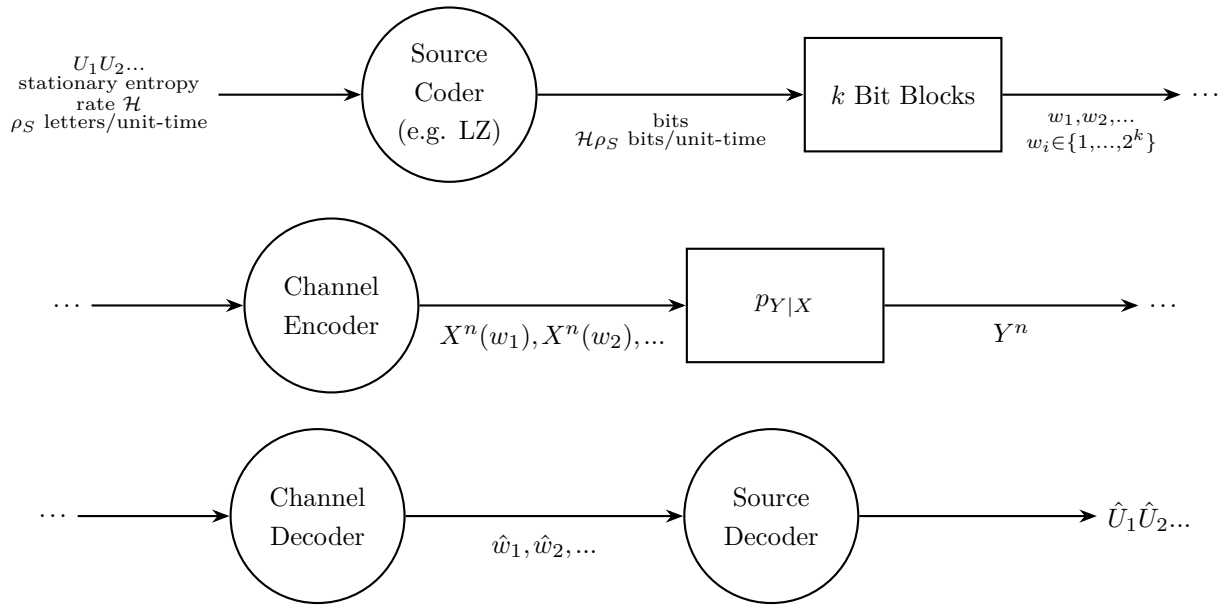
$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n} b(X_i(w)) &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\mathbb{1}_{\{b(X_i(w))\}}\right] \\ &= \frac{1}{n}\sum_{x\in\mathcal{X}}\sum_{i=1}^{n}\mathbb{1}_{\{X_i(w)=x\}}b(x) \\ &= \sum_{x\in\mathcal{X}} b(x)p_X(x)(1\pm\delta) = \mathbb{E}(b(X))\pm\delta\mathbb{E}(|b(X)|) = \beta\pm\epsilon.\end{aligned}$$

□

Good news theorems tells us that, given $p_{Y|X}$ we can construct a scheme



such that $(1/n)\log_2(m) \approx c(p_{Y|X})$ [or $c(p_{Y|X}, \beta)$] and $\mathbb{P}(\hat{w} \neq w|w) < \epsilon$, i.e. with high probability we recover the file without any errors. We can combine this with the data compression results to find:



Now, for a time T, with $k = T\rho_S\mathcal{H}$ and $n = T\rho_C$, if we make sure that $k/n = (\rho_S\mathcal{H})/\rho_C < c(p_{Y|X})$ this will ensure $\hat{U}_1\hat{U}_2... = U_1U_2...$ with high probability (i.e. we recover the message). In conclusion, a **modular design** works as long as $\rho_S\mathcal{H} < \rho_C c(p_{Y|X})$. We also know that if $\rho_S\mathcal{H} > \rho_C c(p_{Y|X})$ then **no** design (modular or not) can work reliably. Thus if we have to choose a design, we must choose a modular design.

Remark 6.3.4 (Relationship between $c(p_{Y|X})$ and $c(p_{Y|X}, \beta)$)**.** *Recall that* $c(p_{Y|X}) = \max_{p_X} I(X;Y)$*. Then in general:*

$$c(p_{Y|X}, \beta) = \max_{\substack{p_X \\ \mathbb{E}(b(X))=\beta}} I(X;Y) \leq \max_{p_X} I(X;Y) = c(p_{Y|X}).$$

Example 6.3.5. *Consider the binary erasure channel (BEC) defined in Example (6.1.4). Now, it is not difficult to verify that* $I(X;Y) = (1-p)H(X)$ *and* $c(p_{Y|X}) = 1-p$*. Suppose now that we have input costs* $b(0) = 0$ *and* $b(1) = 1$*. Then* $\beta = \mathbb{E}(b(X)) = \mathbb{P}(X = 1)$ *and so,*

$$\max_{\substack{p_X \\ \mathbb{E}(b(X))=\beta}} I(X;Y) = h_2(\beta)(1-p).$$

The plot of this function is the one of the standard binary entropy function $h_2(\beta)$ *scaled by* $1-p$*, and thus the maximum is attained at* $\beta = 1/2$ *with value* $1-p$*.*

As a final note, in this section, we have seen that $c(p_{Y|X}, \beta) \leq c(p_{Y|X})$ and $\max_\beta c(p_{Y|X}, \beta) = c(p_{Y|X})$. Another result that holds in general is that for any $p_{Y|X}$ the function $\beta \mapsto c(p_{Y|X}, \beta)$ is always concave.

6.4 The Capacity of a Channel and a Maximisation Problem

The results from the previous section justifies giving $c(p_{Y|X})$ the name **capacity of the channel** $\mathbf{p_{Y|X}}$. As we will see in this section, we can view the notion of capacity as part of a well-defined maximisation problem. This will rely on studying the general problem $\max_{p_X} I(X;Y)$ and look at the so-called KKT conditions [see Appendix (A.2)]. To this end, we first note the following:

$$\begin{aligned}\max_{p_X} I(X;Y) &= \max_{p_X} \sum_{x,y} p_X(x) p_{Y|X}(y|x) \log\left(\frac{p_{X,Y}(x,y)}{p_X(x)p_Y(y)}\right) \\ &= \max_{p_X} \underbrace{\sum_{x,y} p_X(x) p_{Y|X}(y|x) \log\left(\frac{p_{Y|X}(y|x)}{p_Y(y)}\right)}_{=:F_X(p_X)},\end{aligned}$$

where $P_Y(y) = \sum_{x'} p_{Y|X}(y|x')p_X(x') = \sum_{x'} p_{X,Y}(x',y)$. The underbraced part, which we denote by $F_X(p_X)$, may be viewed as a function of p_X (indeed we remark the p_X dependence in the expression as well as in the formula for p_Y). Note that we already know $p_X \mapsto F_X(p_X)$ is a concave function.

6.4.1 Capacity and KKT Conditions

Theorem 6.4.1. *Given* $p_{Y|X}$*,* p_X *maximises* $I(X;Y)$ *if and only if there exists* λ *such that*

$$\sum_y p(y|x) \log\left(\frac{p_Y(y|x)}{p_Y(y)}\right) \leq \lambda \quad \forall x \tag{6.17}$$

with equality for all x *if* $p_X(x) > 0$*. Here* $P_Y(y) = \sum_x p_X(x)p(y|x)$*. Moreover:*

$$\lambda = c(p_{Y|X}). \tag{6.18}$$

Proof. Recall $c(p_{Y|X}) = \max_{p_X} I(X;Y)$ and suppose without loss of generality that $\mathcal{X} = \{1, ..., k\}$. Thus, using the same notation as in the Appendix (A.2) for the KKT conditions, we have

$$\max_{p_X} I(X;Y) = \max_{p \in S_k} f(p),$$

where $f(p) = f((p_1,...,p_k)) = \sum_{i=1}^k \sum_y p_i p(y|i) \log\Big(p(y|i)/\sum_{j=1}^k (p_j p(y|j))\Big)$. For simplicity, we will work with the natural logarithm $\log \equiv \ln$. It is straightforward to see that for the first component,

$$\frac{\partial f}{\partial p_1} = \sum_{i=1}^k \sum_y \left[\mathbb{1}_{\{x=y\}} p(y|i) \ln\left(\frac{p(y|i)}{\sum_j p_j p(y|j)}\right) - p_i p(y|i) \frac{p(y|1)}{\sum_j p_j p(y|j)} \right] = \sum_y p(y|1) \ln\left(\frac{p(y|1)}{p(y)}\right) - 1.$$

As a result, for an arbitrary component and going back to an arbitrary logarithm,

$$\frac{\partial f}{\partial p_i}(p) = \sum_y p(y|i) \log\left(\frac{p(y|i)}{p(y)}\right) - \log(e).$$

Now, KKT tells us that $\sum_y p(y|i)\log(p(y|i)/p(y)) \le \lambda$ for all i with equality for all i when $p_i > 0$. Also, if we multiply both sides by p_i we obtain, by summing over i,

$$c(p_{Y|X}) = I(X;Y) = \sum_i p_i \sum_y p(y|i) \log\left(\frac{p(y|i)}{p(y)}\right) = \sum_i \lambda p_i = \lambda.$$

□

Example 6.4.2. *Suppose $f(p_1,p_2,p_3) = p_1^2 p_2 (2+p_3)$. We want*

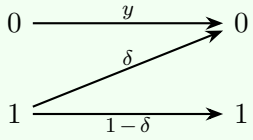
$$\max_{p \in S_3} f(p) \equiv \max_{p \in S_3} \ln(f(p)).$$

Thus, $\ln(f(p_1,p_2,p_3)) = 2\ln(p_1) + \ln(p_2) + \ln(2+p_3)$ and hence $\partial_{p_1} \ln(f) = 2/p_1$, $\partial_{p_2} \ln(f) = 1/p_2$ and $\partial_{p_3} \ln(f) = 1/(2+p_3)$.

• If at the maxima all the p_i are strictly positive then $p_1 = 2p_2$ and $p_3 = p_2 - 2$. Now, $1 = \sum_i p_i = 4p_2 - 2$ implying that $p_2 = 3/4$ and thus $p_3 < 0$. Doesn't work!

• So let us check when $p_1 > 0$, $p_2 > 0$ and $p_3 = 0$. Then $p_1 = 2p_2$ and $p_3 = 0$, and so $p_2 = 1/3$. As a result $p_1 = 2/3$, $p_2 = 1/3$ and $p_3 = 0$, and then $\partial_{p_1} \ln(f) = \partial_{p_2} \ln(f) = 3$ and $\partial_{p_3} \ln(f) = 1/2 \le 3$, where here we have that $\lambda = c = 3$.

Example 6.4.3 (Z-channel)**.** *Below, $x \in \{0,1\}$ and $y \in \{0,1\}$.*



For the capacity c, let p_0, p_1 be the input distributions and let q_0, q_1 be the corresponding output distributions. In particular, $q_1 = (1-\delta)p_1$ and $q_0 = 1 - q_1 = p_0 + \delta p_1$. We use KKT (with $p_0, p_1 > 0$), where:

$$\frac{\partial f}{\partial p_0} = \log\left(\frac{1}{q_0}\right) = \frac{\partial f}{\partial p_1} = \delta \log\left(\frac{\delta}{q_0}\right) + (1-\delta)\log\left(\frac{1-\delta}{q_1}\right) = c.$$

Now, we must have that

$$\log\left(\frac{1}{q_0}\right) = \delta \log\left(\frac{\delta}{q_0}\right) + (1-\delta)\log\left(\frac{1-\delta}{q_1}\right) \iff (1-\delta)\log\left(\frac{q_1}{q_0}\right) = -h_2(\delta) \iff \frac{q_1}{q_0} = 2^{-\frac{h_2(\delta)}{1-\delta}}.$$

Thus:

$$q_1 = \frac{2^{-\frac{h_2(\delta)}{1-\delta}}}{1+2^{-\frac{h_2(\delta)}{1-\delta}}} \quad \textit{and} \quad q_0 = \frac{1}{1+2^{-\frac{h_2(\delta)}{1-\delta}}}.$$

We thus achieve a capacity of,

$$c = \log_2\left(1+2^{-\frac{h_2(\delta)}{1-\delta}}\right).$$

6.4.2 Capacity and the Conditional KL-Divergence

We now give another point of view for the capacity from the conditional KL-divergence. Recall that for $p_Y(y) = \sum_x p_X(x)p(y|x)$,

$$I(X;Y) = \sum_x p_X(x) \sum_y p(y|x) \log\left(\frac{p(y|x)}{p_Y(y)}\right).$$

We can now do the following:

$$\begin{aligned} I(X;Y) &= \sum_x p_X(x) \sum_y p(y|x) \log\left(\frac{q(y)}{p_Y(y)}\right) + \sum_x p_X(x) \sum_y p(y|x) \log\left(\frac{p(y|x)}{q(y)}\right) \\ &= \underbrace{-\sum_y p_Y(y) \log\left(\frac{p_Y(y)}{q(y)}\right)}_{-D(p_Y\|q)\leq 0} + \sum_x p_X(x) \sum_y p(y|x) \log\left(\frac{p(y|x)}{q(y)}\right) \\ &\leq \sum_x p_X(x) \sum_y p(y|x) \log\left(\frac{p(y|x)}{q(y)}\right), \end{aligned}$$

with equality if and only if $q = p_Y$. Thus $I(X;Y)$ may be expressed as the following minimisation problem:

$$I(X;Y) = \min_{q\in\Pi} \sum_x p_X(x) D(p_{Y|X}(\cdot|x)\|q), \tag{6.19}$$

where Π is the set of all distributions on $\mathcal{Y}$. Note that this final sum is often called the **conditional KL-divergence**, and is denoted by $D(p_{Y|X}\|q|p_X)$. Consequently:

$$c(p_{Y|X}) = \max_{p_X} I(X;Y) = \max_{p_X} \min_{q_Y} \sum_x p_X(x) D(p_{Y|X}(\cdot|x)\|q_Y).$$

Note that the RHS is a function of p_X and q_Y. In particular, it is linear in p_X (and thus concave) and it is convex in q_Y. Because of this **only**, we can safely switch min and max (n.b. recall in general that $\max\min \leq \min\max$). Hence:

$$c(p_{Y|X}) = \min_{q_Y} \max_{p_X} \sum_x p_X(x) D(p_{Y|X}(\cdot|x)\|q_Y).$$

Furthermore, observe that

$$\sum_x p_X(x) D(p_{Y|X}(\cdot|x)\|q_Y) \leq \max_x D(p_{Y|X}(\cdot|x)\|q_Y)$$

with equality if $p_X(x) = 1$. Thus we reduce the problem to:

$$c(p_{Y|X}) = \min_{q_Y} \max_x D(p_{Y|X}(\cdot|x)\|q_Y). \tag{6.20}$$

The solution q_Y will be a convex combination of $p_{Y|X=1}, ..., p_{Y|X=k}$ with $p_1, ..., p_k$ some weights such that:

$$D(p_{Y|X}(\cdot|i)\|q_Y) = c \quad \text{for all } i \text{ such that } p_i > 0,$$

$$D(p_{Y|X}(\cdot|i)\|q_Y) \leq c \quad \text{for all } i \text{ such that } p_i = 0.$$

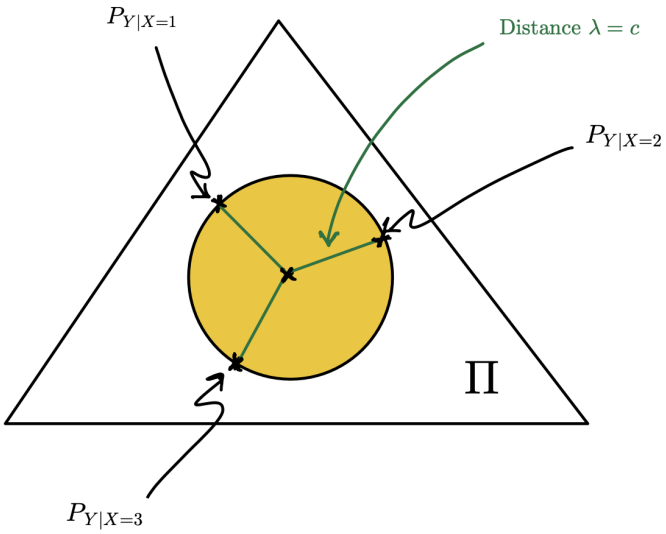


Figure 6.1: Representation of the convex combination of $p_{Y|X=1}, ..., p_{Y|X=k}$ where the goal is to minimise the region having all the $p_{Y|X=i}$ (i.e. region in orange). For Π the set of all distributions on Y, the minimal q_Y will be the one at the intersection of the three branches. q_Y is at a distance $\lambda = c$ away from the $p_{Y|X=i}$.

An Application to Universal Data Compression

Let $\mathcal{Y} = \mathcal{U}$ a source alphabet. For each $x \in \mathcal{X}$ we have a source distribution $p(u|x)$ [we have $|\mathcal{X}|$ distributions on $\mathcal{U}$]. We now design a source code for $\mathcal{U}$ with $q(u) = 2^{-\text{length}(c(U))}$ being a distribution on $\mathcal{U}$. If the true distribution on $\mathcal{U}$ is $p(u|x)$, then

$$\mathbb{E}[\text{length}|x] = \sum_x p(u|x) \log\left(\frac{1}{q(u)}\right).$$

Subtracting $H(U|x)$ on both sides yields:

$$\mathbb{E}[\text{length}|x] - H(U|x) = \sum_x p(u|x) \log\left(\frac{1}{q(u)}\right) - \sum_x p(u|x) \log\left(\frac{1}{p(u|x)}\right) = D(p(\cdot|x)\|q).$$

So, $\min_q \max_x D(p_{U|X}(\cdot|x)\|q)$ solves the worst case average regret source code problem.

6.5 Additive Gaussian Noise Channel

Consider an input $x \in \mathbb{R}$, an output $Y \in \mathbb{R}$ and $Y = x + Z$ for $Z \sim \mathcal{N}(0, \sigma^2)$. We consider a cost $b(x) = x^2$. Recall that the capacity is defined as,

$$c(\beta) = \max_{\substack{p_X \\ \mathbb{E}(X^2)=\beta}} I(X;Y).$$

Theorem 6.5.1. *For the channel $x \in \mathbb{R} \to x + Z \in \mathbb{R}$ for $Z \sim \mathcal{N}(0, \sigma^2)$ with cost function $b(x) = x^2$,*

$$c(\beta) = \frac{1}{2} \log\left(1 + \frac{\beta}{\sigma^2}\right), \tag{6.21}$$

where β/σ^2 is the signal-to-noise ratio (SNR).

Proof. First note that,

$$I(X;Y) = h(Y) - h(Y|X) = h(X+Z) - h(X+Z|X) = h(X+Z) - h(Z).$$

Also, since $\mathbb{E}(X^2) = \beta$ we have that,

$$\mathbb{E}((X+Z)^2) = \mathbb{E}(X^2) + \mathbb{E}(Z^2) + 2\mathbb{E}(XZ) = \beta + \sigma^2 + 2\mathbb{E}(X)\mathbb{E}(Z) = \beta + \sigma^2.$$

Thus, $I(X;Y) \leq \frac{1}{2}\log(2\pi e(\beta + \sigma^2))$ with equality if and only if $X + Z \sim \mathcal{N}(0, \beta + \sigma^2)$, which is equivalent to requiring $X \sim \mathcal{N}(0, \beta)$. □

Remark 6.5.2. *We could have also defined the cost with respect to a cost β as follows:*

$$c_{\leq}(\beta, \text{channel}) = \max_{\substack{p_X \\ \mathbb{E}(X^2)\leq\beta}} I(X;Y).$$

Note that since $c(\beta)$ is increasing in the previous Theorem, then in this case $c_{\leq} = c$.

Our model $x \to x + Z$ has continuous valued input/output, however time is still discrete. In the real world we have,



where $x(t)$ is a real-valued function of time and $\mathcal{N}(t)$ is a white Gaussian noise. Here, this white Gaussian noise is a stationary zero-mean Gaussian process with a well-defined covariance function given by $R(\tau) = \mathbb{E}[\mathcal{N}(\tau)\mathcal{N}(t+\tau)] = \sigma^2\delta(\tau)$, where $\delta(\tau)$ is the Delta function.

Chapter 7

Coding Theory

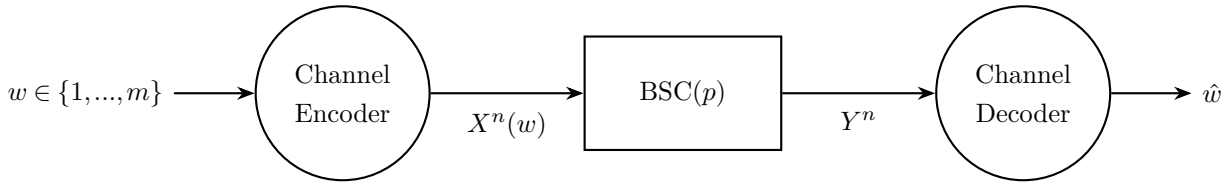
We recall the Binary Symmetric Channel (BSC(p)) with $p \leq 1/2$ without loss of generality. Now,

$$\mathbb{P}(Y^n = y^n | X^n = x^n) = (1-p)^{\#\ \{\ i \text{ such that } x_i = y_i\}} \times p^{\#\ \{\ i \text{ such that } x_i \neq y_i\}},$$

where we set $d_H = \#\ \{\ i$ such that $x_i \neq y_i\}$. Now, we define the **Hamming distance** for two inputs $x, y \in \{0,1\}$ to be $d_H(x,y) := \mathbb{1}_{\{x \neq y\}}$ and also for $x^n, y^n \{0,1\}^n$ so that $d_H(x^n, y^n) := \sum_{i=1}^n \mathbb{1}_{\{x_i \neq y_i\}}$. We note that this is a well-defined distance satisfying the usual assumptions (i) $d_H(x,y) = d_H(y,x)$, (ii) $d_H(x,y) \geq 0$ with equality if and only if $x = y$ and finally the Triangle Inequality (iii) $d_H(x,z) \leq d_H(x,y) + d_H(y,z)$ (indeed, if $x = z$ then LHS$= 0$ and RHS$= 2d_H(x,y) \geq 0$. Otherwise, for $x \neq z$ the RHS$= 1$ and either $d_H(x,y) = 1$ or $d_H(y,z) = 1$). These properties generalize for any $x^n, y^n, z^n \in \{0,1\}^n$. As a result,

$$\mathbb{P}(Y^n = y^n | X^n = x^n) = p^{d_H(x^n, y^n)}(1-p)^{n - d_H(x^n, y^n)} = \left(\frac{p}{1-p}\right)^{d_H} (1-p)^n.$$

Suppose now that we are given an encoder such that:



where the output $\hat{w}$ is defined by the **Maximum-Likelihood rule** (ML rule) $\hat{w} = \arg\max_w \mathbb{P}(Y^n = y^n | X^n = x^n(w))$. Then, the final decoding rule $\hat{w}$ minimizes

$$\frac{1}{m} \sum_{w=1}^{m} \mathbb{P}(\text{error} | w \text{ is the message}).$$

However remark that the ML rule is equivalent to minimum distance decoding as,

$$\hat{w} = \arg\max_w \mathbb{P}(Y^n = y^n | X^n = x^n(w)) = \arg\max_w d_H(y^n, x^n(w)).$$

Example 7.0.1. *Let $m=2$ and $X^n(1)=00000$, $X^n(2)=11111$. Consider the decoding rule,*

$$\text{decoding}(y^n)=\begin{cases}1, & d_H(y^n,00000)\leq 2\\ 2, & d_H(y^n,00000)\geq 3\end{cases}$$

Now suppose that $X^n(1)=0\ldots0$ ($n:=2j+1$ times) and also for $X^n(2)=1\ldots1$. Then:

$$\begin{aligned}\mathbb{P}(\text{error}|1)=\mathbb{P}(d_H(y^n,0\ldots0)>j|x^n=0\ldots0)&=\sum_{\substack{y^n\\ d_H(y^n,0\ldots0)>j}}(1-p)^{n-d_H(y^n,0\ldots0)}p^{d_H(y^n,0\ldots0)}\\ &=\sum_{d=j+1}^{n}\binom{n}{d}(1-p)^{n-d}p^d=\mathbb{P}(\text{error}|2).\end{aligned}$$

Given $X^n(1),\ldots,X^n(m)$ we may define the following quantity:

$$d_{\min}:=\min_{w\neq w'}d_H(x^n(w),x^n(w')),\tag{7.1}$$

so that if a message w is such that $X^n(w)$ is sent, and $d_H(x^n(w),y^n)<(1/2)d_{\min}$, then y^n will be closer to $x^n(w)$ than to any other $x^n(w')$. As a result, if $\mathcal{C}=\{x^n(w):w=1,...,m\}$, we may use the following quantity as a *figure of merit* of the code $\mathcal{C}$:

$$d_{\min}(\mathcal{C}):=\min_{\substack{w,w'\\ w\neq w'}}d_H(x^n(w),x^n(w')).\tag{7.2}$$

7.1 Some Bounds on $d_{\min}$ and Hamming Codes

In this section, we are interested in the following question: given n,m, what is the largest possible $d_{\min}(\mathcal{C})$ where $\mathcal{C}\subseteq\{0,1\}^n$ and $|\mathcal{C}|=m$? We first start with a simple example.

Example 7.1.1 ($m=2$)**.** *We have that,*

$$\max_{\substack{|\mathcal{C}|=2\\ \mathcal{C}\subseteq\{0,1\}^n}}d_{\min}(\mathcal{C})=n,$$

which is achieved by the repetition codes 00000 and 11111 (only works for $m=2$!).

In fact, this problem is still open for general n and m. Let us now investigate some bounds on $d_{\min}$.

Theorem 7.1.2 (Singleton Bound)**.** *Given $n,m>2^k$ where $k\leq n$, then any code $\mathcal{C}$ with $|\mathcal{C}|=m$ and $\mathcal{C}=\{0,1\}^n$ we have,*

$$d_{\min}(\mathcal{C})\leq n-k.\tag{7.3}$$

Proof. Consider the function $\texttt{hash}(x^n)=x^k$ (e.g. $\texttt{hash}(10110)=101$ for $n=5$ and $k=3$). Now, $\texttt{hash}:\{0,1\}^n\to\{0,1\}^k$ of size $=2^k$. Since there are 2^k hash values but $m>2^k$ codewords, there

exists $w \neq w'$ such that $\texttt{hash}(x^n(w)) = \texttt{hash}(x^n(w'))$. Now note:

$$d_H(x^n(w), x^n(w')) \leq n-k \Longrightarrow d_{\min}(\mathcal{C}) \leq n-k.$$

□

Corollary 7.1.3. *If $|\mathcal{C}| = 2^k$ and $\mathcal{C} \subseteq \{0,1\}^n$ then $d_{\min}(\mathcal{C}) \leq n-k+1$.*

Theorem 7.1.4 (Sphere Packing Bound)**.** *Suppose $\mathcal{C} \subseteq \{0,1\}^n$ and consider $d_{\min}(\mathcal{C}) = d$. Then for $m = |\mathcal{C}|$ satisfies*

$$m \sum_{i=0}^{\lfloor \frac{d-1}{2} \rfloor} \binom{n}{i} \leq 2^n,$$

which can be turned into a bound on $d_{\min}(\mathcal{C})$ in terms of $n, m = |\mathcal{C}|$.

Proof. Let $B(x^n, r)$ be the Hamming ball around x^n of radius r, i.e. $B(x^n, r) = \{y^n : d_H(x^n, y^n) \leq r\}$. Observe:

$$|B(x^n, r)| = \binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{r} = \sum_{i=0}^{r} \binom{n}{i}.$$

Observe also that $\{B(x^n(w), \lfloor \frac{d-1}{2} \rfloor), w = 1, \ldots, m\}$ is a disjoint collection of Hamming balls with $\mathcal{C} = (x^n(1), \ldots, x^n(m))$. Indeed $d_H(x^n(w), y^n) \leq \lfloor \frac{d-1}{2} \rfloor$ and $d_H(x^n(w'), y^n) \leq \lfloor \frac{d-1}{2} \rfloor$, thus

$$d_H(x(w), x(w')) \leq d-1.$$

So, $\bigcup_{w=1}^{m} B(x^n(w), r) \subseteq \{0,1\}^n$ for $r = \lfloor \frac{d-1}{2} \rfloor$. As a result:

$$2^n \geq \left| \bigcup_{w=1}^{m} B(x^n(w), r) \right| = \sum_{w=1}^{m} |B(x^n(w), r)| = m \sum_{i=0}^{r} \binom{n}{i}.$$

□

Example 7.1.5 ($(7,4,3)$-Hamming code)**.** *Consider $(n, k, d_{\min}) = (7,4,3)$ for $m = 2^k$. Note that $2^n = 128$, $m = 16$ and $\sum_{i=0}^{(d-1)/2} \binom{n}{i} = \sum_{i=0}^{1} \binom{7}{i} = 8$. Remark that $128 = 16 \times 8$ and thus we have a perfect tiling of the Hamming space.*

Example 7.1.6 ($(n = 2^\alpha - 1, k = 2^\alpha - \alpha - 1, d = 3)$-Hamming code)**.** *Consider the Hamming code $(n = 2^\alpha - 1, k = 2^\alpha - \alpha - 1, d = 3)$ for $m = 2^k$. Then $|B(0,1)| = 1 + n = 2^\alpha$. Also, $2^n = 2^{2^\alpha - 1}$ and $m = 2^{2^\alpha - 1 - \alpha}$ thus $2^n = m|B(0,1)|$. We again obtain a perfect tiling of the Hamming space.*

Now that we have established a perfect tiling, let us justify the existence of such Hamming codes.

Proposition 7.1.7. *The $(7,4,3)$-Hamming code exists and is well-defined.*

Proof. Consider $\mathbb{F}_2^7$ to be the field equipped with the modulo 2 sum. Then set,

$$\mathcal{C} = \left\{ \mathbf{x} \in \mathbb{F}_2^7 : H\mathbf{x} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\},$$

where H is the parity check matrix such that each column is a binary vector of dimension 3. The corresponding equations are,

$$\begin{cases} x_1 = x_5 + x_6 + x_7 \\ x_2 = x_4 + x_6 + x_7 \\ x_3 = x_4 + x_5 + x_7 \end{cases}$$

Thus for each choice of (x_4, x_5, x_6, x_7) we have a codeword since choosing such a set determines (x_1, x_2, x_3). And so $|\mathcal{C}| = 2^4 = 16$. For each $\mathbf{y} \in \mathbb{F}_2^7$, consider $s = H\mathbf{y} \in \mathbb{F}_2^3$ where

$$\mathbb{F}_2^3 = \{y_1, ..., y_8\} := \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

We may thus assign the decoding rule to be:

$$\mathbf{y} = \begin{cases} \mathbf{y} & \text{if } s = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}^T \\ \mathbf{y} + \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T & \text{if } s = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}^T \\ \mathbf{y} + \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T & \text{if } s = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}^T \\ \vdots \qquad\qquad \vdots & \qquad\qquad \vdots \\ \mathbf{y} + \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}^T & \text{if } s = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T \end{cases}$$

This decoding rule corrects any single channel error. Thus Hamming balls of radius 1 around each codeword must be disjoint. As a result $d_{\min} \geq 3$. Also, by the Sphere Packing Bound, we have that $d_{\min} \leq 3$ giving us that $d_{\min} = 3$. This concludes the existence of such codes. □

Proposition 7.1.8. *The $(2^\alpha - 1, 2^\alpha - \alpha - 1, 3)$-Hamming code exists and is well-defined.*

Proof. Consider the Hamming code $(n, \log(m), d_{\min}) = (2^\alpha - 1, n - \alpha, 3) = \{\mathbf{x} \in \mathbb{F}_2^n : H\mathbf{x} = \mathbf{0}\}$, where $H = \begin{pmatrix} \mathbb{I}_\alpha & | & A \end{pmatrix}_{\alpha \times n}$ has as all its columns all (nm)-zero vectors in $\mathbb{F}_2^\alpha$. Thus,

$$\mathbf{x} \in \mathcal{C} \iff \begin{pmatrix} x_1 \\ \vdots \\ x_\alpha \end{pmatrix} = A \begin{pmatrix} x_{\alpha+1} \\ \vdots \\ x_n \end{pmatrix},$$

so there are $2^{n-\alpha}$ codewords in $\mathcal{C}$. Also, if $\mathbf{x} \in \mathcal{C}$, $\mathbf{x}' \in \mathcal{C}$ and $\mathbf{x} \neq \mathbf{x}'$, we have $H\mathbf{x} = \mathbf{0}$, $H\mathbf{x}' = \mathbf{0}$. As a result $H(\mathbf{x} + \mathbf{x}') \equiv H\mathbf{z} = \mathbf{0}$ with $z \neq 0$. The number of 1's in $\mathbf{z}$ is $d_H(\mathbf{x}, \mathbf{x}')$. Now, $H\mathbf{z}$ is a column of $H \neq \mathbf{0}$ thus $\mathbf{z}$ can have a single 1. Also, $H\mathbf{z}$ is the sum of two columns of H and so $H\mathbf{z} \neq \mathbf{0}$. In

conclusion, $\mathbf{z}$ must have three or more 1's implying that $d_H(\mathbf{x},\mathbf{x}') \geq 3$. Hence $d_{\min}(\mathcal{C}) \geq 3$, and by Sphere Packing Bound we also have $d_{\min}(\mathcal{C}) \leq 3$. Overall $d_{\min} = 3$ giving us the existence of such codes. □

Theorem 7.1.9 (Gilbert & Varshamov)**.** *Given n, d, there is a code $\mathcal{C} \subseteq \{0,1\}^n$ with $d_{\min}(\mathcal{C}) \geq d$ and $m = |\mathcal{C}|$ satisfying*

$$m \sum_{i=0}^{d-1} \binom{n}{i} \geq 2^n.$$

Proof. Consider the following algorithm:

1: set $\mathcal{C} = \varnothing$, $\mathcal{A} = \{0,1\}^n$ and $w = 0$
2: **while** $\mathcal{A} \neq \varnothing$ **do**
3: • pick $a \in \mathcal{A}$
4: • set $w \leftarrow w+1$, $x(w) = a$, $\mathcal{C} \leftarrow \mathcal{C} \cup \{a\}$
5: • $\mathcal{A} \leftarrow \mathcal{A} \setminus B(a, d-1)$
6: **end while**

□

From a practical point, given $\mathcal{C} = (x^n(1), \ldots, x^n(m)) \in \{0,1\}^n$, the encoder assigns $w \to x^n(w)$ and even with *moderate* values of n, m (e.g. $n = 100$, $m = 2^{50}$) just storing $\mathcal{C}$ can become problematic. The decoder yields $y^n \to \arg\min_w d_H(x^n(w), y^n)$ and thus an unstructured code $\mathcal{C}$ is **not** practical. We also know: for every $R < 1 - h_2(p) = c(\text{BSC}(p))$, $\epsilon > 0$, there exists $n, m \geq 2^{nR}$ and a code $\mathcal{C} \subseteq \{0,1\}^n$ with $|\mathcal{C}| = m$ such that $\mathbb{P}(\text{error of the code on BSC}(p)) < \epsilon$.

7.2 Binary Linear Codes

A specific family of binary codes are binary linear codes. We will say that an encoder $w \mapsto x^n(w)$ is a **linear encoder** if $w \in \mathbb{F}_2^k$ and the map $w \mapsto x^n(w)$ is a linear map (i.e. if $w_1, w_2 \in \mathbb{F}_2^k$ and $a_1, a_2 \in \mathbb{F}_2$, $x^n(a_1 w_1 + a_2 w_2) = a_1 x^n(w_1) + a_2 x^n(w_2)$ for aw the pointwise modulo 2 multiplication). Thus if we specify the following collection of x^n:

$$\{e_1, e_2, \ldots, e_k\} := \left\{ x^n \left[\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \right], x^n \left[\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \right], \ldots, x^n \left[\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right] \right\},$$

then $x^n \left[\begin{pmatrix} a_1 \ldots a_k \end{pmatrix}^T \right] = \sum_{i=1}^k a_i x(e_i)$. As a result $x^n(w) = \begin{pmatrix} x(e_1) & | & \ldots & | & x(e_k) \end{pmatrix}_{n \times k} \cdot w$ and the matrix $G = \begin{pmatrix} x(e_1) & | & \ldots & | & x(e_k) \end{pmatrix}_{n \times k}$ is the **Generator matrix**. We have thus seen:

$$w \in \mathbb{F}_2^k \mapsto x(w) \in \mathbb{F}_2^n \text{ is linear} \equiv \exists G_{n \times k} \text{ such that } x(w) = Gw.$$

Consider now $\mathcal{C} = (x^n(1), \ldots, x^n(m))$, and we will say that $\mathcal{C}$ is a **linear subspace** of $\mathbb{F}_2^n$ if for all $x, x' \in \mathcal{C}$, we have that $x + x' \in \mathcal{C}$. We can also show that if $\mathcal{C}$ is a linear subspace of $\mathbb{F}_2^n$ then (i) $\exists k$ such

that $|\mathcal{C}| = 2^k$, (ii) $\exists G_{n\times k}$ such that $\mathcal{C} = \{Gw : w \in \mathbb{F}_2^n\}$ implying that any code for which $\mathcal{C}$ is a linear subspace can be implemented by a linear encoder, and (iii) $\exists H_{(n-k)\times n}$ such that $\mathcal{C} = \{x : Hx = 0\}$ where H is the parity check matrix.

Definition 7.2.1. *For a code $\mathcal{C}$, define* $\texttt{weight}_{\min}(\mathcal{C})$ *with $w_H(x) = d_H(0,x)$ the # {1's in x},*

$$\texttt{weight}_{\min}(\mathcal{C}) := \min_{\substack{x\in\mathcal{C}\\ x\neq 0}} w_H(x). \tag{7.4}$$

Theorem 7.2.2. *If $\mathcal{C}$ is linear then $d_{\min}(\mathcal{C}) = \texttt{weight}_{\min}(\mathcal{C})$.*

Proof. As $0 \in \mathcal{C}$, then $w_{\min}(\mathcal{C}) \geq d_{\min}(\mathcal{C})$. On the other hand, if $x, x' \in \mathcal{C}$ such that $d_H(x,x') = d_{\min}(\mathcal{C})$ then $x + x' \in \mathcal{C}$ and,

$$\texttt{weight}_{\min}(\mathcal{C}) \leq w_H(x+x') = d_H(0,x+x') = d_H(x,x') = d_{\min}(\mathcal{C}).$$

□

Example 7.2.3 ($(n = 2^\alpha - 1, k = 2^\alpha - \alpha - 1)$-Hamming codes are linear codes)**.** *These Hamming codes are linear codes with $H = \begin{pmatrix} \mathbb{I}_\alpha & | & A \end{pmatrix}_{\alpha\times n}$, where columns of A are all binary vectors of weight $w \geq 2$. Remember that,*

$$x \in \mathcal{C} \iff \begin{pmatrix} x_1 \\ \vdots \\ x_\alpha \end{pmatrix} = A \begin{pmatrix} x_{\alpha+1} \\ \vdots \\ x_n \end{pmatrix} \iff \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} A \\ - \\ \mathbb{I}_{n-\alpha} \end{pmatrix} \begin{pmatrix} x_{\alpha+1} \\ \vdots \\ x_n \end{pmatrix} \iff \mathcal{C} = \{Gw : w \in \mathbb{F}_2^{n-\alpha}\},$$

where $G = \begin{pmatrix} A & | & \mathbb{I} \end{pmatrix}^T$. In particular, for the $(7,4)$-Hamming code we have:

$$G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Example 7.2.4 (Single parity check code)**.** *Parity check matrix $H = \begin{pmatrix} 1 & \dots & 1 \end{pmatrix}$ and $\mathcal{C} = \{x \in \mathbb{F}_2^n : \sum_{i=1}^n x_i = 0\}$. Thus we have that $Gw = \begin{pmatrix} w & | & \sum_i w_i \end{pmatrix}^T$ where:*

$$G = \begin{pmatrix} 1 & & \\ & \ddots & \\ 1 & \dots & 1 \end{pmatrix}_{n\times(n-1)}.$$

Example 7.2.5 (Non-binary - Reed-Solomon codes)**.** *Consider a field $\mathbb{F}_q$ with $n \leq q$ and $k \leq n$ (all known a-priori). We pick $\alpha_1, ..., \alpha_n$ on n-distinct elements of $\mathbb{F}_q$. We construct an encoder as follows: let $w = (w_0, ..., w_{k-1}) \in \mathbb{F}_q^k$ with $w_i \in \mathbb{F}_q$ such that $w \mapsto x(w) \in \mathbb{F}_q^n$, where*

$$w \rightsquigarrow w(\alpha) = w_0 + w_1\alpha + \ldots + w_{k-1}\alpha^{k-1} \text{ for } \alpha \in \mathbb{F}_q \rightsquigarrow x^n(w) = \begin{pmatrix} w(\alpha_1) \\ \vdots \\ w(\alpha_n) \end{pmatrix} \in \mathbb{F}_q^n.$$

This is a linear map since,

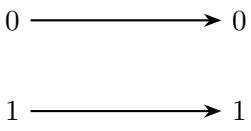
$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 & \ldots & \alpha_1^{k-1} \\ \vdots & \vdots & \vdots & \ldots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \ldots & \alpha_n^{k-1} \end{pmatrix} \begin{pmatrix} w_0 \\ \vdots \\ w_{k-1} \end{pmatrix}.$$

One can then show that $d_{\min}(\mathcal{C}) = \min_{w \neq w'} d_H(x(w), x(w')) \geq n - k + 1$. By the Singleton bound we also have that $d_{\min}(\mathcal{C}) \leq n - k + 1$. Thus, $d_{\min}(\mathcal{C}) = n - k + 1$.

7.3 Polar Coding

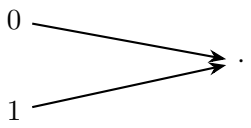
Consider the binary input channel $W : \mathbb{F}_2 \mapsto \mathcal{Y}$ such that $W(y|x)$ has $y \in \mathcal{Y}$ and $x \in \mathbb{F}_2$. We have some situations where transmission of data is relatively simple:

- If W is the *perfect* channel, which has capacity $c = 1$:

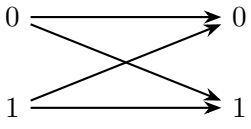


- We may also consider the channel W by requiring $W(y|0)W(y|1) = 0$ for all $y \in \mathcal{Y}$. Equivalently we may require $\sum_y \sqrt{W(y|0)W(y|1)} = 0$. This corresponds to the generalization of the *perfect* channel where $x = 0$ connects to a set $\mathcal{Y}_1 \subseteq \mathcal{Y}$ and $x = 1$ connects to $\mathcal{Y}_2 \subseteq \mathcal{Y}$ such that $\mathcal{Y}_1 \cap \mathcal{Y}_2 = \varnothing$.

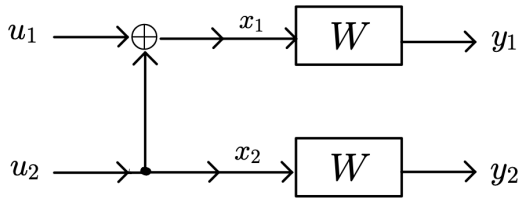
- We also have the following channel with capacity $c = 0$:



- We may also consider the *useless* channel where for all $y \in \mathcal{Y}$, $W(y|0) = W(y|1)$ or equivalently $\sum_y \sqrt{W(y|0)W(y|1)} = 1$. This is considered as a *useless* channel since knowing the output we cannot determine the input.



Let us now consider the construction of circuits by stacking copies of W. We first start by considering two copies of W:



Above, for $x_i \in \mathbb{F}_2, y_i \in \mathcal{Y}, u_i \in \mathbb{F}_2$ thus $x_1 = u_1 \oplus u_2$ and $x_2 = u_2$. Equivalently, we have $u_1 = x_1 \oplus x_2$ and $u_2 = x_2$. Let us choose u_1, u_2 i.i.d. Ber(1/2) such that,

$$\mathbb{P}(U_1U_2 = 00) = \mathbb{P}(U_1U_2 = 01) = \mathbb{P}(U_1U_2 = 10) = \mathbb{P}(U_1U_2 = 11) = \frac{1}{4}.$$

Also,

$$\begin{cases} \mathbb{P}(X_1X_2 = 00) = 1/4 & \text{for } u_1u_2 = 00 \\ \mathbb{P}(X_1X_2 = 01) = 1/4 & \text{for } u_1u_2 = 01 \\ \mathbb{P}(X_1X_2 = 10) = 1/4 & \text{for } u_1u_2 = 10 \\ \mathbb{P}(X_1X_2 = 11) = 1/4 & \text{for } u_1u_2 = 11 \end{cases}$$

and as a result $u_1, u_2 \sim \text{Ber}(1/2)$ implies x_1, x_2 i.i.d. Ber(1/2). Now on the one hand:

$$I(X_1X_2; Y_1Y_2) = I(X_1; Y_1) + I(X_2; Y_2) = 2I(W),$$

where in the first equality we have used the fact that the channel is memoryless, X_1, X_2 independent and the notation $I(W) = I(X;Y)$ for $p_{Y|X} = W$ such that $p_X \sim \text{Ber}(1/2)$. On the other hand,

$$I(X_1X_2; Y_1Y_2) \overset{(i)}{=} I(U_1U_2; Y_1Y_1) \overset{(ii)}{=} I(U_1; Y_1Y_2) + I(U_2; Y_1Y_2|U_1) \overset{(iii)}{=} \underbrace{I(U_1; Y_1Y_2)}_{I(W^-)} + \underbrace{I(U_2; Y_1Y_2U_1)}_{I(W^+)},$$

where (i) is because $U_1U_2 \longleftrightarrow X_1X_2$ is a 1-to-1 map, (ii) is the chain rule, and (iii) is again the chain rule and the fact that U_1, U_2 are independent. We have introduced two new notations:

- We set $\mathbf{W}^- : U_1 \mapsto Y_1Y_2$ such that:

$$\begin{aligned} W^-(y_1y_2|u_1) &= \frac{1}{2}\mathbb{P}(Y_1Y_2 = y_1y_2|U_1 = u_1, U_2 = 0) + \frac{1}{2}\mathbb{P}(Y_1Y_2 = y_1y_2|U_1 = u_1, U_2 = 1) \\ &= \frac{1}{2}\mathbb{P}(Y_1Y_2 = y_1y_2|X_1X_2 = u_10) + \frac{1}{2}\mathbb{P}(Y_1Y_2 = y_1y_2|X_1X_2 = u_11) \\ &= \frac{1}{2}W(y_1|u_1)W(y_2|0) + \frac{1}{2}W(y_1|u_1)W(y_2|1). \end{aligned}$$

As a result, $W^- : U_1 \mapsto Y_1Y_2$ is defined to be:

$$W^-(y_1y_2|u_1) = \frac{1}{2}\sum_{u_2 \in \mathbb{F}_2} W(y_1|u_1 \oplus u_2)W(y_2|u_2).$$

- We set $\mathbf{W}^+ : U_2 \mapsto Y_1Y_2U_1$ such that by Bayes' Theorem:

$$W^+(y_1y_2u_1|u_2) = \mathbb{P}(U_1 = u_1, Y_1 = y_1, Y_2 = y_2|U_2 = u_2) = \frac{1}{2}\mathbb{P}(Y_1Y_2 = y_1y_2|U_1U_2 = u_1u_2).$$

As a result, $W^+ : U_2 \mapsto Y_1Y_2U_1$ is defined to be:

$$W^+(y_1y_2u_1|u_2) = \frac{1}{2}W(y_1|u_1 \oplus u_2)W(y_2|u_2).$$

Example 7.3.1 (Binary Erasure Channel)**.** *Consider the Binary Erasure Channel* $W = \mathrm{BEC}(p)$ *(cf. Example 6.1.4). Now,*

$$W^- : U_1 \mapsto Y_1Y_2 = \begin{cases} x_1, x_2 & \text{with } \mathbb{P} = (1-p)^2 \\ x_1, ? & \text{with } \mathbb{P} = (1-p)p \\ ?, x_2 & \text{with } \mathbb{P} = p(1-p) \\ ?, ? & \text{with } \mathbb{P} = p^2 \end{cases} = \begin{cases} u_1 \oplus u_2, u_2 & \text{with } \mathbb{P} = (1-p)^2 \equiv u_1 \\ u_1 \oplus u_2, ? & \text{with } \mathbb{P} = (1-p)p \equiv ? \\ ?, u_2 & \text{with } \mathbb{P} = p(1-p) \equiv ? \\ ?, ? & \text{with } \mathbb{P} = p^2 \equiv ? \end{cases}$$

where the $\equiv$ *stands for the information we can recover in each case. As a result* $W^- = \mathrm{BEC}(p^2 + 2p(1-p)) = \mathrm{BEC}(p(2-p))$*. For* W^+ *we have,*

$$W^+ : U_2 \mapsto Y_1Y_2U_1 = \begin{cases} u_1 \oplus u_2, u_2, u_1 & \text{with } \mathbb{P} = (1-p)^2 \equiv u_2 \\ u_1 \oplus u_2, ?, u_1 & \text{with } \mathbb{P} = (1-p)p \equiv u_2 \\ ?, u_2, u_1 & \text{with } \mathbb{P} = p(1-p) \equiv u_2 \\ ?, ?, u_1 & \text{with } \mathbb{P} = p^2 \equiv ? \end{cases}$$

As a result we conclude that $W^+ = \mathrm{BEC}(p^2)$*. Thus* $I(W) = 1-p$*,* $I(W^-) = 1-p(2-p)$*,* $I(W^+) = 1-p^2$ *and so we indeed have,*

$$2I(W) = I(W^-) + I(W^+).$$

Also note the following interesting fact: $I(W^-) \leq I(W) \leq I(W^+)$*.*

Proposition 7.3.2.

$$I(W^-) \leq I(W) \leq I(W^+) \tag{7.5}$$

Proof. In general we have that $I(W^+) = I(U_2; Y_1Y_2U_1) \geq I(U_2; Y_2) = I(X_2; Y_2) = I(W)$. And thus, since $2I(W) = I(W^-) + I(W^+)$ we may conclude $I(W^-) \leq I(W) \leq I(W^+)$. □

So we have seen that from two independent uses of the channel W we have synthesized two channels W^- and W^+ with $I(W^-) + I(W^+) = 2I(W)$ and also $I(W^-) \leq I(W) \leq I(W^+)$. However note that $W^- : U_1 \mapsto Y_1Y_2$ is in the form input $\mapsto$ output. However $W^+ : U_2 \mapsto Y_1Y_2U_1$ is in the form input $\mapsto$ {output, input} which is not a great situation.

Remark 7.3.3 (W^+ is a well-defined channel)**.** *Let us convince ourselves that W^+ is a well-defined channel. Consider successive decision making with and without a helper. We have observation Z and we want to recover $U_1U_2\dots U_n$. With the helper, we do this as follows:*

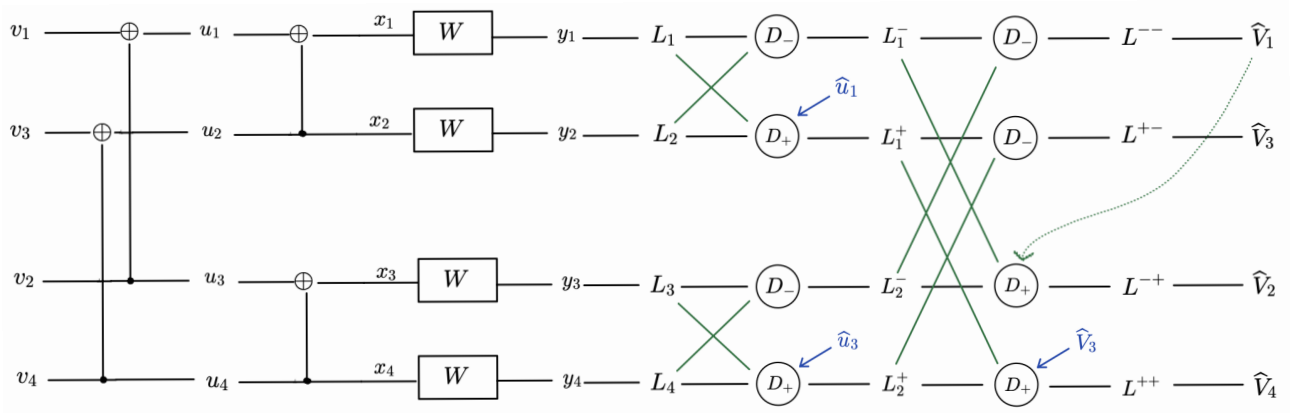
$$\widehat{u}_1=\phi_1(Z) \quad ; \quad \widehat{u}_2=\phi_2(Z,U_1) \quad ; \quad \widehat{u}_3=\phi_3(Z,U_1U_2) \quad ; \quad \dots \quad ; \quad \widehat{u}_n=\phi_n(Z,U^{n-1}).$$

Without helper, we do this as follows:

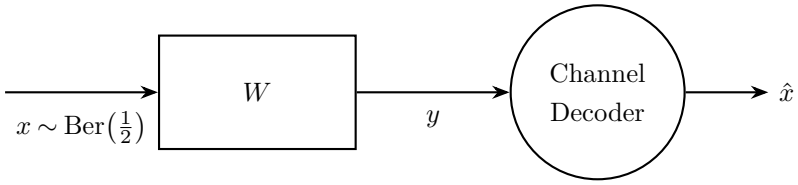
$$\widetilde{u}_1=\phi_1(Z) \quad ; \quad \widetilde{u}_2=\phi_2(Z,\widetilde{U}_1) \quad ; \quad \widetilde{u}_3=\phi_3(Z,\widetilde{U}_1\widetilde{U}_2) \quad ; \quad \dots \quad ; \quad \widetilde{u}_n=\phi_n(Z,\widetilde{U}^{n-1}).$$

Let us now look and compare the events $A:=\{\widehat{u}^n=u^n\}$ and $B:=\{\widetilde{u}^n=u^n\}$.
• If A, then $\widehat{u}_1=u_1 \Longrightarrow \widetilde{u}_1=\widehat{u}_1=u_1 \Longrightarrow \widetilde{u}_2=\widehat{u}_2=u_2 \Longrightarrow \dots \Longrightarrow \widetilde{u}_n=\widehat{u}_n=u_n$. Thus if A occurs then B occurs also.
• If <u>not</u> A, then there exists i such that $\widehat{u}^{i-1}=u^{i-1}$ and $\widehat{u}_i\neq u_i$, thus $\widetilde{u}^{i-1}=u^{i-1},\widetilde{u}_i=\widehat{u}_i\neq u_i \Longrightarrow$ <u>not</u> B.
Overall, A and B are the same events, and we conclude $\mathbb{P}(\widetilde{U}^n\neq U^n)=\mathbb{P}(\widehat{U}^n=U^n)$. Here, $\mathbb{P}(\widetilde{U}^n\neq U^n)$ is the error probability of the system without helper, and $\mathbb{P}(\widehat{U}^n=U^n)$ is the error probability of the system with helper.

Considering 4 channels now yields the following circuit:



We thus have $W^-:U_1\mapsto Y_1Y_2$, $W^-:U_3\mapsto Y_3Y_4$, $W^+:U_2\mapsto Y_1Y_2U_1$ and $W^+:U_4\mapsto Y_3Y_4U_3$. Then we set the *new* channels as $W^{--}:V_1\mapsto Y^4$, $W^{-+}:V_2\mapsto Y^4V_1$, $W^{+-}:V_3\mapsto Y^4U_1U_3\equiv Y^4V^2$ and $W^{++}:V_4\mapsto Y^4U_1U_3V_3\equiv Y^4V^3$. We can the continue further with 8 channels leading to $\{W^{---},\dots,W^{+++}\}$, 16 channels with $\{W^{----},\dots,W^{++++}\}$, and so on for the number of channels in powers of 2.
At the receiver end we have,



The $\widehat{x}$ with $\mathbb{P}(\widehat{x}\neq x)$ being as small as possible is given by the y such that $\widehat{x}=\mathbb{1}_{\{W(y|0)<W(y|1)\}}$. We may thus consider the following **decision statistic** to allow us to make an optimal decision:

$$L_W(y)=\frac{W(y|0)}{W(y|1)}.$$

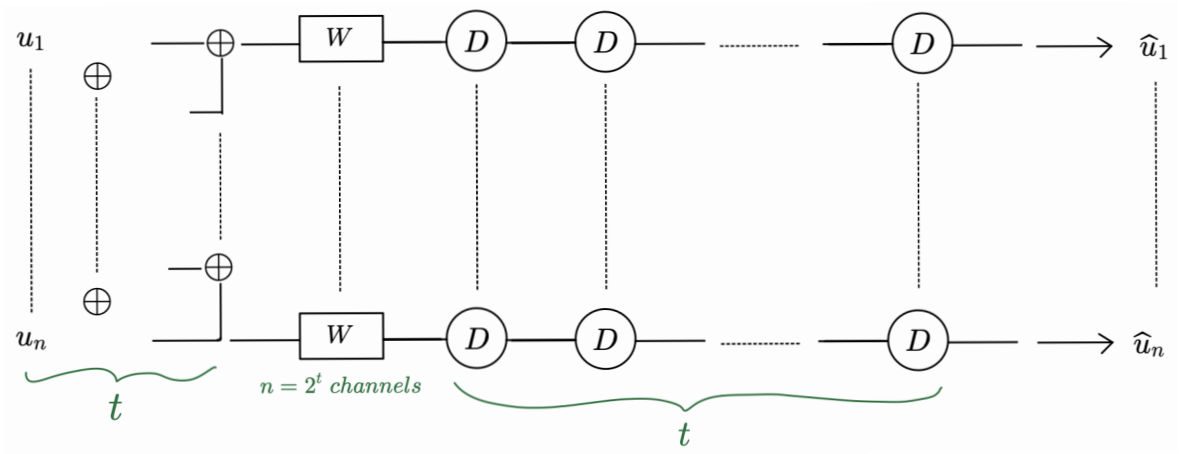
Now recall $W^- : U_1 \mapsto Y_1Y_2$ and $W^+ : U_2 \mapsto Y_1Y_2U_1$ such that $W^-(y_1y_2|u_1) = \frac{1}{2}\sum_{u_2\in\mathbb{F}_2} W(y_1|u_1 \oplus u_2)W(y_2|u_2)$ and $W^+(y_1y_2u_1|u_2) = \frac{1}{2}W(y_1|u_1 \oplus u_2)W(y_2|u_2)$. Now set,

$$L_{W^-}(y_1y_2) = \frac{W^-(y_1y_2|u_1=0)}{W^-(y_1y_2|u_1=1)} = \frac{W(y_1|0)W(y_2|0)+W(y_1|1)W(y_2|1)}{W(y_1|1)W(y_2|0)+W(y_1|0)W(y_2|1)} = \frac{L_W(y_1)L_W(y_2)+1}{L_W(y_2)+L_W(y_1)}$$

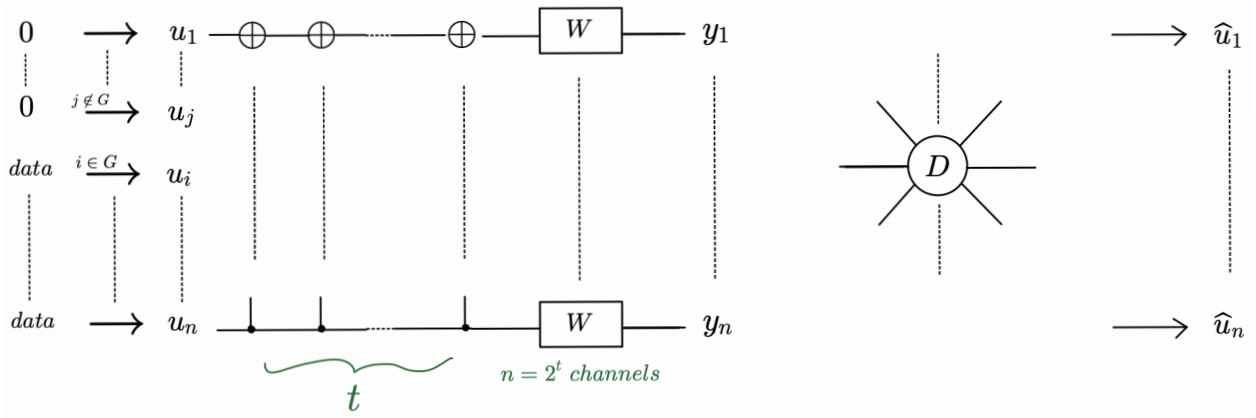
and similarly,

$$L_{W^+}(y_1y_2u_1) = \begin{cases} L_W(y_1)L_W(y_2), & u_1 = 0 \\ L_W(y_1)/L_W(y_2), & u_1 = 1 \end{cases}.$$

In greater generality we consider a general circuit of the form:

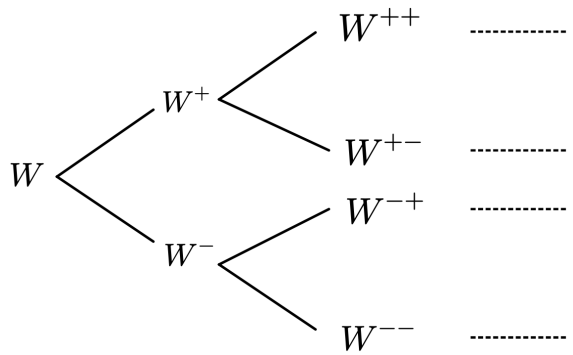


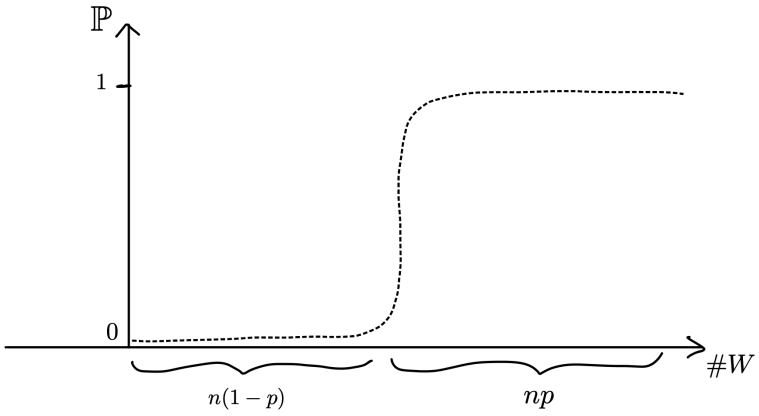
In conclusion, the encoding effort of $u^n \to X^n$ is of the order $\mathcal{O}(t2^t) = \mathcal{O}(n\log(n))$. Also, the decoding effort $Y^n \to \widehat{u}^n$ is of the order $\mathcal{O}(t2^t) = \mathcal{O}(n\log(n))$. This is not such a big effort!
Numerical experiment with the BEC suggests that if $W = \text{BEC}(p)$ and we perform t polarization steps $[n = 2^t$ W's gives us $(W^{-\cdots-}, \ldots, W^{+\cdots+})$ for $n = 2^t$ synthetic BECs$]$. If $p_1, \ldots, p_n$ are the erasure probabilities of the synthetic channels with $p_1 \le \ldots \le p_n$ then (i) $\#\{i$'s such that $p_i \approx 0$ is about $n(1-p)\}$, $\#\{i$'s such that $p_i \approx 1$ is about $np\}$, $\#\{i$'s such that p_i is away of 0 and 1 is about $0\}$, and (ii) if $R < 1-p$ then $\sum_{i=1}^{nR} p_i \approx 0$. Thus the *good* channels are actually very good since they have a very small erasure probability. This suggests the following coding method: for $W = \text{BEC}(p)$, given $R < 1-p$, $\epsilon > 0$ pick t large enough (# of polarization stages) such that $n = 2^t$, then $\sum_{i=1}^{nR} p_i < \epsilon$. Then:



Of the $u_1 \ldots u_n$, there are nR of them which are travelling in *good* synthetic channels. Let $\{u_i : i \in G\}$ the $u_i's$ which have the nR best channels (i.e. with smallest erasure probability). For $i \notin G$, let u_i be fixed to zero. When the decoder estimates $\widehat{u}_1, \widehat{u}_2, \ldots, \widehat{u}_n$, if $i \notin G$ then $\widehat{u}_i = 0$. This method will transmit

nR data bits using the channel W for n times and its error probabilities $\mathbb{P}(\widehat{\text{data}} \neq \text{data}) \leq \sum_{i=1}^{nR} p_i < \epsilon$.





Proposition 7.3.4. *For the BEC(p), the experiment suggests:*
(i) For t large,

$$\frac{1}{2^t}\#\{s^t \in \{+,-\}^t : \text{erasure } \mathbb{P} \text{ of } W^{s^t} \in \{\epsilon, 1-\epsilon\}\} \to 0. \tag{7.6}$$

(ii) For some (ϵ_t) such that $\epsilon_t \to 0$ and $2^t\epsilon_t \to 0$,

$$\frac{1}{2^t}\#\{s^t \in \{+,-\}^t : \text{erasure } \mathbb{P} \text{ of } W^{s^t} < \epsilon_t\} \to 1-p, \tag{7.7}$$

so that the sum of erasure probabilities of the nR best channels is small if $R < 1-p$.

Now for $W = \text{BEC}(p)$ we define $\texttt{med}(W) := \sqrt{p(1-p)}$ to be the **mediocre function** such that $\mathbb{1}_{\{p\in(\epsilon,1-\epsilon)\}}\sqrt{\epsilon(1-\epsilon)} \leq \texttt{med}(W)$. Now, for $W = \text{BEC}(p)$ we have seen that $W^+ = \text{BEC}(p^2)$ and $W^- = \text{BEC}(p(2-p))$. As a result, $\texttt{med}(W) = \sqrt{p(1-p)}$, $\texttt{med}(W^+) = \sqrt{p^2(1-p^2)} = \texttt{med}(W)\sqrt{p(1+p)}$ and finally $\texttt{med}(W^-) = \sqrt{p(2-p)(1-p)^2} = \texttt{med}(W)\sqrt{(2-p)(1-p)}$. Thus the average mediocrity is given by,

$$\frac{1}{2}\big(\texttt{med}(W^+) + \texttt{med}(W^-)\big) = \texttt{med}(W)\frac{1}{2}\left(\sqrt{p(1+p)} + \sqrt{(2-p)(1-p)}\right) \leq \sqrt{\frac{3}{4}}\texttt{med}(W).$$

Thus the average mediocrity of $\{W^-, W^+\}$ is always less than the mediocrity of W. Now, for the next generation,

$$\frac{1}{4}\big(\mathtt{med}(W^{--})+\mathtt{med}(W^{-+})+\mathtt{med}(W^{+-})+\mathtt{med}(W^{++})\big)\leq\left(\sqrt{\frac{3}{4}}\right)^2\mathtt{med}(W).$$

Repeating this procedure over t generations yields,

$$\frac{1}{2^t}\sum_{s^t\in\{+,-\}^t}\mathtt{med}(W^{s^t})\leq\mathtt{med}(W)\left(\sqrt{\frac{3}{4}}\right)^t\leq\frac{1}{2}\left(\frac{3}{4}\right)^{\frac{t}{2}}.$$

Thus the fraction of ϵ-mediocre channels at generation t is bounded as follows,

$$\frac{1}{2^t}\sum_{s^t\in\{+,-\}^t}\mathbb{1}_{\{\text{erasure }\mathbb{P}\text{ of }W^{s^t}\in(\epsilon,1-\epsilon)\}}\leq\frac{1}{2^t}\sum_{s^t\in\{+,-\}^t}\frac{\mathtt{med}(W^{s^t})}{\sqrt{\epsilon(1-\epsilon)}}\leq\frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{t}{2}}\xrightarrow{t\text{ large}}0.$$

This limit converges exponentially fast. This proves that there is a very small number of channels in the phase transition.

Remark 7.3.5. *Remark that the easy proof could have been to say that if $\epsilon_t=\rho^t$ for $\rho>\frac{3}{4}$, then*

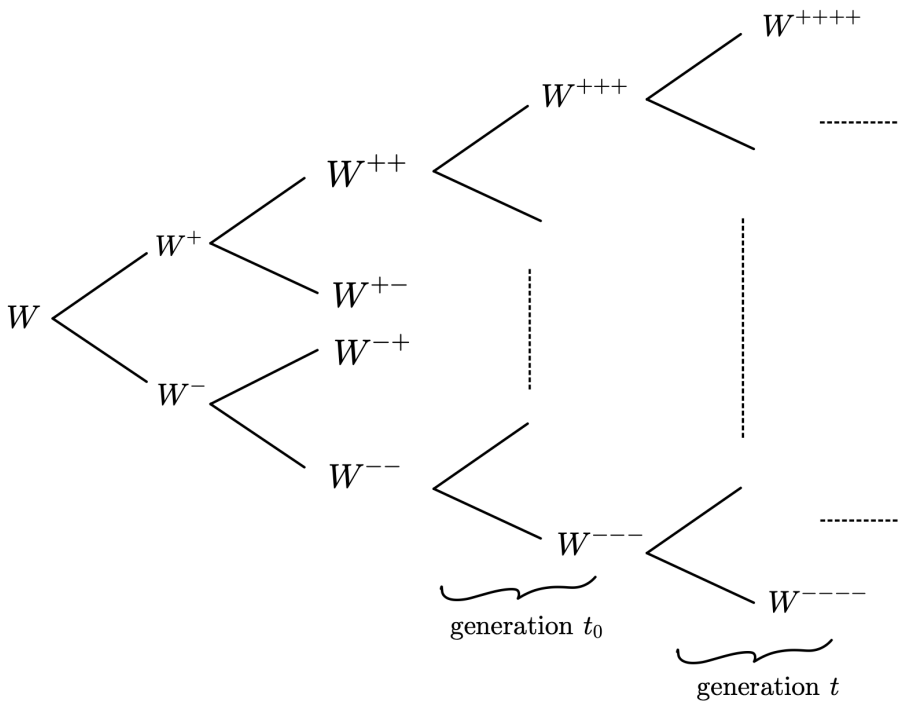
$$\frac{\left(\frac{3}{4}\right)^{\frac{t}{2}}}{\sqrt{\epsilon_t(1-\epsilon_t)}}\xrightarrow{t\text{ large}}0.$$

However:

$$\sum_{\text{best }nR\text{ channels}}\mathbb{P}(\text{erasure})\leq\rho^t\cdot2^t\cdot R=\underbrace{(2\rho)^t}_{>1}R<\infty,$$

thus it is not useful to prove our claim, hence we need to work harder.

We thus present a new idea and look at the following generational tree:



For every ϵ-mediocre channel in generation$[t_0, t]$ eliminates all its descendants from the tree. Thus the survivors at generation t will not have ϵ-mediocre ancestors up to generation t_0. The question now is to know the fraction Γ at generation t that did not survive. The answer is,

$$\Gamma \leq \frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{t_0}{2}} + \frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{t_0+1}{2}} + \ldots + \frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{t}{2}} < \sum_{i=t_0}^{\infty} \frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{i}{2}}$$
$$= \frac{1}{2\sqrt{\epsilon(1-\epsilon)}}\left(\frac{3}{4}\right)^{\frac{t_0}{2}} \frac{1}{1-\sqrt{\frac{3}{4}}}$$
$$\xrightarrow{t_0 \text{ large}} 0.$$

Thus take $t_0 = \lfloor \sqrt{t} \rfloor$ then the survivors at generation t will have all non-ϵ-mediocre ancestors up to generation t_0 and then generation t will contain at least $2^t(1-\delta_t)$ survivors with

$$\delta_t = \left(\frac{3}{4}\right)^{\sqrt{t}/2} \frac{1}{2\sqrt{\epsilon(1-\epsilon)}} \frac{1}{1-\sqrt{\frac{3}{4}}}.$$

Assume ϵ to be small, for example $\epsilon < 1/10$. We will say that W is **ϵ-good** if the erasure probability is $< \epsilon$ and say that W is **ϵ-bad** if the erasure probability is $> 1-\epsilon$. Thus, W is ϵ-good gives that W^+ is ϵ-good and W is ϵ-bad gives that W^+ is not ϵ-good. Also, if W is ϵ-good then W^- is not ϵ-bad and if W is ϵ-bad then W^- is ϵ-bad. In conclusion, in one generation, one cannot change from ϵ-bad to ϵ-good or from ϵ-good to ϵ-bad.

Corollary 7.3.6. *If W^{s^t} is an ϵ-good survivor at generation t then for all $t_0 \leq i \leq t$, W^{s^i} must be ϵ-good.*

A channel W^{s^t} is said to be **ϵ-unlucky** if,

$$\frac{1}{t}\#\{t_0 \leq i \leq t : s_i = +\} < \frac{1}{2} - \epsilon.$$

Now, the fraction of ϵ-unlucky channels at generation t is equal to $\mathbb{P}(\sum_{i=t_0}^{t} \mathbb{1}_{\{s_i=+\}} < t(\frac{1}{2}-\epsilon))$ where $s_i \overset{\text{iid}}{\sim} \{\pm\}$ with probability $\frac{1}{2}$. By the Law of Large Numbers,

$$\mathbb{P}\left(\frac{1}{t-t_0}\sum_{i=t_0}^{t} \mathbb{1}_{\{s_i=+\}} < \frac{1}{t-t_0}\left(\frac{1}{2}-\epsilon\right)t\right) \xrightarrow{t\to\infty} 0,$$

where the sum converges to $\frac{1}{2}$ by LLN and $\frac{1}{2}-\epsilon < \frac{1}{2}$. So, the number of channels at generation t which are (i) ϵ-good, (ii) survivor and (iii) ϵ-lucky is at least equal to

$$2^t\Big(\underbrace{\text{fraction of } \epsilon\text{-good channels}}_{\to 1-p} - \underbrace{\delta_t}_{\to 0} - \underbrace{\delta'_t}_{\to 0}\Big) \approx (1-p)2^t.$$

Now let us examine the erasure probability $p(s^t)$ of such a channel W^{s^t} satisfying (i), (ii) and (iii). We are interested in $p(s^t) \leq \epsilon$, $p(s^{t-1}) \leq \epsilon, \ldots, p(s^{t_0}) \leq \epsilon$ (because of (ii) and (iii)). Then,

$$p(s^i) \leq p(s^{i-1})^2 \mathbb{1}_{\{s_i=+\}} + 2p(s^{i-1})\mathbb{1}_{\{s_i=-\}} \leq p(s^{i-1})^2 \mathbb{1}_{\{s_i=+\}} + p(s^{i-1})^{1-\eta}\mathbb{1}_{\{s_i=-\}}, \quad \eta = -(\log(\epsilon))^{-1}.$$

Indeed, for $p<\epsilon$ we have $2\le p^{-\eta}$ which is equivalent to $2\le(1/p)^{\eta}$, again equivalent to $1\le\eta\log(1/p)$. Thus we take $\eta=-(\log(\epsilon))^{-1}$. Take the logarithm,

$$\log_2\left(\frac{1}{p(s^i)}\right)\ge 2\log_2\left(\frac{1}{p(s^{i-1})}\right)\mathbb{1}_{\{s_i=+\}}+(1-\eta)\log_2\left(\frac{1}{p(s^{i-1})}\right)\mathbb{1}_{\{s_i=-\}}.$$

As a result,

$$\log_2\log_2\left(\frac{1}{p(s^i)}\right)\ge\log_2\log_2\left(\frac{1}{p(s^{i-1})}\right)+\mathbb{1}_{\{s_i=+\}}+\underbrace{\log_2(1-\eta)\mathbb{1}_{\{s_i=-\}}}_{\text{small negative number }:=-\epsilon'}.$$

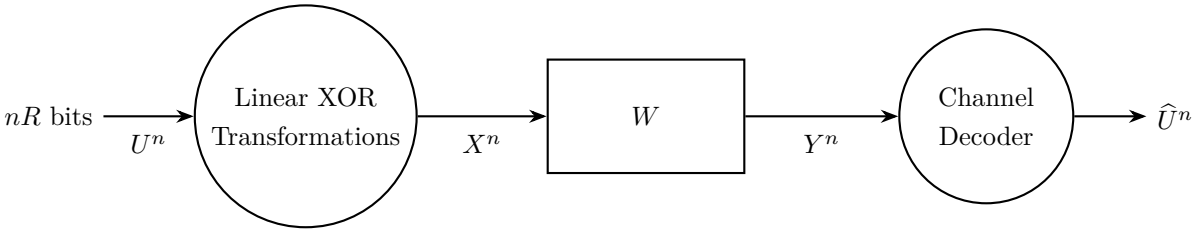
And so,

$$\log_2\log_2\left(\frac{1}{p(s^t)}\right)\ge\log_2\log_2\left(\frac{1}{p(s^{t_0})}\right)-t\epsilon'+\underbrace{\frac{t}{2}(1-\epsilon)}_{W\text{ is }\epsilon\text{-lucky}}\ge\frac{t}{2}(1-\underbrace{(\epsilon+2\epsilon')}_{\text{small}})\approx\log_2(\sqrt{n}).$$

Overall, $p(s^t)\approx 2^{-\sqrt{n}}$ thus the erasure probability of good channels is *very* small. As a result, for $R<1-p$, (1) there are for large $t, n=2^t$, nR good lucky survivors. Also, (2) all of these have erasure probability less than $2^{-n^{\frac{1}{2}-\delta}}$. Thus the sum of the erasure probabilities converges to zero.

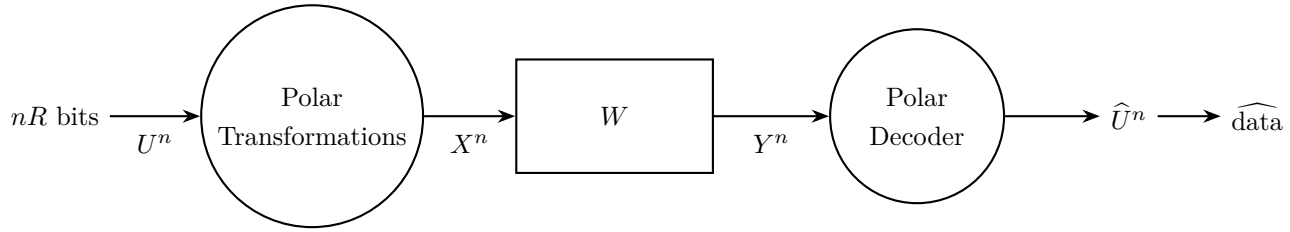
Conclusion

In polar coding we have proved that for $W=\mathrm{BEC}(p)$ for any $\delta>0$ and any $R<1-p$ we can construct a system for all large enough t,



with $U^n=\begin{pmatrix}\mathbb{I}_{nR} & \mathbf{0}\end{pmatrix}^T\begin{pmatrix}nR\text{ bit data}\end{pmatrix}^T$ where this first matrix accounts for permutations. Note also that the linear XOR transformations and channel decoder have an $n\log(n)$ complexity, where also $\mathbb{P}(\widehat{\text{data}}\neq\text{data})\le 2^{-n^{\frac{1}{2}-\delta}}$ for $n=2^t$.

One can also prove that for any binary input $W, \delta>0, R<I(W)$, we can construct for large enough t the following system,

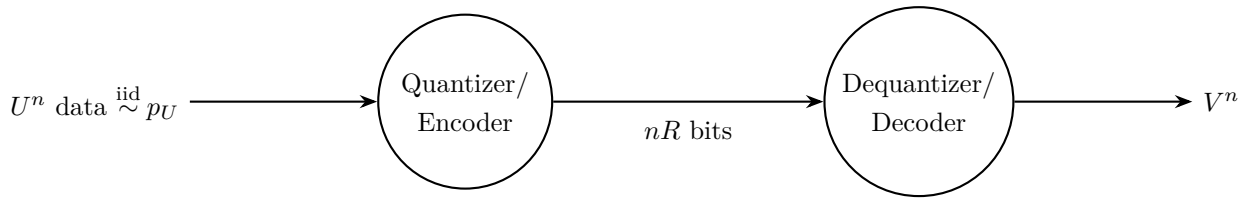


with $U^n=\begin{pmatrix}\text{permutations of }\mathbb{I}_n\end{pmatrix}^T\begin{pmatrix}nR\text{ bit data} & | & \text{known fixed values}\end{pmatrix}^T$ where this first matrix accounts for permutations. Note again that the polar transformations and polar decoder have an $n\log(n)$ complexity, where also $\mathbb{P}(\widehat{\text{data}}\neq\text{data})\le 2^{-n^{\frac{1}{2}-\delta}}$ for $n=2^t$.

Chapter 8

Lossy Compression

We end these lecture notes by tying back to the notion of data compression. We present the notion of Quantization/Lossy compression (use of inexact approximations and partial data discarding to represent the content) from Rate-Distortion theory. The setup is the following:



We consider a **distortion function** $d : \mathcal{U} \times \mathcal{V} \to \mathbb{R}$ and set,

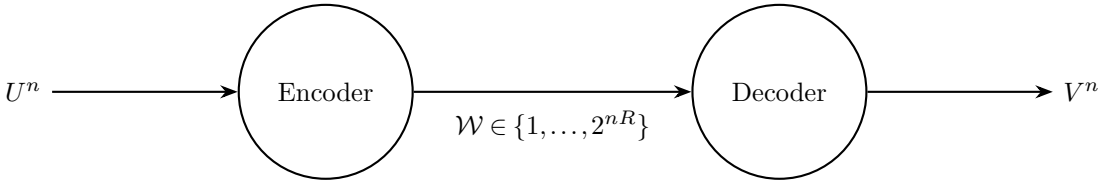
$$D := \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} d(U_i, V_i)\right] \equiv \mathbb{E}\left[d(U^n, V^n)\right].$$

Example 8.0.1. *$U_1, U_2, \ldots \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ with $\mathcal{U} = \mathcal{V} = \mathbb{R}$ and $d(u,v) = (u-v)^2$.*

Example 8.0.2. *$U_1, U_2, \ldots \overset{\text{iid}}{\sim} \text{Ber}(p)$ with $\mathcal{U} = \mathcal{V} = \{0,1\}$ and $d(u,v) = \mathbb{1}_{\{u \neq v\}}$.*

Theorem 8.0.3. *In a system as above, there exists $p_{V|U}$ such that (i) $R \geq I(U;V)$ and (ii) $D = \mathbb{E}(d(U,V))$.*

Proof. Given a system,



induces $p_{U^n V^n}$ (e.g. $p_{U^n V^n}(u^n, v^n) = p_{U^n}(u^n) \mathbb{1}_{\{v^n = \text{dec}(\text{enc}(u^n))\}}$ for i.i.d. U^n) and we thus know

$p_{U_1V_1}, \ldots, p_{U_nV_n}$ where each $p_{U_iV_i}(u,v) = p_U(u)p_{V_i|U_i}(v|u)$. We may also define,

$$p_{UV}(u,v) := \frac{1}{n}\sum_{i=1}^{n} p_{U_iV_i}(u,v) \equiv p_U(u)p_{V|U}(v|u).$$

We now compute:
(i)

$$D = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} d(U_i,V_i)\right] = \frac{1}{n}\sum_{i=1}^{n}\sum_{u,v} p_{U_iV_i}(u_i,v_i)d(u,v) = \sum_{u,v}\left[\frac{1}{n}\sum_{i=1}^{n} p_{U_iV_i}(u_i,v_i)\right]d(u,v) = \mathbb{E}(d(U,V)).$$

(ii) $nR \geq H(\mathcal{W}) \geq I(\mathcal{W};V^n) \geq I(U^n;V^n)$ holds by the Data Processing inequality on the Markov chain $U^n \to \mathcal{W} \to V^n$ and,

$$\begin{aligned} I(U^n;V^n) = H(U^n) - H(U^n|V^n) &\overset{U_i}{=}\overset{\text{iid}}{\sim} \sum_{i=1}^{n} H(U_i) - H(U^n|V^n) \\ &= \sum_{i=1}^{n}\left(H(U_i) - H(U_i|U^{i-1},V^n)\right) && \rightsquigarrow \text{chain rule} \\ &\geq \sum_{i=1}^{n}\left(H(U_i) - H(U_i|V_i)\right) = \sum_{i=1}^{n} I(U_i;V_i). \end{aligned}$$

As a result $R \geq \frac{1}{n}\sum_{i=1}^{n} I(U_i;V_i)$. Now $I(U_i;V_i) \equiv \texttt{function}(p_{U_i}, p_{V_i|U_i})$ is concave in p_{U_i} and convex in $p_{V_i|U_i}$. Then:

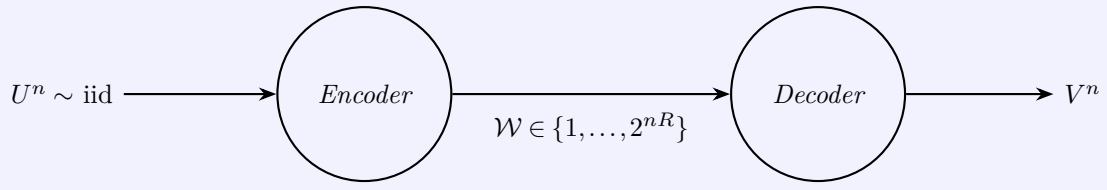
$$R \geq \texttt{function}(p_U, p_{V|U}) = I(U;V) \qquad \rightsquigarrow \text{i.i.d. and convexity.}$$

□

Corollary 8.0.4. *In any system with (R,D), the **rate-distortion function** $R(D)$ satisfies:*

$$R \geq \min_{\substack{p_{V|U} \\ D=\mathbb{E}(d(U,V))}} I(U;V) := R(D). \tag{8.1}$$

Theorem 8.0.5. *Given $p_U, d(U,V)$, for any $p_{V|U}, \epsilon > 0$, there is a design*



such that,

$$\left|\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} d(U_i,V_i)\right] - \mathbb{E}[d(U,V)]\right| \leq \epsilon \tag{8.2}$$

for $V^n = \text{dec}(\text{enc}(U^n))$ such that $U^n \overset{\text{iid}}{\sim} p_U$. We also have,

$$R < I(U;V) + \epsilon. \tag{8.3}$$

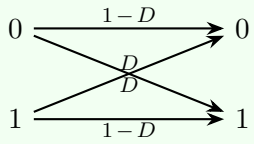
Example 8.0.6. • *Let $\mathcal{U}=\mathcal{V}=\{0,1\}$, $U\sim\text{Ber}(\frac{1}{2})$ and the distortion $d(u,v)=\mathbb{1}_{\{u\neq v\}}$. As a result $\mathbb{E}(d(U,V))=\mathbb{P}(U\neq V)$. Then, we consider a 1-to-1 transformation $U\oplus V=\mathbb{1}_{\{u\neq v\}}$:*

$$I(U;V)=H(U)-H(U|V)=1-H(U\oplus V|V)\geq 1-H(U\oplus V)=1-h_2(D).$$

Thus:

$$\min_{\substack{p_{V|U}:\\ \mathbb{E}(d(U;V))=D}} I(U;V)\geq 1-h_2(D).$$

Furthermore, if we choose $p_{V|U}$ to be the transition of a $BSC(D)$:



Then $\mathbb{P}(U\neq V)=D$ and $I(U;V)=1-h_2(D)$ which yields $R(D)=1-h_2(D)$ (i.e. the minimum is attained).

• *In the case where $U\sim\text{Ber}(p)$ we have that $R(D)\geq[h_2(p)-h_2(D)]^+$ with equality for some clever choices of the transition probabilities of the channel. Thus $R(D)=[h_2(p)-h_2(D)]^+$.*

Example 8.0.7. *Let $\mathcal{U}=\mathcal{V}=\mathbb{R}$, $U\sim\mathcal{N}(0,\sigma^2)$ and $d(u,v)=(u-v)^2$. Now, considering a 1-to-1 transformation $U-V$,*

$$I(U;V)=h(U)-h(U|V)=h(U)-h(U-V|V)\geq h(U)-h(U-V).$$

Recall that $\mathbb{E}((U-V)^2)=D$ implies $h(U-V)\leq\frac{1}{2}\log(2\pi eD)$ and so $I(U;V)\geq\frac{1}{2}\log(\sigma^2/D)$. Then:

$$R(D)=\min_{\substack{p_{V|U}:\\ \mathbb{E}((U-V)^2)=D}} I(U;V)\geq\left[\frac{1}{2}\log\left(\frac{\sigma^2}{D}\right)\right]^+.$$

To show that this lower-bound is indeed tight for $D\leq\sigma^2$ we need to find V such that (i) $Z:=U-V\sim\mathcal{N}(0,D)$ and (ii) $Z:=U-V$ is independent of V. Note $U=V+(U-V)=V+Z$ such that $U\sim\mathcal{N}(0,\sigma^2)$, $Z\sim\mathcal{N}(0,D)$ and V,Z are independent, it must be that $V\sim\mathcal{N}(0,\sigma^2-D)$. So,

$$\begin{pmatrix}U\\V\end{pmatrix}\sim\mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix},\begin{pmatrix}\sigma^2 & \sigma^2-D\\ \sigma^2-D & \sigma^2-D\end{pmatrix}\right)\Longrightarrow R(D)=\frac{1}{2}\log\left(\frac{\sigma^2}{D}\right)\quad\text{for } D\leq\sigma^2.$$

For $D\geq\sigma^2$ we choose V independent of U and V to be zero mean and $\mathbb{E}(V^2)=D-\sigma^2$. Thus $I(U;V)=0$ and $\mathbb{E}((U-V)^2)=\sigma^2+D-\sigma^2=D$. As a result:

$$R(D)=\left[\frac{1}{2}\log\left(\frac{\sigma^2}{D}\right)\right]^+.$$

Proof of Theorem (8.0.5). Pick n large (to be specified later) and δ small (also to be determined later). Let $R = I(U;V) + \epsilon$. Generate

$$\begin{aligned} V^n(1) &= V_{11} \dots V_{1n} \\ V^n(2) &= V_{21} \dots V_{2n} \\ &\vdots \\ V^n(m) &= V_{m1} \dots V_{mn} \end{aligned}$$

where $m := \lfloor 2^{nR} \rfloor$ and $V_{ij} \overset{\text{iid}}{\sim} p_V$. Reveal the table to both the encoder and decoder:

```
enc(u^n): look for i such that (u^n, V^n(i)) ∈ T(n, p_UV, δ)
set i = 0 and start with an empty tree
if found then
    enc(u^n) = i
    print "Success"
else
    enc(u^n) ≡ randomly chosen from {1, ..., m}
    print "Failure"
end if
dec(i) = V^n(i)
```

For the choice $m = \lfloor 2^{nR} \rfloor$ we have $(1/n)\log(m) < R = I(U;V) + \epsilon$ and thus (8.3) holds. Now suppose that the encoder succeeds, i.e. $(u^n, V^n(i)) \in T(n, p_{UV}, \delta)$ where we will denote $V^n(i) \equiv V^n = V_1 \dots V_n$ for simplicity. Then,

$$\frac{1}{n}\sum_{j=1}^{n} d(U_j, V_j) = \sum_{\alpha \in \mathcal{U}} \sum_{\beta \in \mathcal{V}} \underbrace{\frac{1}{n}\sum_{j=1}^{n} \mathbb{1}_{\{U_j \equiv \alpha, V_j \equiv \beta\}}}_{\text{By typicality} = p_{UV}(\alpha,\beta)(1 \pm \delta)} d(\alpha,\beta) = \mathbb{E}(d(U,V)) \pm \delta \underbrace{\mathbb{E}(|d(U,V)|)}_{< \epsilon/2},$$

where we choose δ small enough so that the final term is less than $\epsilon/2$. Now, for $U^n \sim$ i.i.d. and $V^n = \text{dec}(\text{enc}(U^n))$ we have,

$$\begin{aligned} \mathbb{E}\left[\frac{1}{n}\sum_{j=1}^{n} d(U_j, V_j)\right] &= \mathbb{E}\left[\frac{1}{n}\sum_{j=1}^{n} d(U_j, V_j) \Big| \text{success}\right] \mathbb{P}(\text{success}) + \mathbb{E}\left[\frac{1}{n}\sum_{j=1}^{n} d(U_j, V_j) \Big| \text{failure}\right] \mathbb{P}(\text{failure}) \\ &= \left(\mathbb{E}(d(U,V)) \pm \frac{\epsilon}{2}\right) \mathbb{P}(\text{success}) + (\pm D_{\max}) \mathbb{P}(\text{failure}) \\ &= \mathbb{E}(d(U,V)) \pm \epsilon \quad \text{if } \mathbb{P}(\text{failure}) \approx 0 \end{aligned}$$

and where $D_{\max} := \max_{u,v} |d(u,v)|$. For $\{U^n = u^n\}$ failure means that for every $i \in \{1, \dots, m\}$, $(u^n, V^n(i)) \notin T(n, p_{UV}, \delta)$. Suppose then $u^n \notin T(n, p_U, \delta/2)$ (this is a rare event for i.i.d. U^n), we will say $\mathbb{P}(\text{failure}) \leq 1$. Now for $u^n \in T(n, p_U, \delta/2)$:

$$\mathbb{P}(\text{failure}) = \prod_{i=1}^{m} \mathbb{P}\big((u^n, V^n(i)) \notin T(n, p_{UV}, \delta)\big) = \big[\mathbb{P}\big((u^n, V^n) \notin T(n, p_{UV}, \delta)\big)\big]^m$$
$$= \big[1 - \mathbb{P}\big((u^n, V^n) \in T(n, p_{UV}, \delta)\big)\big]^m,$$

where $U^n \in T(n, p_U, \delta/2)$ and $V^n \overset{\text{iid}}{\sim} p_V$. Note that $1 - x \leq e^{-x}$ and thus, for $u^n \in T(n, p_U, \delta/2)$,

$$\mathbb{P}(\text{failure}) \leq e^{-m\mathbb{P}\left((u^n, V^n) \in T(n, p_{UV}, \delta)\right)}.$$

If we are able to show that $(*)$ $\mathbb{P}\big((u^n, V^n) \in T(n, p_{UV}, \delta)\big) \geq 2^{-n[I(U;V)+\epsilon/2]}$ then we will be able to conclude

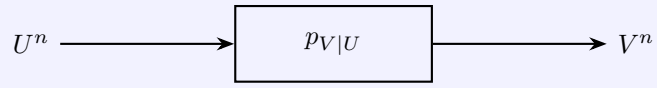
$$m\mathbb{P}\big((u^n, V^n) \in T(n, p_{UV}, \delta)\big) \geq 2^{n[R - I(U;V) - \epsilon/2]} \geq 2^{\frac{n\epsilon}{2}} \to \infty.$$

And so $\mathbb{P}(\text{failure}) \leq e^{-m\mathbb{P}\left((u^n, V^n) \in T(n, p_{UV}, \delta)\right)} \to 0$ for $u^n \in T(n, p_U, \delta/2)$. So the overall failure probability is given by,

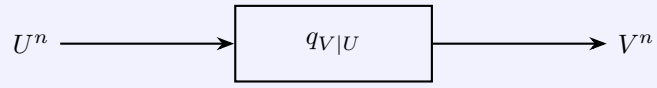
$$\mathbb{P}(\text{failure}) \leq \underbrace{\mathbb{P}(u^n \notin T(n, p_U, \delta/2))}_{\to 0} + \underbrace{e^{-m\mathbb{P}\left((u^n, V^n) \in T(n, p_{UV}, \delta)\right)}}_{\to 0} \longrightarrow 0.$$

To show $(*)$ we may consider the following Lemma:

Lemma 8.0.8. *Given p_{UV}, given $0 < \epsilon' < \epsilon$ and $u^n \in T(n, p_U, \epsilon')$. Let $S(u^n) := \{v^n : (u^n, v^n) \in T(n, p_{UV}, \epsilon)\}$. Then if*



we obtain the following results:
(i) $\mathbb{P}(V^n \in S(u^n)) \to 1$ as $n \to \infty$.
(ii) For every $v^n \in S(u^n)$ we have $\mathbb{P}_{V|U}(v^n | s^n) = 2^{-nH(V|U)(1 \pm \epsilon)}$.
(iii) $|S(u^n)| \leq 2^{nH(V|U)(1+\epsilon)}$ and for n large $|S(u^n)| \geq (1-\epsilon)2^{nH(V|U)(1-\epsilon)}$.
(iv) If we define a new channel



then for remainder terms $\epsilon(\text{junk})$,

$$\mathbb{P}(V^n \in S(u^n)) \leq 2^{-n\left\{\left[\sum_{u,v} p(u,v) \log\left(\frac{p(u|v)}{q(u|v)}\right)\right](1-\epsilon) - \epsilon(\text{junk})\right\}}$$

and for n large,

$$\mathbb{P}(V^n \in S(u^n)) \geq 2^{-n\left\{\left[\sum_{u,v} p(u,v) \log\left(\frac{p(u|v)}{q(u|v)}\right)\right](1+\epsilon) + \epsilon(\text{junk})\right\}}.$$

(v) A special case of (iv) is $q_{V|U} \equiv p_V$ and thus $\mathbb{P}(V^n \in S(u^n)) \approx 2^{-nI(U;V)}$.

By this Lemma, for $\epsilon' = \delta/2$ and $\epsilon = \delta$, $(*)$ and in turn (8.2) follow immediately. □

Proof of Lemma (8.0.8). We prove these statements in turn:
(ii) If $v^n \in S(u^n)$ then $(u^n, v^n) \in T(n, p_{UV}, \epsilon)$. Thus

$$\#\{i : (u_i, v_i) = (\alpha, \beta)\} = np_{UV}(\alpha, \beta)(1 \pm \epsilon).$$

We finally conclude:

$$\begin{aligned} p_{V|U}(v^n|u^n) = \prod_{i=1}^{n} p_{V|U}(v_i|u_i) &= \prod_{\alpha,\beta} p_{V|U}(\beta|\alpha)^{np_{UV}(\alpha,\beta)(1\pm\epsilon)} \\ &= 2^{\sum_{\alpha,\beta} n(1\pm\epsilon)p_{UV}(\alpha,\beta)\log(p_{V|U}(\beta|\alpha))} = 2^{-nH(V|U)(1\pm\epsilon)}. \end{aligned}$$

Note also that if $v^n \in S(u^n)$ then:

$$\begin{aligned} q_{V|U}(v^n|u^n) = 2^{n(1\pm\epsilon)\sum_{\alpha,\beta} p_{UV}(\alpha,\beta)\log(q_{V|U}(\beta|\alpha))} &= 2^{-n(1\pm\epsilon)\left[H(V|U)+\sum_{\alpha,\beta} p_{UV}(\alpha,\beta)\log\frac{p_{V|U}(\beta|\alpha)}{q_{V|U}(\beta|\alpha)}\right]} \\ &= 2^{-n(1\pm\epsilon)\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right]}. \end{aligned}$$

(iii) • We first have,

$$1 \geq \sum_{v^n \in S(u^n)} p_{V|U}(v^n|u^n) \geq \sum_{v^n \in S(u^n)} 2^{-nH(V|U)(1+\epsilon)} = |S(u^n)|2^{-nH(V|U)(1+\epsilon)}.$$

• Suppose (i) is true, then for large n we have,

$$(1-\epsilon) \leq \mathbb{P}(V^n \in S(u^n)) = \sum_{v^n \in S(u^n)} p_{V|U}(v^n|u^n) \leq \sum_{v^n \in S(u^n)} 2^{-nH(V|U)(1-\epsilon)} = |S(u^n)|2^{-nH(V|U)(1-\epsilon)}.$$

(iv) • For the upper-bound,

$$\begin{aligned} \mathbb{P}(V^n \in S(u^n)) = \sum_{v^n \in S(u^n)} q_{V|U}(v^n|u^n) &\leq \sum_{v^n \in S(u^n)} 2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1-\epsilon)} \\ &\leq |S(u^n)|2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1-\epsilon)} \\ &\leq 2^{nH(V|U)(1+\epsilon)}2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1-\epsilon)} \\ &= 2^{-n\left[D(p_{UV}\|p_U q_{V|U})-\epsilon(2H(V|U)+D(p_{UV}\|p_U q_{V|U}))\right]}, \end{aligned}$$

where $\epsilon(\text{junk}) = 2H(V|U) + D(p_{UV}\|p_U q_{V|U})$.

• For the lower-bound and for n large,

$$\begin{aligned} \mathbb{P}(V^n \in S(u^n)) = \sum_{v^n \in S(u^n)} q_{V|U}(v^n|u^n) &\geq \sum_{v^n \in S(u^n)} 2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1+\epsilon)} \\ &= |S(u^n)|2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1+\epsilon)} \\ &\geq (1-\epsilon)2^{nH(V|U)(1-\epsilon)}2^{-n\left[H(V|U)+D(p_{UV}\|p_U q_{V|U})\right](1+\epsilon)} \\ &= (1-\epsilon)2^{-n\left[D(p_{UV}\|p_U q_{V|U})+\epsilon(2H(V|U)+D(p_{UV}\|p_U q_{V|U}))\right]} \end{aligned}$$

where again $\epsilon(\text{junk}) = 2H(V|U) + D(p_{UV} \| p_U q_{V|U})$.

(v) If $V^n \overset{\text{iid}}{\sim} p_V$ is independent of u^n then for the upper-bound,

$$\mathbb{P}((u^n, V^n) \in T(n, p_{UV}, \epsilon)) = \mathbb{P}(V^n \in S(u^n)) \leq 2^{-n\left[I(U;V) + \epsilon(2H(V|U) + I(U;V))\right]}.$$

For the lower-bound,

$$\mathbb{P}((u^n, V^n) \in T(n, p_{UV}, \epsilon)) = \mathbb{P}(V^n \in S(u^n)) \geq (1-\epsilon) 2^{-n\left[I(U;V) - \epsilon(2H(V|U) + I(U;V))\right]}.$$

(i) Without loss of generality, suppose that $\mathcal{V} = \{a, b\}$ and $\mathcal{U} = \{0, 1\}$ such that $u^n = \underbrace{11\ldots11}_{n_1}\underbrace{22\ldots22}_{n_2}$. Then $n_1 = np_U(1)(1 \pm \epsilon')$ and $n_2 = np_U(2)(1 \pm \epsilon')$ thus $V^n = V_1 \ldots V_{n_1} V'_1 \ldots V'_{n_2}$. Now:

$$\begin{aligned}
\{V^n \in S(u^n)\} \equiv \{\#\{i \in [1, \ldots, n_1] : V_i = a\} = np_{UV}(1a)(1 \pm \epsilon)\} \\
\&\{\#\{i \in [1, \ldots, n_1] : V_i = b\} = np_{UV}(1b)(1 \pm \epsilon)\} \\
\&\{\#\{i \in [1, \ldots, n_2] : V'_i = a\} = np_{UV}(2a)(1 \pm \epsilon)\} \\
\&\{\#\{i \in [1, \ldots, n_2] : V'_i = b\} = np_{UV}(2b)(1 \pm \epsilon)\}.
\end{aligned}$$

We want to show that these four events each have a small probability. Let us prove this for the first event, where we consider the complement of this first event:

$$\mathbb{P}\left(\left|\sum_{i=1}^{n_1} \mathbb{1}_{\{V_i = a\}} - np_{UV}(1a)\right| > \epsilon np_{UV}(1a)\right) = \mathbb{P}\left(\left|\frac{1}{n_1}\sum_{i=1}^{n_1} \mathbb{1}_{\{V_i = a\}} - \frac{n}{n_1} p_{UV}(1a)\right| > \epsilon \frac{n}{n_1} p_{UV}(1a)\right).$$

By the Law of Large Numbers, $\frac{1}{n_1}\sum_{i=1}^{n_1} \mathbb{1}_{\{V_i = a\}} \longrightarrow p_{V|U}(a|1)$ and $\frac{n}{n_1} p_{UV}(1a) \longrightarrow \frac{p_{UV}(1a)}{p_U(1)(1 \pm \epsilon')} = \frac{p_{V|U}(a|1)}{(1 \pm \epsilon')}$. So,

$$\left|\frac{1}{n_1}\sum_{i=1}^{n_1} \mathbb{1}_{\{V_i = a\}} - \frac{n}{n_1} p_{UV}(1a)\right| \longrightarrow p_{V|U}(a|1)\left(1 - \frac{1}{1 \pm \epsilon'}\right) = p_{V|U}(a|1)\left(\frac{\pm\epsilon'}{1 \pm \epsilon'}\right) \leq p_{V|U}(a|1)\frac{\epsilon'}{1 - \epsilon'}.$$

However, $\epsilon \frac{n}{n_1} p_{UV}(1a) \geq \frac{\epsilon}{1-\epsilon'} \frac{p_{UV}(1a)}{p_U(1)} = \frac{\epsilon}{1-\epsilon'} p_{V|U}(a|1)$. We conclude that for large n the complement has small probability. Since this also holds for the other three events we conclude that,

$$\mathbb{P}(V^n \notin S(u^n)) \xrightarrow{n \to \infty} 0.$$

□

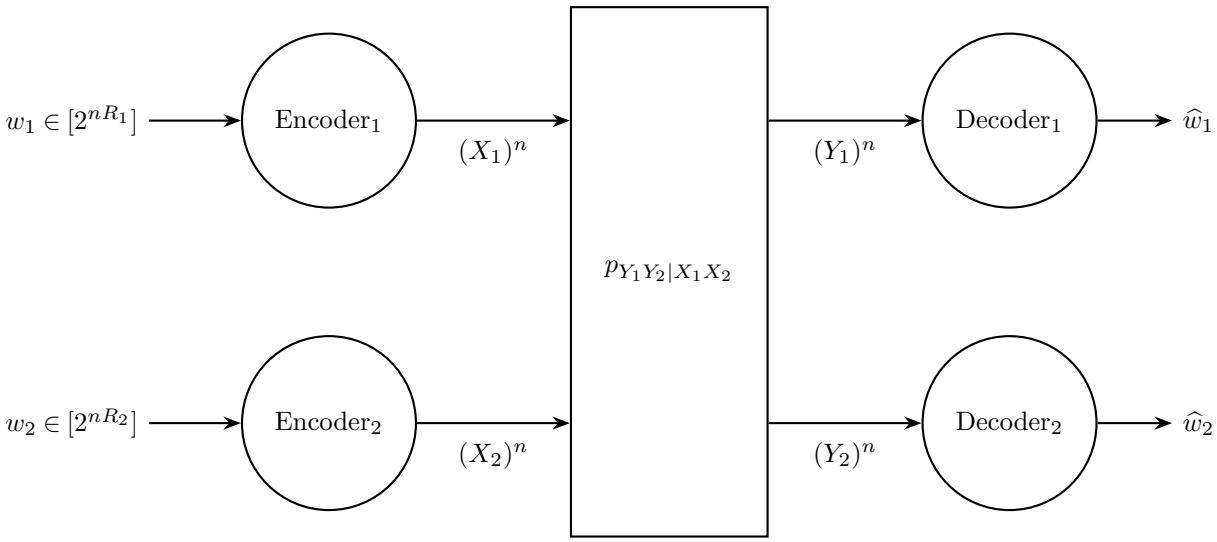
Chapter 9

Summary and Open Problems

Throughout these notes, we have seen multiple concepts. The first was lossless source coding which is characterized by the entropy $H(U)$ [minimum # of hits to describe data exactly]. We also saw a universal version where if a long sequence of data is to be described then we have distribution agnostic schemes (source statistics which perform almost as well as schemes that knows source statistics). We finally ended on lossy source coding characterized by $R(D) := \min I(U;V)$ [minimum # of hits within a given fidelity]. We then went on to describe the process of data transmission over a noisy channel, such that its capacity is set to be $c(p_{Y|X}) = \max I(X;Y)$ [maximum # of data hits per channel use that can be reliably transmitted]. We also saw that for any $\mathcal{X}, \mathcal{Y}, p_X, R$ we can (almost always) design an encoder/decoder of rate R such that for all channels $p_{Y|X}$ with $I(X;Y)|_{p_X} > R$ the encoder/decoder will work.

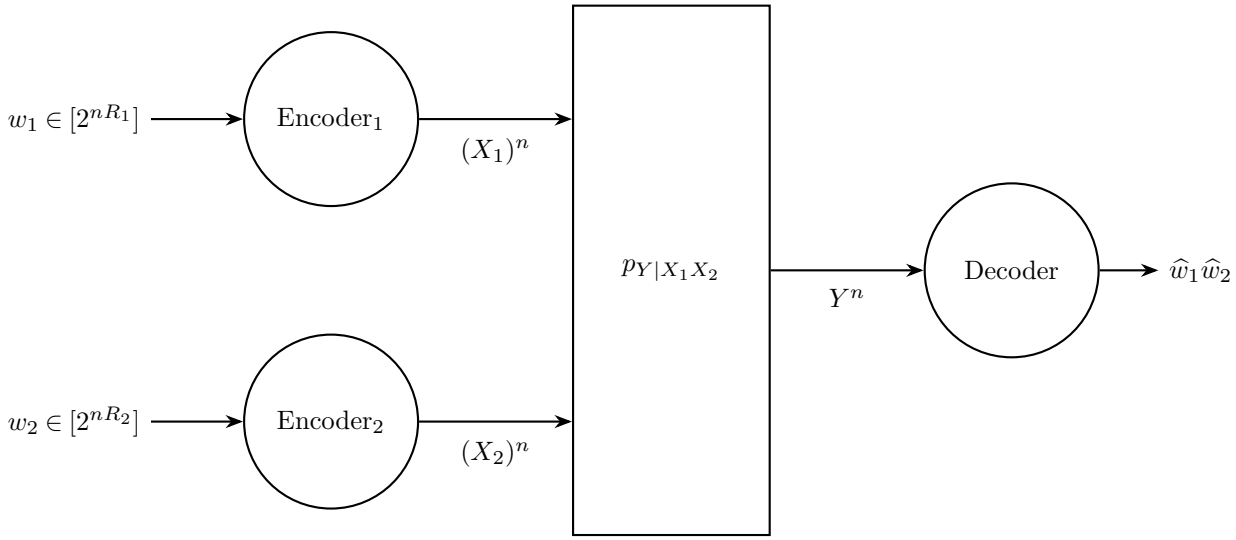
Let us now look at some open problems for the question: for which pairs (R_1, R_2) will ensure that $\mathbb{P}((\widehat{w}_1, \widehat{w}_2) \neq (w_1, w_2))$ can be made small?

- **Interference Channel**

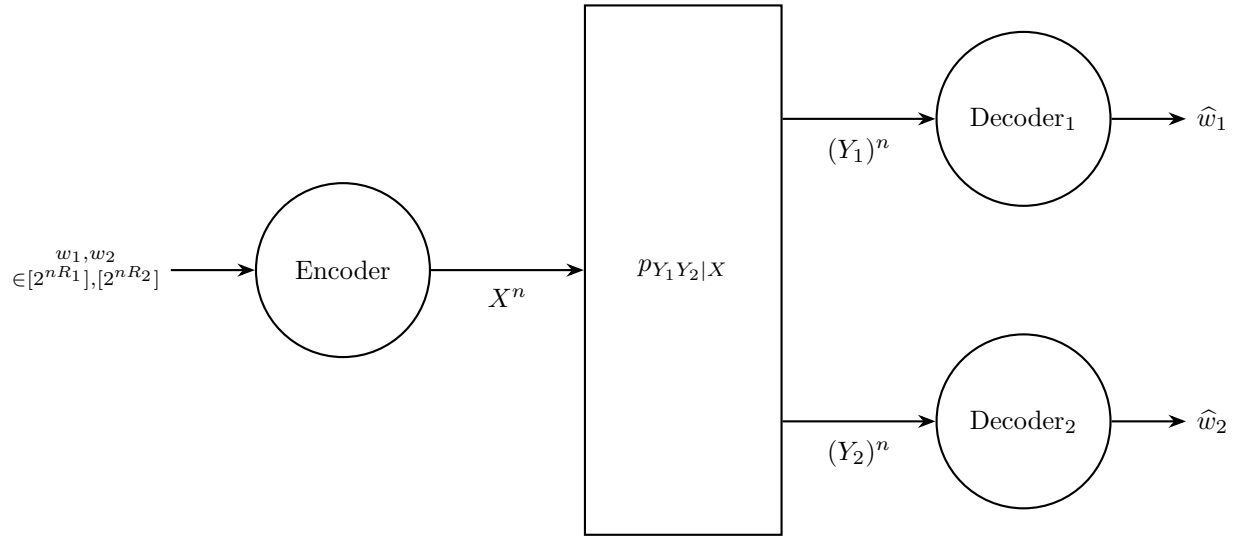


The question we want an answer to (also for all subsequent problems) is the following: given $p_{Y_1Y_2|X_1X_2}$ for which pairs (R_1, R_2) will ensure that $\mathbb{P}((\widehat{w}_1, \widehat{w}_2) \neq (w_1, w_2))$ can be made small? For instance, this question for the interference channel above is still unknown. The general idea is, for δ small, to make $\frac{1}{2^{nR}} \sum_{w \in [2^{nR}]} \mathbb{P}(\text{error}|w) < \delta$ and $\max_{w \in [2^{nR}]} \mathbb{P}(\text{error}|w) < \delta$, and also to be able to distinguish them.

- **Multiple Access Channel**

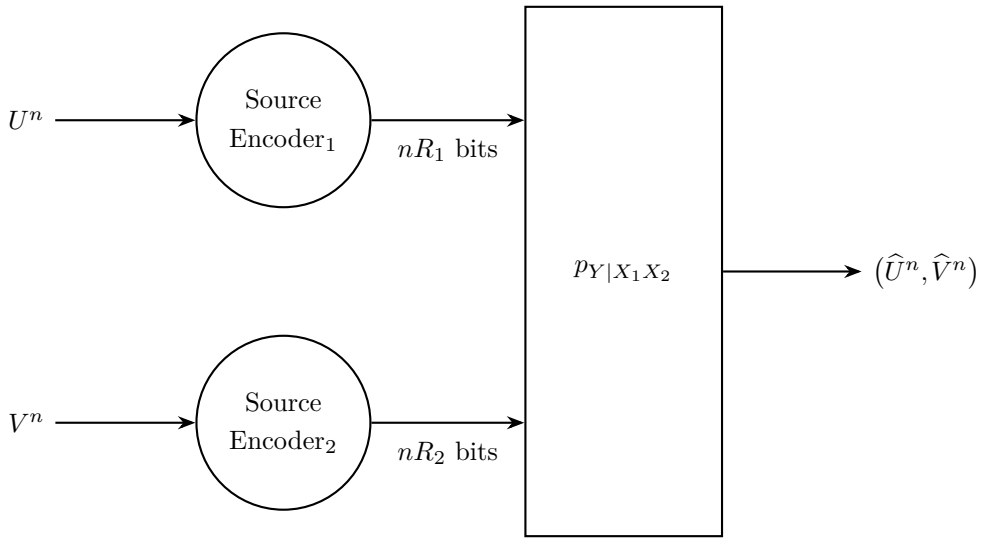


- **Broadcast Channel**

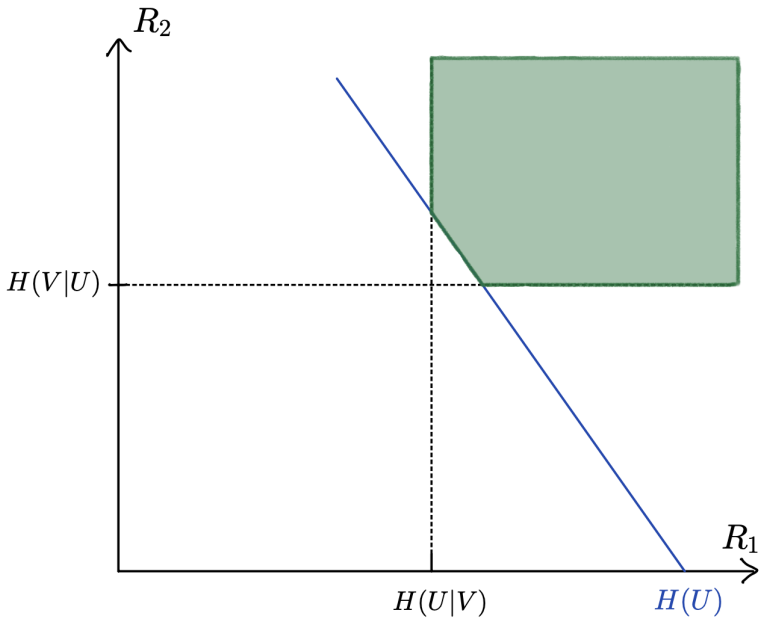


Again in this case, the problem of finding the optimal (R_1, R_2) is still unknown (except in the case where one channel is nosier than the other).

- **Source Coding**



Here we have that $\mathbb{P}(\widehat{U}^n\widehat{V}^n \neq U^nV^n)$ is small almost surely. Here is a plot of (R_1, R_2) where the admissible region is represented in green.



However, replacing the final conditions by $\frac{1}{n}\sum_{j=1}^n d_1(\widehat{U}_j, U_j) \approx D_1$ and $\frac{1}{n}\sum_{j=1}^n d_2(\widehat{V}_j, V_j) \approx D_2$ yields that finding such a region is still an unresolved problem.

Appendix A

Appendices

A.1 Convexity and Entropy

In this Appendix, we recall the basic notions of convexity of a set and of general functions before presenting an elegant proof of the fact that the entropy of a random variable is concave.

Definition A.1.1. *A set $S \in \mathbb{R}^n$ is said to be **convex (set)** if for all $x, y \in S$ and $0 \leq \lambda \leq 1$ we have $\lambda x + (1-\lambda)y \in S$.*

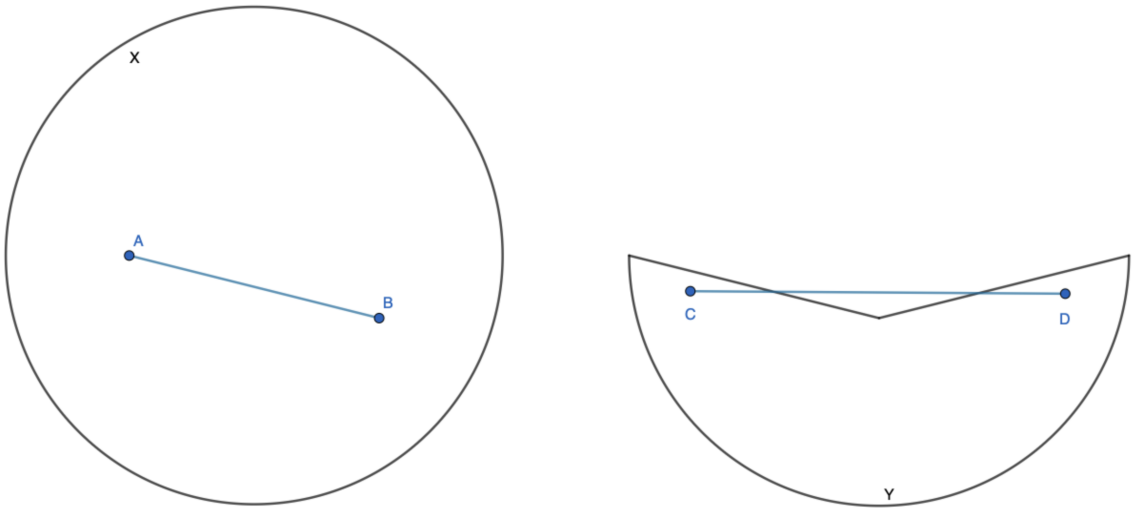


Figure A.1: The set X is a convex set, while the set Y is not.

Definition A.1.2. *Given a convex set S, a function is said to be **convex (function)** if for all $x, y \in S$ and $0 \leq \lambda \leq 1$ we have $\lambda f(x) + (1-\lambda) f(y) \geq f(\lambda x + (1-\lambda)y)$.*

Definition A.1.3. *Given a convex set S, a function is said to be **concave (function)** if and only if $(-f)$ is convex.*

We may now observe three points: (i) $H(U)$ is a function of p_U, (ii) if $|\mathcal{U}| = k$ then we may identify p_U with a point in $\mathbb{R}^k$ and in fact also in any **simplex** $S_k \subseteq \mathbb{R}^k$, $S_k = \left\{(x_1, ..., x_k) \in \mathbb{R}^k : \sum_{i=1}^k x_i = 1, x_i \geq 0\right\}$. Finally, (iii) S_k is a convex set. As a result, we may view $H(\cdot)$ as a $\mathbb{R}$-valued function on S_k.

Proposition A.1.4. *$H : S_k \to \mathbb{R}$ is concave.*

Proof. For $p \in S_k$, $q \in S$ and $0 \leq \lambda \leq 1$ we need to show $\lambda H(p) + (1-\lambda)H(q) \leq H(\lambda p + (1-\lambda)q)$. To do so, we consider a pair of random variables (U,V) for $U \in \{1,2\}$, $V \in \{1,...,k\}$ such that $p_U(1) = \lambda$, $p_U(2) = 1-\lambda$, $p_{V|U}(v|1) = p(v)$, and $p_{V|U}(v|2) = q(v)$. As a result of this construction, we know that $p_V = \lambda p + (1-\lambda)q$. Thus $H(\lambda p + (1-\lambda)q) = H(V)$ and $\lambda H(p) + (1-\lambda)H(q) = H(V|U)$. Then this problem boils down to showing that $H(V) \geq H(V|U)$, which we know is true since conditioning reduces entropy [Theorem (2.1.11)]. □

We now present a final Proposition without proof, but that may be shown in a similar way as the previous result (i.e. using results on entropy).

Proposition A.1.5. *$I(U;V)$ may be viewed as a function of $(p_U, p_{V|U})$. Moreover, by fixing the probability kernel:*
(i) $I(U;V)$ is a concave function of p_U.
(ii) $I(U;V)$ is a convex function of $p_{V|U}$.

A.2 KKT Conditions

Suppose that $\mathcal{X} = \{1,...,k\}$ and $p_X = (p_1,...,p_k)$ with $p_i \geq 0$ and $\sum_i p_i = 1$. From the previous Appendix, we defined the simplex $S_k = \{(p_1,...,p_k) : p_i \geq 0, \sum_i p_i = 1\}$ and saw that S_k is a convex subset of $\mathbb{R}^k$. Let us now formally present the KKT conditions:

Theorem A.2.1 (KKT conditions)**.** *Suppose $f : S_k \to \mathbb{R}$ is smooth (i.e. continuous, C^1 and/or C^2). If $p = (p_1,...,p_k) \in S_k$ maximises f, then there exists a λ such that:*

$$\frac{\partial f}{\partial p_i} \leq \lambda \quad \text{for all } i. \tag{A.1}$$

$$\frac{\partial f}{\partial p_i} = \lambda \quad \text{for all } i \text{ with } p_i > 0. \tag{A.2}$$

Proof. Consider a small $\epsilon > 0$ and i,j with $p_j > 0$. Then set $q_1 = p_1$, $q_i = p_i + \epsilon$, $q_j = p_j - \epsilon$ and $q_k = p_k$ so that $q \in S_k$. Now, consider a Taylor expansion around $q = p$ so that we have

$$f(q) = f(p) + \epsilon \frac{\partial f}{\partial p_i}(p) - \epsilon \frac{\partial f}{\partial p_j}(p) + o(\epsilon) = f(p) + \epsilon \underbrace{\left(\frac{\partial f}{\partial p_i} - \frac{\partial f}{\partial p_j} \right)}_{\leq 0} + o(\epsilon).$$

So for every i,j with $p_j > 0$ we have that $\partial_{p_j} f \geq \partial_{p_i} f$. Note now that if $p_i > 0$ and $p_j > 0$ then we also have that $\partial_{p_i} f \geq \partial_{p_j} f$. Overall we conclude the equality case $\partial_{p_i} f = \partial_{p_j} f$. □

From the KKT conditions, we can prove the following key result:

Theorem A.2.2. *If $f : S_k \to \mathbb{R}$ is smooth, concave and $p = (p_1, ..., p_k) \in S_k$ which maximises f satisfies the KKT conditions, then $f(p) \geq f(q)$ for all $q \in S_k$.*

Proof. For any q and any $0 < \theta \leq 1$, by concavity,

$$f(\theta q + (1-\theta)p) \geq \theta f(q) + (1-\theta) f(p).$$

In particular we have that

$$f(q) \leq \frac{1}{\theta}\left[f(\theta q + (1-\theta)p) - f(p) + \theta f(p)\right] = f(p) + \frac{1}{\theta}\left[f(p + \theta(q-p)) - f(p)\right].$$

Then if we show that $\lim_{\theta \searrow 0} \frac{1}{\theta}\left[f(p+\theta(q-p)) - f(p)\right] \leq 0$ then we are done. To this end, consider the Taylor expansion,

$$f(p_1 + \theta(q_1 - p_1), p_2 + \theta(q_2 - p_2), ..., p_k + \theta(q_k - p_k)) - f(p_1, ..., p_k) = \theta \sum_{i=1}^{k} \frac{\partial f}{\partial p_i}(p)(q_i - p_i) + o(\theta).$$

So, taking a limit we obtain:

$$\begin{aligned}
\lim_{\theta \searrow 0} \frac{1}{\theta}\left[f(p+\theta(q-p)) - f(p)\right] &= \sum_{i=1}^{k} \frac{\partial f}{\partial p_i}(p)(q_i - p_i) \\
&= \sum_{i:p_i>0} \frac{\partial f}{\partial p_i}(p)(q_i - p_i) + \sum_{i:p_i=0} \frac{\partial f}{\partial p_i}(p) \underbrace{(q_i - p_i)}_{\geq 0} \\
&\leq \lambda \sum_{i:p_i>0} (q_i - p_i) + \lambda \sum_{i:p_i=0} (q_i - p_i) \qquad \rightsquigarrow \text{KKT} \\
&= \lambda \sum_{i=1}^{k} (q_i - p_i) = 0.
\end{aligned}$$

□

A.3 Stochastic Processes and Entropy

In this second appendix, we present succinctly some fundamental notions of stochastic processes. In particular, we will concentrate mainly on stochastic processes as general objects and we will see a particular case consisting of Markov processes. The first notion that we will present as a definition is the notion of stationary processes, being a quite restrictive condition when considering general stochastic processes.

Definition A.3.1. *A process $\{U_i\}_{i=1,2,...}$ is said to be a **stationary process** if $\mathbb{P}(U_{i+1} = u_1) = \mathbb{P}(U_i = u_1)$ for all $i \geq 0$. This also holds for a more general condition:*

$$\mathbb{P}(U_{i+1}, ..., U_{i+n} = u^n) = \mathbb{P}(U^n = u^n), \qquad \forall i \geq 0, n \geq 1. \tag{A.3}$$

Proposition A.3.2. *For a stationary process* $\{X_i\}_{i=1,2,...}$*, the following holds:*
(i) $\frac{1}{n}H(X_1,...,X_n) \geq H(X_n|X_{n-1},...,X_1)$.
(ii) $\frac{1}{n}H(X_1,...,X_n) \leq \frac{1}{n-1}H(X_1,...,X_{n-1})$.

Proof. We start with the first point.
(i) We first use the fact that conditioning reduces entropy and then apply stationarity to find, for all $1 \leq i \leq n$:

$$H(X_n|X^{n-1}) \leq H(X_n|X_{n-i+1},X_{n-i+2},...,X_{n-1}) = H(X_i|X^{i-1}).$$

As a result we find that:

$$H(X_n|X^{n-1}) = \frac{1}{n}\sum_{i=1}^{n} H(X_n|X^{n-1}) \leq \frac{1}{n}\sum_{i=1}^{n} H(X_i|X^{i-1}) = \frac{1}{n}H(X_1,...,X_n).$$

(ii) Using the chain rule for entropy we have:

$$\begin{aligned}\frac{1}{n}H(X_1,...,X_n) = \frac{1}{n}\sum_{i=1}^{n} H(X_i|X^{i-1}) &= \frac{1}{n}\left(H(X_n|X^{n-1}) + \sum_{i=1}^{n-1} H(X_i|X^{i-1})\right) \\ &= \frac{1}{n}\Big(H(X_n|X^{n-1}) + H(X_1,...,X_{n-1})\Big).\end{aligned}$$

From stationarity we know that for all $1 \leq i \leq n$: $H(X_n|X^{n-1}) \leq H(X_i|X^{i-1})$, and thus by summing on both sides over $i = 1,...,n-1$ also dividing by $(n-1)$ yields:

$$H(X_n|X^{n-1}) \leq \frac{1}{n-1}\sum_{i=1}^{n-1} H(X_i|X^{i-1}) = \frac{1}{n-1}H(X_1,...,X_{n-1}).$$

Combining the above two equations gives us:

$$\begin{aligned}\frac{1}{n}H(X_1,...,X_n) = \frac{1}{n}\Big(H(X_n|X^{n-1}) + H(X_1,...,X_{n-1})\Big) &\leq \frac{1}{n}\left[\frac{1}{n-1}H(X_1,...,X_{n-1}) + H(X_1,...,X_{n-1})\right] \\ &\leq \frac{1}{n-1}H(X_1,...,X_{n-1}).\end{aligned}$$

□

Proposition A.3.3. *For a stationary process* $\{X_i\}_{i=-\infty}^{\infty}$*, we have:*

$$H(X_0|X_{-1},...,X_{-n}) = H(X_0|X_1,...,X_n). \tag{A.4}$$

Proof. We first use the chain rule for entropy and stationarity to find:

$$\begin{aligned}H(X_0|X_{-1},...,X_{-n}) &= H(X_0,X_{-1},...,X_{-n}) - H(X_{-1},...,X_{-n}) \\ &= H(X_0,X_1,...,X_n) - H(X_1,...,X_n) \\ &= H(X_0|X_1,...,X_n).\end{aligned}$$

□

Now, to specify the statistics of a stochastic process $\{U_i\}_{i=1,2,...}$ we require the knowledge of the conditional distributions $p_{U_1}(u_1)$, $p_{U_2|U_1}(u_2|u_1)$, $p_{U_3|U_2,U_1}(u_3|u_2,u_1)$, and for all $n \geq 1$ we have the conditional transition probability $p_{U_n|U^{n-1}}(u_n|u_1,...,u_{n-1})$. As a result, a **Markov process** is one in which:

$$p_{U_n|U_1,...,U_{n-1}}(u_n|u_1,...,u_{n-1}) = p_{U_n|U_{n-1}}(u_n|u_{n-1}), \tag{A.5}$$

and thus in this case, such a process is fully characterised by the knowledge of $p_{U_n|U_{n-1}}(u_n|u_{n-1})$. In particular, a **Markov chain** $X \leftrightarrow Y \leftrightarrow Z$ is a process for which $p_{XYZ}(x,y,z) = p_X(x)p_{Y|X}(y|x)p_{Z|Y}(z|y) = p_Z(z)p_{Y|Z}(y|z)p_{X|Y}(x|y)$. Equivalently, this means that conditioned on Y, X and Z are independent.

A.4 Interpretations of the KL-Divergence

In this appendix, the aim is to give some form of interpretation to the KL-divergence. We first note that:

$$D_{KL}(q\|p) := \sum_{u\in\mathcal{U}} q(u)\log_2\left(\frac{q(u)}{p(u)}\right) = \sum_{u\in\mathcal{U}} p(u)\frac{q(u)}{p(u)}\log_2\left(\frac{q(u)}{p(u)}\right) =: \sum_{u\in\mathcal{U}} p(u) f\left(\frac{q(u)}{p(u)}\right), \tag{A.6}$$

where $f : \mathbb{R}^+ \longrightarrow \mathbb{R}$ such that $\xi \mapsto \xi\log(\xi)$. Note in this case that $f(1) = 0$ and f is convex. Thus, we understand that we could modify the definition of the KL-divergence to other functions f provided that they satisfy the right conditions. The study of the generalization of the divergence is known as the **f-Divergence Theory**, where its purpose is to define new divergences $D_f(q\|p)$ for some convex function f with $f(1) = 0$.

Example A.4.1. *The following points are examples of functions f satisfying the required conditions:*
(1) For $f(\xi) = |\xi - 1|$ we obtain the L_1-distance $\sum_{u\in\mathcal{U}}|p(u) - q(u)|$.
(2) For $f(\xi) = \xi^2 - 1$ we obtain the χ^2-divergence.

Note now, that there exists a relationship between the mutual information of two random variables and the KL-divergence:

$$I(U;V) = \sum_{u,v} p(u,v)\log_2\left(\frac{p(u,v)}{p(u)p(v)}\right) = D_{KL}(p_{U,V}\|p_U p_V). \tag{A.7}$$

This represents the likelihood that we mistake the independent distribution to be in fact the joint distribution.

Let us now look at yet another interpretation of the KL-divergence. Suppose $U \sim p$ however we are wrongly told that U has statistics given by q, and we design a code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ believing q. In this case, the ideal codeword length would be given by $\text{length}(c(u)) = \log_2\left(\frac{1}{q(u)}\right)$. Thus:

$$\mathbb{E}[\text{length}(c(U))] = \sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{1}{q(u)}\right) = \sum_{u\in\mathcal{U}} p(u)\log_2\left(\frac{p(u)}{q(u)p(u)}\right) = H(p) + D_{KL}(p\|q). \tag{A.8}$$

In this setting, $H(p)$ represents the expected length had we used p in our design, and $D_{KL}(p\|q)$

represents the *penalty* we pay for using q instead of the true distribution p.
This suggests a possible framework to handle uncertain knowledge. Suppose we know that $U \sim p$ but we do not know p explicitly, except that $p \in \mathcal{P}$ where $\mathcal{P}$ is some class of distributions. Nevertheless, we are asked to come up with a code $c : \mathcal{U} \longrightarrow \{0,1\}^*$ which is equivalent to picking a distribution q for which $\text{length}(c(u)) = \log_2\left(\frac{1}{q(u)}\right)$. In this instance, we could for example look at a quantity for the evaluation of the design:

$$\text{eval}(q, \mathcal{P}) := \max_{p \in \mathcal{P}} \mathbb{E}_p[\text{length}(c(U))] - H(p), \tag{A.9}$$

such that it represents the *penalty* defined in (A.8). Now, we can pick q that will minimize this maximal *penalty*:

$$\min_q \max_{p \in \mathcal{P}} D_{KL}(p\|q). \tag{A.10}$$

However, an alternative formulation will replace $D_{KL}(p\|q)$ with $\max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right)$, which is of course larger than $D_{KL}(p\|q)$. Indeed, since $D_{KL}(p\|q) \geq 0$ we have:

$$\begin{aligned} 0 \leq D_{KL}(p\|q) = \sum_{u \in \mathcal{U}} p(u) \log_2\left(\frac{p(u)}{q(u)}\right) \leq \sum_{u \in \mathcal{U}} p(u) \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right) &= \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right)\left(\sum_{u \in \mathcal{U}} p(u)\right) \\ &= \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right). \end{aligned}$$

We can thus choose a q to minimize the following quantity,

$$\max_{p \in \mathcal{P}} \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right), \tag{A.11}$$

so that the problem becomes the minimization of the following expression:

$$\min_q \max_{p \in \mathcal{P}} \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right). \tag{A.12}$$

There exists some other formulations of this problem, but remark that the optimal minimal q that will solve the problem above is $q^* = p$ with value zero. Note that, throughout this optimization problem, we have assumed without loss of generality that $p(u) > 0$ for all $u \in \mathcal{U}$.

We consider now an even more general problem, where for $f : \mathcal{U} \to \mathbb{R}^+$ we have $S := \sum_{u \in \mathcal{U}} f(u)$ and $p(u) = \frac{f(u)}{S}$. Then:

$$\min_q \max_{u \in \mathcal{U}} \log_2\left(\frac{f(u)}{q(u)}\right) = \min_q \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)S}{q(u)}\right) = \log_2(S) + \min_q \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right). \tag{A.13}$$

This problem is also optimally solved by setting $q^* = p$ but this time with a value of $\log_2(S)$. Thus, problem (A.12) in this case will be given by:

$$\min_q \max_{p \in \mathcal{P}} \max_{u \in \mathcal{U}} \log_2\left(\frac{p(u)}{q(u)}\right) = \min_q \max_{u \in \mathcal{U}} \max_{p \in \mathcal{P}} \log_2\left(\frac{p(u)}{q(u)}\right) = \min_q \max_{u \in \mathcal{U}} \log_2\left(\frac{f(u)}{q(u)}\right), \tag{A.14}$$

with $f(u) = \max_{p \in \mathcal{P}} p(u)$ where in this case f is not yet a probability distribution. As a result, we consider the optimal distribution to be $q^* = \frac{f(u)}{S}$ where $S = S(\mathcal{P}) = \sum_{u \in \mathcal{U}} \max_{p \in \mathcal{P}} p(u)$ and such that the value of the above problem is $\log_2(S(\mathcal{P}))$. Note that $S(\mathcal{P})$ is the so-called **Shtarkov sum**.

Example A.4.2. *Let $\mathcal{U}=\{0,1\}^n$ and $\mathcal{P}$ be the class of i.i.d. distributions on $\mathcal{U}$. Namely, these are distributions indexed by $\theta\in[0,1]$ such that,*

$$p_\theta(u^n)=\theta^k(1-\theta)^{n-k}, \tag{A.15}$$

where $k=\{\#$ of 1's in $u^n\}$. Consider then:

$$f(u^n)=\max_{0\le\theta\le1}\theta^k(1-\theta)^{n-k}=\left(\frac{k}{n}\right)^k\left(1-\frac{k}{n}\right)^{n-k}. \tag{A.16}$$

As a result, the Shtarkov sum is given by:

$$S(\mathcal{P})=\sum_{u^n\in\{0,1\}^n}f(u^n)=\sum_{k=0}^n\binom{n}{k}\left(\frac{k}{n}\right)^k\left(1-\frac{k}{n}\right)^{n-k}. \tag{A.17}$$

Now note that, for any $\alpha\in[0,1]$, $1=(\alpha+(1-\alpha))^n=\sum_{k=0}^n\binom{n}{k}\alpha^k(1-\alpha)^{n-k}\ge\binom{n}{k}\alpha^k(1-\alpha)^{n-k}$. We may bound above the Shtarkov sum as:

$$S(\mathcal{P})=\sum_{k=0}^n\binom{n}{k}\left(\frac{k}{n}\right)^k\left(1-\frac{k}{n}\right)^{n-k}\le\sum_{k=0}^n1=(n+1). \tag{A.18}$$

So, with $q(u^n)=\frac{f(u^n)}{S}$ such that $\text{length}(c(u^n))=\left\lceil\log_2\left(\frac{1}{q(u^n)}\right)\right\rceil\le\log_2\left(\frac{1}{q(u^n)}\right)+1$, we know that there exists a prefix-free code $c:\mathcal{U}^n\to\{0,1\}^*$ where the per source letter regret is upper-bounded by:

$$\frac{1}{n}\left[\text{length}(c(u^n))-\log_2\left(\frac{1}{p_\theta(u^n)}\right)\right]\le\frac{1}{n}\left[1+\log_2(n+1)\right]. \tag{A.19}$$

Now, $\binom{n}{k}\le\left(\frac{k}{n}\right)^{-k}\left(1-\frac{k}{n}\right)^{-(n-k)}\iff\frac{1}{n}\log_2\binom{n}{k}\le h_2\left(\frac{k}{n}\right)$, and a coding scheme is given by:

$$c(u^n)=\underbrace{[\text{describe } k\in\{0,\dots,n\} \text{ in binary}]}_{\text{at most } \lceil\log_2(n+1)\rceil \text{ bits}}\underbrace{\left[\text{describe which of the } \binom{n}{k} \text{ sequences it is}\right]}_{\text{can be done in } \le\left\lceil\log_2\binom{n}{k}\right\rceil \text{ bits}}.$$

Thus, setting $k=\{\#$ of 1's in $u^n\}$, we finally obtain:

$$\text{length}(c(u^n))\le2+\log_2(n+1)+\log_2\binom{n}{k}\le2+\log_2(n+1)+nh_2\left(\frac{k}{n}\right). \tag{A.20}$$